

Calabi–Yau Quantum Groups

Factorization, Hall Algebras, Chiral Centres, Quantum Doubles,
and Automorphic Characters

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Abstract

This monograph gives a single complete language for Calabi–Yau categories, local quantum fields, ordered and associative chiral algebras, Hall and BPS algebras, quantum groups, centres, anomalies, and automorphic products. Its basic input is an integral Calabi–Yau quantum-group datum containing the categorical, BV, boundary, Hall, Hopf, and character layers together with their comparison morphisms. The cyclic Deligne and shifted-symplectic theorems provide the universal categorical avatar. The quantum master equation provides factorization observables. Boundary restriction provides the ordered algebra and bulk-to-centre map. Oriented extension correspondences provide Hall multiplication. A paired triangular Hopf datum provides the completed double and exchange. The K3 chapters construct the universal action envelope generated by Heisenberg, LLV, and local ADE correspondences. The $K3 \times E$ chapters construct the universal Igusa Hall–Borcherds target and state the exact parity-refined criterion for geometric equivalence. The final BV and Einstein completion identifies the deformation algebra, field equations, Bianchi identities, propagators, anomalies, and gravitational sectors of the same mixed boundary datum.

The theorem spine

The work is organized by the order in which geometry creates algebra:

collision $\longrightarrow$ ordered fusion $\longrightarrow$ centre $\longrightarrow$ Hall extension
 $\longrightarrow$ quantum double $\longrightarrow$ character.

Each arrow is constructed in a fixed category. Each inverse limit uses a proper positive filtration. Each physical interpretation is the image of a displayed open–closed, Hall, current, trace, or determinant morphism. The central examples are calculated in generators and relations before their abstract generalization.

Notation and normalization

The ground field has characteristic zero. Stable categories are presentable and idempotent-complete. Tensor products preserve colimits separately. Completions are separated inverse limits of finite weight or proper-height quotients. The Igusa normalization is

$$\Delta_5^\theta = 64D_5, \quad \chi_{10} = D_5^2,$$

with D_5 monic. Physical charge is additive; Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

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BOOK **I**

The invariant architecture

The complete datum

A Calabi–Yau category, a local BV theory, a Hall correspondence, and a quantum group occupy different mathematical categories. The theory begins by writing all four objects and every comparison map.

Definition 1.1 (Integral Calabi–Yau quantum-group datum). An integral Calabi–Yau quantum-group datum is

$$\mathfrak{D} = (C, \theta_{CY}; X, \mathcal{L}, \omega, S, Q; b, A_b; \mathcal{M}, \mathcal{E}, p, q, \varphi, o; H^+, H^0, H^-, \Delta, \langle -, - \rangle; \mathcal{Z}),$$

where:

1. (C, θ_{CY}) is a smooth or proper Calabi–Yau category with its negative-cyclic class;
2. $(X, \mathcal{L}, \omega, S, Q)$ is an actual cyclic elliptic BV theory and an anomaly-cancelled quantization;
3. b is a boundary generator and $A_b = \mathrm{RHom}(b, b)$ is its ordered factorization endomorphism algebra;
4. $(\mathcal{M}, \mathcal{E}, p, q, \varphi, o)$ is an oriented Hall moduli problem with coefficient theory φ ;
5. $(H^+, H^0, H^-, \Delta, \langle -, - \rangle)$ is a complete paired triangular Hopf datum;
6. $\mathcal{Z}$ is a character, determinant, or automorphic section of the constructed algebraic object.

All comparison morphisms are part of the datum and preserve their stated filtration, grading, and operadic structure.

The formal categorical avatar

A d -Calabi–Yau category universally supplies framed E_2 Hochschild cochains and, on its representable moduli stack, a $(2 - d)$ -shifted symplectic form. These two structures determine a formal local avatar prior to the choice of a spacetime field theory.

Definition 2.1 (Formal factorization envelope). Fix a presentable formal-disk operadic category. For a framed E_2 algebra B , let $\text{FactEnv}_{\text{form}}(B)$ be the free formal factorization object generated by B in that category. This construction is the relative operadic left adjoint to the formal-disk restriction functor.

Theorem 2.2 (Canonical formal Calabi–Yau avatar). A Calabi–Yau category (C, θ_{CY}) determines

$$CC^*(C), \quad \text{FactEnv}_{\text{form}}(CC^*(C)), \quad \mathcal{M}_C^{(2-d)\text{-symp}}.$$

A morphism preserving the Calabi–Yau class induces the corresponding morphisms of framed Hochschild algebras, formal factorization envelopes, and shifted-symplectic moduli whenever the moduli functor is representable.

Proof. The cyclic Deligne theorem gives the framed E_2 structure. Relative operadic induction gives the formal envelope. Pantev–Toën–Vaquié–Vezzosi transgression gives the shifted-symplectic form. Naturality follows from functoriality of Hochschild theory, the adjunction, and shifted-symplectic transgression. $\square$

The formal avatar is the universal categorical layer. A geometric BV theory realizes it on spacetime through a supplied open–closed map. A Hall moduli problem realizes its extension algebra through a supplied comparison. The distinction makes every construction composable.

The assembly theorem

Theorem 3.1 (Integral assembly). *Every integral Calabi–Yau quantum-group datum determines functorially:*

1. *a quantum factorization algebra of observables on X ;*
2. *an ordered boundary algebra A_b and a bulk-to-centre morphism*

$$\mathrm{Obs}_Q \longrightarrow Z_{\mathrm{ord}}(A_b);$$

3. *an associative Hall algebra*

$$H^+ = \bigoplus_{\gamma} H_*(\mathcal{M}_{\gamma}, \varphi_{\gamma} \otimes o_{\gamma});$$

4. *a completed double $D(H^+, H^0, H^-)$ with universal R -matrix on every convergent continuous module category;*
5. *a local–wrapped current object after a projection to a curve and an elliptic-degree decomposition;*
6. *a character $\mathcal{Z}$ obtained from the chosen trace or automorphic construction.*

The construction commutes with morphisms of integral data.

Proof. The quantum master equation gives the observable differential and factorization maps. Boundary restriction gives A_b , and the action on every boundary module gives the centre map. The two-step extension correspondence and base change give associativity of Hall convolution. The Hopf pairing gives the completed double and its canonical tensor. Factorization pushforward and the local–wrapped colour give the curve object. The selected trace gives the character. Naturality is inherited at every stage. $\square$

The physical experiment

The mathematics is read from one experiment. Operators approach along a complex curve and produce a singular chiral coefficient. Defects approach along an oriented line and compose in order. Motion in a second topological direction produces exchange. Closing a defect produces a trace. Extending one brane by another produces Hall multiplication. Integrating out a field produces a determinant line. A Calabi–Yau pairing identifies incoming and outgoing states. The resulting structures are collision, fusion, braiding, trace, extension, anomaly, and duality. The assembly theorem records them in their native categories and connects them by explicit maps.

Book entrance: Ordered and Associative Chiral Geometry

*Collision on a complex curve, order on a transverse line, exchange in a plane,
and the three corresponding bar constructions.*

BOOK **II**

The invariant object

Two directions of locality

5.1 The local experiment

Let X be a smooth complex algebraic curve and let $I \subset \mathbb{R}$ be an oriented interval. Place two protected observables at (z_1, t_1) and (z_2, t_2) in $X \times I$. The limits $z_1 \rightarrow z_2$ and $t_1 \rightarrow t_2$ carry independent information. The first limit produces a singular distribution along the diagonal of X^2 . The second composes operators in the order selected by the orientation of I .

Principle 5.1 (Two-direction principle). Holomorphic collision belongs to the factorization coefficient on X . Ordered fusion belongs to an internal E_1 multiplication. Exchange belongs to an additional two-dimensional transport structure.

This separation gives five geometric carriers:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered fusion and bar construction
oriented plane	braid transport
circle	Hochschild trace
stable curve	modular sewing.

Each later structure is introduced by the datum carried by its row.

5.2 Carrier and operation

For the diagonal embedding $\Delta : X \hookrightarrow X^2$ and its complement $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$, a Beilinson–Drinfeld chiral Lie operation is a morphism

$$\mu : j_* j^*(\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_* \mathfrak{g}$$

in the chiral pseudo-tensor category of right D -modules [BD04]. The D -module is the carrier. The morphism μ is the operation. A factorization object packages carriers on all powers X^I together with their disjoint-locus structure maps. The complete operator product is the pair consisting of that system and its multiplication maps.

In a formal coordinate z near the diagonal, the pole filtration of $\mathcal{O}_{X^2}(*\Delta)$ has

$$\mathrm{gr}_m^F \mathcal{O}_{X^2}(*\Delta) \cong \mathcal{O}_\Delta(m\Delta) \cong N_{\Delta/X^2}^{\otimes m}, \quad m \geq 1.$$

A principal part of order m therefore has an intrinsic normal symbol. Coordinate changes act triangularly on the filtered object. Residues recover the first symbol; normal jets recover the higher symbols.

The Ran coefficient category

6.1 The fixed six-functor setting

Fix a presentable stable category $\mathcal{D}_X$ of right D -modules on $\mathrm{Ran}(X)$ with the following operations:

1. the pointwise tensor product $\otimes^!$;
2. chiral convolution $\otimes^{\mathrm{ch}}$ along disjoint union;
3. base change and projection formulae for the disjoint-union correspondences;
4. geometric realizations and totalizations preserved by the tensor products used below.

The standard renormalized D -module setting of chiral Koszul duality has these operations [FG12]. Every construction below is formed in this fixed setting.

Let

$$\mathrm{Ran}(X) \times \mathrm{Ran}(X) \xleftarrow{j} (\mathrm{Ran}(X) \times \mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{u} \mathrm{Ran}(X)$$

be the disjoint-union correspondence. Define

$$F \otimes^{\mathrm{ch}} G = u_! j^!(F \boxtimes G).$$

The pointwise product is induced by the diagonal of $\mathrm{Ran}(X)$.

Definition 6.1 (Aligned-decomposition projector). For a finite set I , let $\mathrm{Dec}(I)$ be the finite groupoid of ordered decompositions $I = A \sqcup B$ and let

$$a_I : \mathrm{Dec}(I) \longrightarrow \mathrm{Dec}(I) \times \mathrm{Dec}(I), \quad d \longmapsto (d, d)$$

be the diagonal. In the linear species model define

$$e_I = a_{I,!} a_I^!.$$

Thus e_I projects a pair of decompositions onto the summands in which the two decompositions agree. In the six-functor Ran model the same formula denotes extraordinary pullback to the diagonal correspondence followed by the available renormalized pushforward.

Theorem 6.2 (Aligned sector in the finite decomposition model). *Let $\mathcal{S}_I$ be a stable category indexed by ordered decompositions of a finite set I , and assume the direct-sum decomposition*

$$\mathcal{M}_I \simeq \bigoplus_{d_1, d_2 \in \text{Dec}(I)} \mathcal{M}_{d_1, d_2}.$$

The projection

$$e_I : \mathcal{M}_I \longrightarrow \mathcal{M}_I, \quad e_I|_{\mathcal{M}_{d_1, d_2}} = \begin{cases} \text{id}, & d_1 = d_2, \\ 0, & d_1 \neq d_2, \end{cases}$$

is idempotent. Its image is the aligned sector

$$\mathcal{M}_I^{\text{al}} = \bigoplus_{d \in \text{Dec}(I)} \mathcal{M}_{d, d}.$$

The family e_I is natural under bijections and compatible with iterated disjoint union. A mixed distributive law in this model is a natural transformation defined on $\mathcal{M}_I^{\text{al}}$ whose two restrictions to the triple diagonal $d_1 = d_2 = d_3$ agree.

Proof. Idempotence is checked on each direct summand. Naturality follows because a bijection carries an ordered decomposition to an ordered decomposition and preserves equality of the two indices. For three products, both iterated alignments project

$$\bigoplus_{d_1, d_2, d_3} \mathcal{M}_{d_1, d_2, d_3} \quad \text{onto} \quad \bigoplus_d \mathcal{M}_{d, d, d}.$$

The two projectors are therefore equal. This equality is precisely the triple-alignment coherence. $\square$

Theorem 6.3 (Global aligned projector). *In the fixed six-functor Ran category, the common-decomposition inclusion is a finite clopen inclusion on every finite-set chart. The functors*

$$e_I = a_{I,!} a_I^!$$

therefore assemble to a natural idempotent e_{al} on the mixed correspondence object. Its image is the common-decomposition correspondence. The two iterated projectors on three decomposition indices coincide, and Beck–Chevalley transports this equality through every Ran transition map.

Proof. The decomposition index is a finite discrete groupoid. Equality of two decomposition indices selects a union of connected components, so the diagonal inclusion is clopen and its extension-by-zero followed by restriction is an idempotent. A surjection of finite sets pulls equal decompositions to equal decompositions, which gives naturality. For three indices, both iterated projectors select the same clopen union $d_1 = d_2 = d_3$. The six-functor base-change isomorphisms identify the corresponding pull-push composites. $\square$

The aligned image carries the canonical interchange obtained by applying the projection formula to one common disjoint-union correspondence. Locality states that transverse multiplication is a morphism for this factorization coproduct. A strong interchange occurs on the full mixed object exactly when every off-aligned summand vanishes.

6.2 Factorization coalgebras

A chiral commutative coalgebra is a commutative coalgebra object for $\otimes^{\text{ch}}$. It is factorizing when every coproduct becomes an equivalence on the relevant disjoint locus. Let

$$\text{Fact}_D(X) \subset \text{CoAlg}_{\text{ch}}(\mathcal{D}_X)$$

be the full subcategory of factorizing objects. Chiral Koszul duality identifies the corresponding Lie-theoretic and factorization descriptions on the completed Ran category [FG12, Heu24].

Proposition 6.4 (Pointwise closure). *The componentwise tensor product of two factorization objects is a factorization object. Thus $\text{Fact}_D(X)$ is symmetric monoidal under $\otimes^!$ whenever the fixed Ran enhancement is closed under that tensor product.*

Proof. For a decomposition $I = I_1 \sqcup I_2$, restrict to its disjointness locus. The factorization equivalences of F and G give

$$\begin{aligned} j^!(F_I \otimes^! G_I) &\simeq j^! F_I \otimes^! j^! G_I \\ &\simeq (F_{I_1} \boxtimes F_{I_2}) \otimes^! (G_{I_1} \boxtimes G_{I_2}) \\ &\simeq (F_{I_1} \otimes^! G_{I_1}) \boxtimes (F_{I_2} \otimes^! G_{I_2}). \end{aligned}$$

Compatibility under refinement and permutation follows componentwise from the corresponding compatibilities for F and G . $\square$

The two noncommutative chiral objects

7.1 Transverse order

A transversely ordered chiral algebra is an E_1 -algebra in the pointwise coefficient category. Its multiplication composes two operators that occupy the same chiral support and are ordered in an independent real direction. This is the structure carried by a boundary or line defect in a holomorphic–topological theory on $X \times \mathbb{R}$.

Definition 7.1 (Transversely ordered chiral algebra). A transversely ordered chiral algebra is an augmented object

$$A^\perp \in \text{Alg}_{E_1}^{\text{aug}}(\text{Fact}_D(X), \otimes^!).$$

The superscript $\perp$ records the transverse oriented direction. The shorter phrase *ordered chiral algebra* will always mean this object.

7.2 Associativity inside chiral convolution

Chiral convolution combines disjoint finite subsets before allowing them to collide. Associativity for this product is intrinsic to the curve and has a different source and a different bar construction.

Definition 7.2 (Associative chiral algebra). An associative chiral algebra is an augmented algebra

$$A_{\text{Ass}}^{\text{ch}} \in \text{Alg}_{E_1}^{\text{aug}}(\mathcal{D}_X, \star^{\text{ch}}).$$

A chiral Lie algebra is a Lie algebra object

$$\mathfrak{g}^{\text{ch}} \in \text{Alg}_{\text{Lie}}(\mathcal{D}_X, \star^{\text{ch}}).$$

The commutator of an associative chiral algebra is a chiral Lie algebra.

The operation $m^\perp : A \otimes^! A \rightarrow A$ composes transverse intervals. The operation $m^{\text{ch}} : A \star^{\text{ch}} A \rightarrow A$ composes chiral configurations. They have different domains, different units, and different deformation complexes.

Theorem 7.3 (Independence of the two associative structures). *Let M be an object for which the split square-zero extensions below exist. A pointwise associative product $\mu_\perp : M \otimes^! M \rightarrow M$ defines a transversely ordered algebra on $\mathbf{1}_\perp \oplus M$, while the reduced chiral-convolution product may independently be zero. A chiral-convolution product $\mu_{\text{ch}} : M \star^{\text{ch}} M \rightarrow M$ defines an associative chiral algebra on $\mathbf{1}_{\text{ch}} \oplus M$, while the reduced pointwise product may independently be zero. Hence neither associative structure determines the other.*

A common carrier equipped with both products is a mixed object after a mixed distributive datum has been supplied and its unit, binary, and triple coherences have been verified.

Proof. Adjoining the unit to an associative product on the augmentation ideal gives a unital split extension. The zero product is associative in either monoidal category. These two constructions vary independently. On a common carrier, compatibility is additional structure because the mixed composites have different a priori factorizations through the two monoidal products. $\square$

Definition 7.4 (Mixed associative chiral datum). A mixed associative chiral datum is a factorization object A equipped with

1. a pointwise multiplication $m^\perp : A \otimes^! A \rightarrow A$ and a unit map $\eta_\perp : \mathbf{1}_\perp \rightarrow A$;
2. a chiral-convolution multiplication $m^{\text{ch}} : A \star^{\text{ch}} A \rightarrow A$ and a unit map $\eta_{\text{ch}} : \mathbf{1}_{\text{ch}} \rightarrow A$;
3. an aligned mixed distributive law identifying the two composites on $\text{im}(e)$.

The coherence diagrams are the two color-3 faces of the mixed forest complex. A bare symbol $\mathbf{1}$ below denotes the unit of the monoidal category in which the displayed construction is formed.

7.3 Formal-disc realization

Fix a formal coordinate on a disc. Translation-equivariance turns a chiral-convolution product into a state-field operation $Y(a, z)b$ with coefficients in Laurent series. The following theorem packages the exact relationship with the vertex-algebra language.

Theorem 7.5 (Formal-disc realizations). *Let V be a translation-equivariant state–field system with vacuum and injective state–field map.*

1. *If its fields are weakly associative in the sense of Li, then V is a nonlocal vertex algebra.*
2. *If, in addition, V is topologically free over $\mathbb{k}[[\hbar]]$, carries a unitary rational quantum Yang–Baxter operator $S(z)$, and satisfies S -locality and the hexagon compatibility, then V is a quantum vertex algebra.*
3. *A translation-equivariant associative chiral-convolution algebra produces such a state–field system whenever its formal-disc restriction satisfies the reconstruction, lower-truncation, and weak-associativity hypotheses. Conversely, a nonlocal or quantum vertex algebra produces the corresponding formal-disc factorization system by its products of fields.*

Proof. The first assertion is Li’s reconstruction theorem for quasi-compatible fields [Li05]. The second is the quantum-vertex reconstruction theorem: unitarity gives inverse exchange, the quantum Yang–Baxter equation compares the two three-field exchange paths, and the hexagon relates exchange to the vertex operation [Li09, DSGK19]. For the third assertion, the formal-disc chiral product supplies the iterated field products; the stated analytic-formal hypotheses are exactly those used by the reconstruction theorems. The converse assigns to a configuration of formal points the iterated field product and uses weak associativity, or braided associativity, for factorization under refinement. $\square$

7.4 Three separation laws

The complete principal-part filtration, transverse order, and planar exchange form a strict hierarchy of supplied data. The m th normal symbol of a pole is recovered by the m th normal-jet map. An oriented line supplies contractible order chambers and hence E_1 fusion. A plane supplies braid paths and hence exchange. These positive constructions locate each operation in the first geometry that carries it.

Transversely ordered chiral algebras

Definition 8.1 (Ordered chiral algebra). An ordered chiral algebra on X is an augmented E_1 -algebra object

$$A \in \text{Alg}_{E_1}^{\text{aug}}(\text{Fact}_D(X), \otimes^!).$$

Equivalently, A is a factorization object equipped with multiplication maps

$$m_I : A_I \otimes^! A_I \longrightarrow A_I$$

and a unit, compatible with all diagonal transition maps and all disjoint-locus factorization equivalences.

For every decomposition $I = I_1 \sqcup I_2$, locality is the componentwise identity

$$\Delta_{I_1, I_2} \circ m_I = (m_{I_1} \boxtimes m_{I_2}) \circ \sigma \circ (\Delta_{I_1, I_2} \otimes^! \Delta_{I_1, I_2}),$$

where σ reorders the middle two factors. This equation says that ordered fusion commutes with separation of chiral supports.

Theorem 8.2 (Locality recognition). *To give an ordered chiral algebra is equivalent to giving a factorization family A_I , an associative unital multiplication m_I on every component, and the displayed locality identities, compatibly with refinements, permutations, and diagonal transitions.*

Proof. An E_1 -algebra in the symmetric monoidal category $\text{Fact}_D(X)$ is, by definition, an associative multiplication and unit whose structure maps are morphisms in $\text{Fact}_D(X)$. A morphism in $\text{Fact}_D(X)$ is exactly a componentwise morphism compatible with the transition and factorization maps. Expanding this compatibility for a binary decomposition gives the displayed identity; the higher decompositions follow by iteration. $\square$

8.1 Product field theories

Let $\text{Fact}_{\text{hol,lc}}(X^{\text{an}} \times \mathbb{R})$ denote factorization algebras that are holomorphic along X and locally constant along $\mathbb{R}$. Product additivity gives

$$\text{Fact}_{\text{hol,lc}}(X^{\text{an}} \times \mathbb{R}) \simeq \text{Alg}_{\mathbb{E}_1}(\text{HolFact}(X^{\text{an}}))$$

for product-generated objects. Algebraization by regular holonomic Riemann–Hilbert then gives ordered chiral algebras in $\text{Fact}_D(X)$ whenever the coefficient system is regular holonomic and the comparison preserves the required colimits.

Theorem 8.3 (Holomorphic–topological recognition). *Let $\mathcal{F}$ be a product-generated factorization algebra on $X^{\text{an}} \times \mathbb{R}$ that is holomorphic on X and locally constant on $\mathbb{R}$. Then the restriction of $\mathcal{F}$ to a transverse point is an $\mathbb{E}_1$ -algebra in holomorphic factorization algebras on X . A regular-holonomic algebraization produces an ordered chiral algebra on X .*

Proof. An ordered family of disjoint intervals in $\mathbb{R}$ acts on the restriction by factorization. Contractibility of each ordered configuration chamber gives the coherent $\mathbb{E}_1$ structure. Factorization in the X direction is preserved because product embeddings commute. Riemann–Hilbert transports the structure in the regular-holonomic subcategory. $\square$

Block-ordered operations

The preceding definition is intrinsic. A geometric model arises after one chooses an infinity-operad Ch_X representing the selected chiral factorization theory. The Boardman–Vogt tensor product

$$\mathrm{OrdCh}_X = \mathrm{E}_1 \otimes \mathrm{Ch}_X$$

then represents compatible ordered and chiral operations.

Theorem 9.1 (Operadic exponential law). *Let P and Q be infinity-operads whose Boardman–Vogt tensor product exists, and let $\mathcal{V}$ be a compatible presentable symmetric monoidal infinity-category. Then*

$$\mathrm{Alg}_{P \otimes Q}(\mathcal{V}) \simeq \mathrm{Alg}_P(\mathrm{Alg}_Q(\mathcal{V})).$$

In particular, after a chiral operad Ch_X has been constructed, the tensor product $\mathrm{E}_1 \otimes \mathrm{Ch}_X$ classifies E_1 -algebra objects in Ch_X -algebras.

Proof. The tensor product of infinity-operads is characterized by the displayed universal property [Lur17]. The second statement is the specialization $P = \mathrm{E}_1$ and $Q = \mathrm{Ch}_X$. This formal equivalence applies to each geometric model of Ch_X after that model and its operadic tensor product have been constructed. $\square$

A profile is an ordered list $(I_1, \dots, I_r)$ of finite chiral label sets. Its geometric carrier is $X^{I_1} \times \dots \times X^{I_r}$. Chiral morphisms refine and collide labels inside one block. Transverse morphisms merge adjacent blocks. The tensor-product operad imposes their mixed squares.

9.1 Mixed forests

Fix a cofibrant dg-operad $Q_X \rightarrow \text{Ch}_X$. Apply the functorial W -construction to $E_1 \otimes Q_X$. A cell is represented by a levelled forest with transverse and chiral internal edges. Put

$$\text{or}(F) = \det(E_\perp(F)) \otimes \det(E_{\text{ch}}(F)).$$

The differential is the sum of the screen differential and one-edge contractions.

Theorem 9.2 (Two-color forest differential). *Let C_* be the chain complex freely generated by oriented two-color forests, with one determinant-line generator for every transverse edge and one for every chiral edge. Suppose contraction of an edge is functorial on the coefficient system and every codimension-two contraction square commutes before orientations are inserted. With the product orientation, the transverse and chiral contraction operators satisfy*

$$d_\perp^2 = 0, \quad d_{\text{ch}}^2 = 0, \quad d_\perp d_{\text{ch}} + d_{\text{ch}} d_\perp = 0.$$

Thus $d = d_{\text{int}} + d_\perp + d_{\text{ch}}$ squares to zero whenever the internal differential anticommutes with both contraction operators.

Proof. Every term of $d_\perp^2$ contracts the same pair of transverse edges in two orders. The underlying coefficient maps agree by functoriality, while exchanging the two transverse determinant generators reverses the sign. The terms cancel in pairs. The same argument proves $d_{\text{ch}}^2 = 0$. For one edge of each color the coefficient maps again form a commuting square and exchanging the two determinant generators contributes the Koszul sign -1 , proving the mixed anticommutator identity. The internal terms vanish by the chain-map assumption. $\square$

This construction supplies a geometric model whenever a concrete Q_X and its screen coefficients are available. The intrinsic definition of ordered chiral algebra remains independent of that choice.

BOOK **III**

Bar, cobar, and the boundary state

Three bar constructions

The phrase *the chiral bar* denotes three functors once its monoidal product and algebraic operad have been named.

Definition 10.1 (Transverse, associative-chiral, and Lie-chiral bars). For augmented objects of the indicated types define

$$\mathrm{Bar}^\perp(A^\perp) := \mathbf{1} \otimes_{A^\perp}^{\mathbb{L},!} \mathbf{1},$$

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(A_{\mathrm{Ass}}^{\mathrm{ch}}) := \mathbf{1} \star_{A_{\mathrm{Ass}}^{\mathrm{ch}}}^{\mathbb{L}} \mathbf{1},$$

and

$$\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}) := C_*^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}}[1])$$

with its chiral Chevalley differential. The first bar is formed with $\otimes^!$, the second with $\star^{\mathrm{ch}}$, and the third is the Lie-to-commutative-coalgebra Koszul functor.

Theorem 10.2 (Typed bar theorem). *The three bars have the following chain models.*

1. $\mathrm{Bar}^\perp(A^\perp)$ is the tensor coalgebra $T_{\otimes^!}^c(s\bar{A})$ with coderivation induced by $m^\perp$.
2. $\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(A_{\mathrm{Ass}}^{\mathrm{ch}})$ is the tensor coalgebra $T_{\star^{\mathrm{ch}}}^c(s\bar{A})$ with coderivation induced by m^{ch} .
3. $\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g}^{\mathrm{ch}})$ is the chiral symmetric coalgebra with differential induced by the chiral Lie bracket.

Each differential squares to zero by the defining relation of its own operad. The outputs live in different coalgebra categories and are compared by a displayed monoidal or PBW morphism.

Proof. Normalize the two simplicial two-sided bars in their respective monoidal categories. Degeneracies remove units, leaving the two tensor coalgebras, and faces contract adjacent

factors using the relevant multiplication. The Lie formula is the ordinary operadic bar of the Lie operad internal to chiral convolution. Associativity and Jacobi are exactly the quadratic identities that make the corresponding coderivations square to zero. $\square$

Definition 10.3 (PBW-admissible chiral Lie algebra). A chiral Lie algebra $\mathfrak{g}^{\text{ch}}$ is PBW-admissible when its chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$ has a complete separated PBW filtration, the canonical map

$$\text{Sym}^{\text{ch}}(\mathfrak{g}) \longrightarrow \text{gr}_{\text{PBW}} U^{\text{ch}}(\mathfrak{g})$$

is an equivalence, and the bar filtration is strongly convergent in the chosen nilcomplete Ran category.

Theorem 10.4 (Filtered chiral PBW comparison). *Let $\mathfrak{g}$ be a chiral Lie algebra in a complete filtered chiral category. Assume:*

1. *its chiral enveloping algebra $U^{\text{ch}}(\mathfrak{g})$ exists;*
2. *the PBW filtration is exhaustive, complete, and separated;*
3. *the chiral PBW morphism*

$$\text{Sym}^{\text{ch}}(\mathfrak{g}) \longrightarrow \text{gr } U^{\text{ch}}(\mathfrak{g})$$

is an isomorphism;

4. *the induced filtrations on the two bar complexes are strongly convergent.*

Then antisymmetrization induces a filtered quasi-isomorphism

$$\text{Bar}_{\text{Ass}}^{\text{ch}}(U^{\text{ch}}(\mathfrak{g})) \simeq \text{Bar}_{\text{Lie}}^{\text{ch}}(\mathfrak{g}).$$

Proof. The associated graded of the associative chiral bar is the Koszul bar of $\text{Sym}^{\text{ch}}(\mathfrak{g})$. The usual Koszul antisymmetrization identifies this coalgebra with the chiral Chevalley coalgebra. Hypotheses (2)–(4) give a complete, separated, strongly convergent comparison spectral sequence. Since its associated-graded map is a quasi-isomorphism, the filtered map is a quasi-isomorphism. This is the PBW lane of chiral Koszul duality [BD04, FG12]. $\square$

Definition 10.5 (Bar-admissible mixed datum). A mixed associative chiral datum is bar-admissible when its aligned distributive law acts on the full bisimplicial two-sided bar and both geometric realizations commute with the tensor products occurring in that bisimplicial object.

Theorem 10.6 (Mixed Fubini theorem). *Let A be a mixed algebra in a presentable stable category. Assume the mixed distributive datum extends to the bisimplicial two-sided bar, and that both monoidal products preserve its geometric realizations separately. Then the two iterated realizations and the diagonal realization are canonically equivalent:*

$$|p \mapsto |q \mapsto B_{p,q}(A)|_{\text{ch}}|_{\perp} \simeq |\text{diag } B_{\bullet,\bullet}(A)| \simeq |q \mapsto |p \mapsto B_{p,q}(A)|_{\perp}|_{\text{ch}}.$$

Proof. The distributive datum makes the two simplicial directions commute, so $B_{\bullet,\bullet}(A)$ is a genuine bisimplicial object. Geometric realization is a colimit. Since the two tensor products preserve the relevant colimits, Fubini for colimits identifies both iterated realizations with the realization of the diagonal simplicial object. $\square$

The notation used in the remainder of this part is

$$\mathrm{Bar}_X^{\mathrm{ord}} = \mathrm{Bar}^\perp.$$

It always refers to transverse order.

The internal ordered bar

Let A be an augmented ordered chiral algebra with augmentation ideal $\bar{A}$. The simplicial two-sided bar has

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Its realization is

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) = |B_\bullet(\mathbf{1}, A, \mathbf{1})|.$$

Theorem 11.1 (Normalized bar). *In the fixed stable dg enhancement, Dold–Kan normalization identifies the geometric realization with the reduced tensor coalgebra. Explicitly, there is a natural equivalence*

$$\mathrm{Bar}_X^{\mathrm{ord}}(A) \simeq (T^c(s\bar{A}), d_{\mathrm{int}} + b_m),$$

where b_m is the coderivation induced by the multiplication of A . The deconcatenation formula

$$\Delta_{\mathrm{dec}}[a_1 | \cdots | a_n] = \sum_{i=0}^n [a_1 | \cdots | a_i] \otimes [a_{i+1} | \cdots | a_n]$$

makes the bar a conilpotent coassociative coalgebra in $\mathrm{Fact}_D(X)$.

Proof. Normalization removes degeneracies containing the unit and leaves tensor powers of $s\bar{A}$. The simplicial faces give the internal differential and adjacent multiplication. Associativity is precisely the equality that makes the multiplication coderivation square to zero. Deconcatenation is coassociative because a word cut twice has the same three pieces in either order. The bar differential is local with respect to every cut. $\square$

Theorem 11.2 (Double-coalgebra bar). *In every Ran enhancement in which geometric realization preserves chiral commutative coalgebras, $\mathrm{Bar}_X^{\mathrm{ord}}(A)$ carries the deconcatenation coproduct and the chiral factorization coproduct. Deconcatenation is a morphism of chiral coalgebras.*

Proof. Every simplicial operator is a morphism of factorization objects because the multiplication, unit, and augmentation satisfy the componentwise locality identity. The chiral coproduct therefore realizes levelwise. For a fixed decomposition $I = I_1 \sqcup I_2$, deconcatenation cuts the transverse word and factorization separates each letter into its I_1 - and I_2 -components. These two operations act on independent indices, so their two composites agree after the canonical symmetry. $\square$

11.1 Three coproducts

The theorem system uses three coproducts with distinct geometric meanings:

Δ_{fact} : separation of chiral support,

Δ_{dec} : cutting an ordered bar word,

Δ_{Hopf} : fusion of representations.

The first two are present on the transverse bar. The third is additional bialgebra data on an acting algebra.

Proposition 11.3 (Coproduct separation). *Factorization separation is carried by every ordered coefficient. Deconcatenation is generated functorially by the bar construction. A Hopf coproduct is supplied by an independent bialgebra structure. A coalgebra comparison morphism relates the Hopf coproduct to either geometric coproduct whenever a theory identifies their carriers.*

Proof. Factorization separation is part of the coefficient object, and deconcatenation is the canonical cut map on tensor words. A Hopf coproduct is a map from the algebra to a completed tensor square satisfying algebraic coassociativity. Its source, target, and defining universal property differ from the first two, so a relation among them is exactly the data of a comparison morphism. $\square$

11.2 Interval factorization homology

For a right A -module R and a left A -module L , factorization homology of an interval with endpoint labels gives

$$\int_{([0,1], \{0,1\})} (A; R, L) \simeq R \otimes_A^{\mathbb{L}} L.$$

This statement holds internally in every presentable monoidal category supporting factorization homology [AFT17, Lur17]. Taking $R = L = \mathbf{1}$ identifies the interval amplitude with the ordered bar.

Cobar and twisting cochains

Let C be a coaugmented conilpotent coalgebra in $\text{Fact}_D(X)$. The cobar algebra is

$$\text{Cobar}_X^{\text{ord}}(C) = (T(s^{-1}\bar{C}), d_{\text{int}} + d_{\Delta}).$$

A twisting cochain $\tau : C \rightarrow A$ has degree 1 and satisfies

$$d\tau + \tau d + m(\tau \otimes \tau)\bar{\Delta} = 0.$$

The universal twisting cochains are

$$\tau_A : \text{Bar}_X^{\text{ord}}(A) \rightarrow A, \quad \tau_C : C \rightarrow \text{Cobar}_X^{\text{ord}}(C).$$

Theorem 12.1 (Bar–cobar adjunction). *There is a natural adjunction*

$$\text{Cobar}_X^{\text{ord}} : \text{Coalg}_{\text{conil}}(\text{Fact}_D(X)) \rightleftarrows \text{Alg}_{\text{E}_1}^{\text{aug}}(\text{Fact}_D(X)) : \text{Bar}_X^{\text{ord}}.$$

Its mapping space is the space of twisting cochains.

Proof. An algebra map from the tensor algebra $T(s^{-1}\bar{C})$ is determined by its restriction to $s^{-1}\bar{C}$. Compatibility with the cobar differential is the Maurer–Cartan equation above. The same equation describes coalgebra maps from C to the cofree conilpotent coalgebra $T^c(s\bar{A})$. $\square$

Complete bar–cobar reconstruction

Let $A = \prod_{w \geq 0} A_w$ be complete for a positive weight grading, with $A_0 = \mathbf{1}$ and degreewise dualizable pieces. Let $F^p A = \prod_{w \geq p} A_w$. The invariant completion statement is the equivalence between nilcomplete algebras and conilcomplete Koszul-dual coalgebras; this is the maximal equivalence locus of operadic Koszul duality [Heu24]. The positive weight hypotheses below give an explicit sufficient criterion in the present dg setting.

Theorem 13.1 (Complete reconstruction). *Let $A = \prod_{w \geq 0} A_w$ be a positive complete augmented algebra with degreewise dualizable pieces. Assume products and inverse limits are exact on the weight towers used below. Let $C = \prod_{w \geq 0} C_w$ be a complete coaugmented coalgebra whose reduced coproduct is locally conilpotent in every weight, whose coradical filtration is exhaustive, and whose completion map to the inverse limit of its finite coradical quotients is an equivalence. Then the counit*

$$\mathrm{Cobar}_X^{\mathrm{ord}} \mathrm{Bar}_X^{\mathrm{ord}}(A) \longrightarrow A$$

is an equivalence, and the unit

$$C \longrightarrow \mathrm{Bar}_X^{\mathrm{ord}} \mathrm{Cobar}_X^{\mathrm{ord}}(C)$$

is an equivalence.

Proof. Filter both objects by total weight. On associated graded objects, the bar–cobar composite is the standard tensor resolution. Its contracting homotopy inserts the coaugmentation at the first admissible cut. The filtration is complete, separated, and bounded below in every total weight. The comparison spectral sequence therefore converges strongly and lifts the associated-graded equivalence. $\square$

13.1 Quadratic recognition

Let $A = T(V)/(R)$ with $R \subset V \otimes V$. Its quadratic dual coalgebra $A^{\text{q!}}$ is cogenerated by sV with corelations s^2R . The canonical map $A^{\text{q!}} \rightarrow \text{Bar}(A)$ gives the Koszul complex.

Theorem 13.2 (Quadratic Koszul criterion). *Assume positive weight and degreewise finite duals. The following conditions are equivalent:*

1. $A^{\text{q!}} \rightarrow \text{Bar}(A)$ is a quasi-isomorphism;
2. $\text{Cobar}(A^{\text{q!}}) \rightarrow A$ is a quasi-isomorphism;
3. the left and right Koszul complexes have homology concentrated in weight zero;
4. $\text{Tor}_i^A(\mathbf{1}, \mathbf{1})$ is concentrated in internal weight i .

Proof. The universal twisting morphism $A^{\text{q!}} \rightarrow A$ is Koszul exactly when the associated twisted tensor products resolve the unit. The fundamental theorem of twisting morphisms identifies this condition with the bar and cobar quasi-isomorphisms. Weight purity is the homological form of the same statement [Pri70, LV12]. $\square$

Modules, comodules, and Morita theory

A twisting cochain $\tau : C \rightarrow A$ defines twisted tensor functors between A -modules and C -comodules. Put $I = \ker(A \rightarrow \mathbf{1})$ and $F^p A = \prod_{w \geq p} A_w$. Write $\text{Mod}_A^{\wedge I}$ for the full derived category of left A -modules M for which

$$M \longrightarrow \varprojlim_p (M \otimes_A^{\mathbb{L}} A / F^p A)$$

is an equivalence. Write

$$D_{\text{conil}, \text{cpl}}^{\text{co}}(C\text{-comod})$$

for the coderived category of coaugmented C -comodules whose coradical filtration is exhaustive and complete. The coderived localization kills coacyclic totalizations and is the natural unbounded target for coalgebra modules [Pos11].

Theorem 14.1 (Koszul module equivalence). *Let $C = \text{Bar}(A)$ and let A satisfy the positive completeness and exact-limit hypotheses of the complete reconstruction theorem. Assume the module and comodule weight towers are degreewise finite or satisfy the corresponding Mittag–Leffler exactness. Then the universal twisting cochain induces an equivalence*

$$\text{Mod}_A^{\wedge I} \simeq D_{\text{conil}, \text{cpl}}^{\text{co}}(C\text{-comod}).$$

For complete right and left modules, relative tensor products correspond to derived cotensor products of the transformed comodules.

Proof. The unit and counit are represented by two-sided twisted bar complexes. Their complete weight filtrations have associated graded complexes equal to the ordinary tensor contractions. In each total weight the filtration is finite; the degreewise-finite or Mittag–Leffler

hypothesis removes derived-limit terms, so the spectral sequences converge strongly. Coacyclic totalizations vanish in the coderived target. The two-sided transform sends the diagonal A -bimodule to the diagonal C -bicomodule, and applying it to a two-sided resolution identifies derived tensor with derived cotensor. This is the complete algebra–coalgebra form of Koszul duality [Pos11, Lur17]. $\square$

Let C be a presentable factorization category and let b be a compact factorization generator. The internal endomorphism object

$$A_b = \mathrm{RHom}_C(b, b)$$

has ordered multiplication by composition.

Theorem 14.2 (Factorization Morita theorem). *Assume b generates each local category and the local Morita equivalences commute with factorization descent. Then*

$$C \simeq \mathrm{Mod}_{A_b}(\mathrm{Fact}_D(X)).$$

Proof. The compact-generator Morita theorem applies on every factorization chart. The descent hypothesis identifies the equivalences on intersections and hence on the Ran colimit. $\square$

Derived centre and bulk action

The ordered Hochschild cochains are

$$Z_{\text{ord}}(A) = \text{RHom}_{A \otimes A^{\text{op}}}(A, A).$$

Theorem 15.1 (Higher Deligne centre). *Let the ambient coefficient category be presentable, stable, symmetric monoidal, closed, and let tensor product preserve geometric realizations separately. Then $Z_{\text{ord}}(A)$ carries a canonical E_2 -algebra structure. It is the universal E_2 algebra acting centrally on A .*

Proof. Hochschild cochains carry the brace operations, and the brace operad is equivalent to chains on little two-disks. This is the Deligne theorem [MS02, KS00]. Universality is the identification of Hochschild cochains with endomorphisms of the identity A -bimodule. $\square$

A boundary condition in a field theory gives a morphism

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(A_{\partial}).$$

Locality makes the action central. A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

For a Calabi–Yau category, Hochschild cochains carry the framed E_2 refinement of the cyclic Deligne theorem [BR23].

Cyclic closure and modular sewing

A cyclic trace becomes a surface theory after dualizability and nondegeneracy are supplied.

Definition 16.1 (Perfect Calabi–Yau ordered algebra). A perfect Calabi–Yau ordered algebra is a dualizable ordered algebra A together with a trace

$$\mathrm{tr} : \mathrm{HH}_0(A) \longrightarrow \mathbf{1}[-d]$$

whose induced pairing $A \otimes A \rightarrow \mathbf{1}[-d]$ is nondegenerate.

Theorem 16.2 (Ribbon-graph realization of a Calabi–Yau algebra). *Let A be a homologically finite, smooth and proper cyclic A_∞ algebra with a nondegenerate pairing of degree d . Then the cyclic bar construction of A defines an open topological conformal field theory. Its operations are indexed by chains on moduli spaces of bordered surfaces, equivalently by the corresponding ribbon-graph complexes. The value on the circle is Hochschild homology $\mathrm{HH}_*(A)$.*

Proof. The nondegenerate cyclic pairing identifies outgoing and incoming boundary labels and contracts the two flags of every internal ribbon edge. The A_∞ identities are exactly the boundary identities for metric ribbon graphs. Smoothness and properness give the dualizability and finite pairing required for all contractions. Costello’s construction then produces the open TCFT and identifies the closed state complex with Hochschild chains [Cos07]. $\square$

Remark 16.3. The trace functional supplies the annular operation. Smoothness, properness, and the nondegenerate Calabi–Yau pairing extend it to the complete ribbon-surface theory of the theorem.

Stable-curve sewing is supplied by a modular operad action. Its contractions form the modular structure carried by the chiral coefficient. The two sewing geometries coexist when their contraction maps supercommute.

BOOK **IV**

Exact noncommutative models

Free and square-zero pairs

Let $M \in \text{Fact}_D(X)$.

$$T_X(M) = \mathbf{1} \oplus M \oplus M^{\otimes 2} \oplus \cdots$$

is the free augmented ordered algebra.

Theorem 17.1 (Free–square-zero duality). *Assume exact tensor products and dualizable M . Then*

$$\text{Bar}(T_X(M)) \simeq \mathbf{1} \oplus sM, \quad \text{RHom}_{T_X(M)}(\mathbf{1}, \mathbf{1}) \simeq \mathbf{1} \oplus M^\vee[-1]$$

with square-zero multiplication on the reduced summand. Conversely, for $A = \mathbf{1} \oplus M$ with $M^2 = 0$,

$$\text{Bar}(A) = T^c(sM), \quad A^\dagger \simeq \widehat{T}(M^\vee[-1]).$$

Proof. For the tensor algebra, every word of positive length has a unique last letter. The standard extra degeneracy contracts every bar simplex with an internal cut, leaving the one-letter class. Dualization gives the square-zero algebra. For the square-zero algebra, the multiplication part of the bar differential vanishes, leaving the cofree tensor coalgebra; its continuous dual is the completed tensor algebra. $\square$

The quantum plane

Let

$$A_q = \mathbb{k}\langle x, y \rangle / (yx - qxy), \quad q \in \mathbb{k}^\times.$$

Give x and y weight one.

Theorem 18.1 (PBW basis of the quantum plane). *The monomials $x^a y^b$, $a, b \geq 0$, form a basis of A_q .*

Proof. Use the rewriting rule $yx \mapsto qxy$. Every reduction decreases the inversion number of a word. The leading word yx has distinct initial and terminal letters, so every overlap decomposes into independent reductions. Bergman's diamond argument therefore gives the ordered monomials as normal forms. $\square$

The quadratic dual is

$$A_q^\dagger = \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Theorem 18.2 (Koszulness). *A_q is Koszul, and its Koszul complex is*

$$0 \rightarrow A_q \otimes \mathbb{k}(\xi\eta) \xrightarrow{d_2} A_q \otimes (\mathbb{k}\xi \oplus \mathbb{k}\eta) \xrightarrow{d_1} A_q \rightarrow \mathbb{k} \rightarrow 0,$$

with

$$d_1(a \otimes \xi + b \otimes \eta) = ax + by, \quad d_2(a \otimes \xi\eta) = ay \otimes \xi - qax \otimes \eta.$$

Proof. Direct substitution gives $d_1 d_2 = ayx - qaxy = 0$. The image of d_1 is the augmentation ideal (x, y) , so the complex is exact at A_q and has cokernel $\mathbb{k}$. We now compute $\ker d_1$.

Write

$$u = \sum_{i,j \geq 0} u_{ij} x^i y^j, \quad v = \sum_{i,j \geq 0} v_{ij} x^i y^j.$$

The relation $y^j x = q^j x y^j$ gives

$$ux + vy = \sum_{a,b \geq 0} (q^b u_{a-1,b} + v_{a,b-1}) x^a y^b,$$

where coefficients with a negative index are zero. Thus $ux + vy = 0$ implies $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Define $c = \sum_{i,j \geq 0} u_{i,j+1} x^i y^j$. Then $u = cy$ and, using $x^i y^j x = q^j x^{i+1} y^j$,

$$v = -qcx.$$

Hence $(u, v) = d_2(c)$. Finally, $d_2(c) = 0$ implies $cy = 0$; the PBW basis and $q \neq 0$ show that A_q is a domain, so $c = 0$. The complex is exact, and the quadratic Koszul criterion proves that A_q is Koszul. $\square$

Theorem 18.3 (Yoneda algebra of the quantum plane). *Use the convention that a Yoneda product $\alpha\beta$ means first β and then α . The augmentation module has*

$$\mathrm{Ext}_{A_q}^*(\mathbb{k}, \mathbb{k}) \cong \mathbb{k}\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi) = A_q^!$$

as a bigraded algebra, where $|\xi| = |\eta| = 1$ in cohomological degree and each generator retains its internal weight one.

Proof. Apply $\mathrm{Hom}_{A_q}(-, \mathbb{k})$ to the minimal Koszul resolution in the preceding theorem. Every matrix entry of d_1 and d_2 lies in the augmentation ideal, so the dual differential vanishes. The cohomology has basis

$$1, \quad \xi, \eta, \quad \xi\eta$$

in cohomological degrees 0, 1, 1, 2. Splicing the two degree-one extensions gives the degree-two class. The syzygy $y \otimes \xi - qx \otimes \eta$ identifies the two splicings by $\xi\eta = -q\eta\xi$, while the absence of degree-two x^2 and y^2 syzygies gives $\xi^2 = \eta^2 = 0$. These relations account for the complete four-dimensional cohomology, and therefore present the Yoneda algebra. $\square$

Tensoring A_q with any commutative factorization coefficient produces a concrete ordered chiral algebra with noncommutative transverse multiplication. Its bar, quadratic dual, and derived augmentation boundary are now explicit in generators and relations.

Quiver path algebras

Let Q be a finite quiver and $S = \bigoplus_{i \in Q_0} \mathbf{1}e_i$. Assign $M_a \in \text{Fact}_D(X)$ to each arrow $a : i \rightarrow j$ and set

$$M = \bigoplus_{a \in Q_1} e_j M_a e_i, \quad T_S(M) = S \oplus M \oplus M \otimes_S M \oplus \cdots.$$

Words are composable paths.

For the quiver $1 \xrightarrow{a} 2$, the path algebra has basis e_1, e_2, a and relation $a = e_2 a e_1$. The simple modules S_1, S_2 satisfy

$$\text{Ext}^1(S_1, S_2) \cong \mathbb{k}, \quad \text{Ext}^1(S_i, S_i) = 0, \quad \text{Ext}^1(S_2, S_1) = 0.$$

The bar differential records concatenation of paths, and the quadratic dual reverses the arrow with cohomological shift.

Theorem 19.1 (Relative path-algebra bar). *For a tensor path algebra $T_S(M)$ over semisimple S ,*

$$S \otimes_{T_S(M)}^{\mathbb{L}} S \simeq S \oplus sM$$

as an S -bicomodule. A quadratic quotient $T_S(M)/(R)$ is Koszul precisely when its relative quadratic complex is exact away from weight zero.

Proof. The last-arrow contraction is S -bilinear and gives the same extra degeneracy as in the free case. The quotient statement follows from the relative twisting-morphism criterion. $\square$

Weyl and Clifford deformations

Let V be finite-dimensional and let $\omega : V \otimes V \rightarrow \mathbb{k}$ be alternating. The Weyl algebra is

$$W_{\hbar}(V, \omega) = T(V)[[\hbar]] / \langle uv - vu - \hbar\omega(u, v) \rangle.$$

For a purely odd space and symmetric ω , the same formula with the supercommutator $uv + vu$ gives a Clifford algebra.

Theorem 20.1 (Filtered PBW theorem). *For alternating ω in the even Weyl case, or symmetric ω in the purely odd Clifford case, the associated graded algebra is $\text{Sym}(V)$ in the Weyl case and $\wedge(V)$ in the Clifford case.*

Proof. Filter by word length. The constant term lowers filtration by two, so the associated graded relation is the commutative or skew-commutative quadratic relation. The cubic ambiguities reduce to the cocycle identity for a central scalar form, which holds because scalars commute. The ordered monomials give the PBW basis. $\square$

The quadratic-linear-constant presentation has a curved Koszul dual. The constant term becomes the curvature functional

$$m_0 : s^2 R \longrightarrow \mathbb{k}[2].$$

This construction is the algebraic origin of the curvature; geometric anomaly curvature enters through a separate comparison morphism.

Associative chiral algebras on the formal disc

Let V be a complete topological vector space with vacuum $\mathbf{1}$ and translation T . A state–field map

$$Y(-, z) : V \longrightarrow \text{End}(V)((z))$$

is a coordinate presentation of an associative chiral-convolution product when its iterated products are expansions of one factorization kernel in the nested regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$.

Proposition 21.1 (Associative convolution from weak associativity). *Suppose $Y(\mathbf{1}, z) = \text{id}$, $Y(a, z)\mathbf{1} \in a + zV[[z]]$, $[T, Y(a, z)] = \partial_z Y(a, z)$, and for every a, b, c there is N such that*

$$(z_1 - z_2)^N Y(a, z_1)Y(b, z_2)c = (z_1 - z_2)^N Y(Y(a, z_1 - z_2)b, z_2)c$$

in the common formal expansion ring. Then the fields define an associative chiral algebra on the formal disc. The converse holds for a vacuum-generated translation-equivariant associative chiral algebra admitting state–field reconstruction.

Proof. The displayed identity is the equality of the two nested restrictions of the three-point chiral product. Vacuum and translation give the unit and coordinate covariance. Factorization on higher configuration spaces is obtained by iterating the equality; coherence follows from associativity of the chiral-convolution product. $\square$

Example 21.2 (Braided current fields). Let $R(z)$ be an invertible solution of the spectral Yang–Baxter equation, normalized by $R^{21}(-z)R(z) = 1$. Generating fields $a_i(z)$ with exchange

$$a_i(z_1)a_j(z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)a_l(z_2)a_k(z_1)$$

form a quantum vertex algebra whenever the vacuum, translation, and weak associativity relations hold. The R -matrix supplies the comparison between the two formal expansion regions and records planar exchange in addition to transverse order.

Zamolodchikov–Faddeev exchange

Let V be finite-dimensional and let $R(u) \in \text{End}(V \otimes V) \otimes \mathbb{k}(u)$ be invertible. Introduce generators $Z_i(u)$ with exchange relation

$$Z_1(u)Z_2(v) = R_{12}(u-v)Z_2(v)Z_1(u).$$

Theorem 22.1 (Yang–Baxter confluence). *The exchange algebra has a consistent normal ordering for three generators precisely when*

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v).$$

Proof. Reverse the word $Z_1(u)Z_2(v)Z_3(w)$. Exchanging positions in the order (12), (13), (23) gives the left side. Exchanging in the order (23), (13), (12) gives the right side. These are the two reductions of the unique length-three overlap. Equality is therefore the diamond condition. $\square$

For the rational permutation solution

$$R(u) = 1 - \frac{\hbar P}{u},$$

the Yang–Baxter equation follows from $P_{ij}^2 = 1$ and $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$.

The RTT Yangian of $\mathfrak{gl}_2$

Let

$$T(u) = 1 + \sum_{r \geq 1} T^{(r)} u^{-r}, \quad R(u) = 1 - \frac{\hbar P}{u}.$$

The RTT Yangian $Y_{\hbar}(\mathfrak{gl}_2)$ is generated by the coefficients $t_{ij}^{(r)}$ subject to

$$R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v).$$

Equivalently,

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar(t_{kj}(u)t_{il}(v) - t_{kj}(v)t_{il}(u)).$$

Proposition 23.1 (Coefficient relations). *Put $t_{ij}^{(0)} = \delta_{ij}$. The RTT identity is equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar(t_{kj}^{(r)}t_{il}^{(s)} - t_{kj}^{(s)}t_{il}^{(r)}), \quad r, s \geq 0.$$

In particular,

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first modes form the enveloping algebra of the copy $\hbar\mathfrak{gl}_2$ inside the Yangian.

Proof. Multiply the generating-series relation by $u-v$ and compare the coefficient of $u^{-r-1}v^{-s-1}$. The two terms contributed by multiplication by $u-v$ produce the two commutators on the left, and the two ordered products on the right are the corresponding coefficients of the RTT numerator. Setting $r=0$ gives the displayed first-mode relation. $\square$

Theorem 23.2 (Hopf structure). *The formulas*

$$\Delta(T(u)) = T(u) \dot{\otimes} T(u), \quad \varepsilon(T(u)) = 1, \quad S(T(u)) = T(u)^{-1}$$

define a Hopf algebra structure in the u^{-1} -adic completion. Coefficientwise,

$$\Delta(t_{ij}^{(r)}) = \sum_{a=1}^2 \sum_{p=0}^r t_{ia}^{(p)} \otimes t_{aj}^{(r-p)}.$$

Proof. Apply the RTT relation in each tensor factor. Matrix multiplication in the auxiliary space gives the RTT relation for $\Delta(T)$. Coassociativity is associativity of matrix multiplication. Expanding the matrix product gives the coefficient formula. The inverse series exists uniquely because $T(u) = 1 + O(u^{-1})$; its defining identities give the antipode axioms. $\square$

The evaluation map

$$\mathrm{ev}_a : T(u) \mapsto 1 + \frac{\hbar E}{u - a}, \quad E = (E_{ij}),$$

lands in $U(\mathfrak{gl}_2)[[u^{-1}]]$ and gives the standard evaluation modules. The PBW theorem identifies the associated graded with $U(\mathfrak{gl}_2[t])$ [Mol07, Dri85].

This example contains three structures: ordered multiplication in the Yangian, coproduct on tensor products of modules, and exchange by the rational R -matrix.

The Hall algebra of the A_2 quiver

Let $Q : 1 \rightarrow 2$ over $\mathbb{F}_q$. Write S_1, S_2 for the simples and M for the unique non-split extension

$$0 \rightarrow S_2 \rightarrow M \rightarrow S_1 \rightarrow 0.$$

In the twisted Hall algebra, multiplication counts subobjects with Euler-form normalization. Direct counting gives

$$[S_1] * [S_2] = v^{-1}[S_1 \oplus S_2] + v^{-1}[M], \quad [S_2] * [S_1] = [S_1 \oplus S_2], \quad v^2 = q.$$

Hence the root vector

$$E_{12} = E_1 E_2 - v^{-1} E_2 E_1$$

is represented by the indecomposable extension.

Theorem 24.1 (Quantum Serre relation). *The Hall generators satisfy*

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0,$$

and the analogous relation with 1 and 2 exchanged.

Proof. Every representation of dimension vector $(2, 1)$ is a linear map $f : \mathbb{F}_q^2 \rightarrow \mathbb{F}_q$ of rank zero or one. Read a word from the top quotient to the bottom quotient. The complete-flag counts are

rk f	112	121	211
0	$q + 1$	$q + 1$	$q + 1$
1	$q + 1$	1	0.

For rank zero every one-dimensional subspace of $\mathbb{F}_q^2$ is admissible, giving $q + 1$ flags in each word. For rank one, the word 121 forces the unique kernel line, the word 211 is incompatible with the final subrepresentation, and the word 112 allows every line at the remaining stage.

The total Euler-form exponents of the three words are respectively $-1, 0, 1$. Their twisted coefficients are therefore $v^{-1}, 1, v$. On the rank-zero stratum the Serre combination has coefficient

$$(q+1)(v^{-1} - (v + v^{-1}) + v) = 0.$$

On the rank-one stratum it has coefficient

$$v^{-1}(q+1) - (v + v^{-1}) = v^{-1}(v^2 + 1) - (v + v^{-1}) = 0.$$

The two isomorphism classes exhaust the dimension vector $(2, 1)$, proving the relation. This is Ringel's realization of $U_v^+(\mathfrak{sl}_3)$ [Rin90, Gre95]. $\square$

Matrix traces and the necklace bracket

Let $F = \mathbb{k}\langle x, y \rangle$ and $F_{\text{cyc}} = F/[F, F]$. Cyclic derivatives are defined by

$$\partial_x^{\text{cyc}}[a_1 \cdots a_n] = \sum_{a_i=x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1}$$

and similarly for y . Define

$$\{f, g\}_{\text{neck}} = [\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g]_{\text{cyc}}.$$

Theorem 25.1 (Trace realization). *For matrices $X, Y \in \mathfrak{gl}_N$ and the canonical symplectic bracket on matrix entries,*

$$\{\text{Tr } f(X, Y), \text{Tr } g(X, Y)\} = \text{Tr}\{f, g\}_{\text{neck}}(X, Y).$$

Proof. Differentiate the trace by moving the varied matrix entry cyclically to the first position. This gives the cyclic derivative. Contract the X derivative of one trace with the Y derivative of the other and subtract the reversed contraction. The matrix indices close into the trace of the cyclic product. $\square$

In the stable commuting-matrix quotient, let

$$p_{ab} = \text{Tr}(X^a Y^b).$$

Then

$$\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1, b+d-1}.$$

For two eigenvalues, the relative invariants

$$A = (x_1 - x_2)^2, \quad B = (x_1 - x_2)(y_1 - y_2), \quad C = (y_1 - y_2)^2$$

satisfy $B^2 = AC$ and carry the standard $\mathfrak{sl}_2$ Poisson bracket after linear rescaling.

BOOK **V**

The nonlocal ladder

Spectral exchange as an associative chiral operation

26.1 The coefficient field and the exchange operator

Let V be a finite-dimensional vector space and let K be a field of meromorphic functions in spectral differences. Fix an invertible operator

$$R(u) \in \text{End}(V \otimes V) \otimes K$$

that satisfies the spectral Yang–Baxter equation

$$R_{12}(u-v)R_{13}(u-w)R_{23}(v-w) = R_{23}(v-w)R_{13}(u-w)R_{12}(u-v). \quad (26.1)$$

The equation states that two routes through the chamber complex of three labelled points give the same transport.

Definition 26.1 (Zamolodchikov–Faddeev algebra). The exchange algebra $\mathcal{Z}_R$ is generated by the components of a current $A(u) \in V \otimes \mathcal{Z}_R$ with relation

$$A_1(u)A_2(v) = R_{12}(u-v)A_2(v)A_1(u). \quad (26.2)$$

A normal word has spectral labels in a fixed chamber $u_1 \succ \cdots \succ u_n$.

Theorem 26.2 (Confluence is Yang–Baxter). *The quadratic rewriting system defined by (26.2) is confluent on words with pairwise distinct spectral labels precisely when (26.1) holds. In that case the normal ordered words form a K -basis of $\mathcal{Z}_R$.*

Proof. Every two-letter inversion is removed by one application of (26.2). The minimal overlap occurs in a word of length three. Moving the leftmost label through the other two gives the coefficient $R_{12}R_{13}R_{23}$, while moving the rightmost label in the opposite sequence gives $R_{23}R_{13}R_{12}$. Equality of the two coefficients is (26.1). Bergman’s diamond lemma then gives the normal-word basis. $\square$

26.2 A diagonal exchange model

Choose a basis $e_1, \dots, e_N$ of V and functions $q_{ij}(u)$ satisfying

$$q_{ij}(u)q_{ji}(-u) = 1, \quad q_{ii}(u) = 1.$$

The diagonal operator

$$R(u)(e_i \otimes e_j) = q_{ij}(u)e_i \otimes e_j$$

satisfies the Yang–Baxter equation because all three factors are scalars. The generators $a_i(u)$ obey

$$a_i(u)a_j(v) = q_{ij}(u-v)a_j(v)a_i(u).$$

Every ordered monomial is a basis vector. The quantum plane is the constant two-generator specialization $q_{12} = q, q_{21} = q^{-1}$.

Calculation 26.3 (Three-particle coefficient). For $u \succ v \succ w$,

$$a_i(w)a_j(v)a_k(u) = q_{ij}(w-v)q_{ik}(w-u)q_{jk}(v-u)a_k(u)a_j(v)a_i(w).$$

The coefficient is independent of the sequence of adjacent exchanges. This scalar identity is the diagonal Yang–Baxter equation.

Proof. Each inversion of two adjacent letters contributes its scalar exchange factor. Both reduced words for the longest permutation invert the same three pairs. Scalar commutativity makes the products identical. $\square$

RTT and reflection algebras

27.1 The RTT bialgebra

Let $T(u) = (t_{ij}(u))$ be a matrix of formal currents. The RTT algebra $\mathcal{T}_R$ is defined by

$$R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v). \quad (27.1)$$

Theorem 27.1 (RTT coproduct). *The formula*

$$\Delta(T(u)) = T_{13}(u)T_{12}(u), \quad \epsilon(T(u)) = I,$$

defines a coassociative bialgebra structure on $\mathcal{T}_R$.

Proof. Apply (27.1) in each tensor factor. The Yang–Baxter equation moves the three R -operators through the three matrix factors. The resulting identity is the RTT relation for $\Delta(T)$. Matrix multiplication is associative, hence the two iterated coproducts both equal $T_{14}T_{13}T_{12}$. $\square$

For the Yang matrix $R(u) = 1 - \hbar P/u$, expansion at $u = \infty$ gives the RTT Yangian. The coefficient of $u^{-1}v^{-1}$ is

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}),$$

so the first current layer is the Rees enveloping algebra of $\mathfrak{gl}_N$.

27.2 Boundary exchange

A reflecting boundary reverses the spectral path. For a constant invertible $R \in \text{End}(V^{\otimes 2})$ satisfying Yang–Baxter, define the reflection algebra $\mathcal{K}_R$ by

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \quad (27.2)$$

Theorem 27.2 (Boundary confluence). *The relation (27.2) gives a coherent exchange law for three boundary insertions. The two reductions of every length-three overlap agree by the Yang–Baxter equation and its inverse.*

Proof. A boundary exchange consists of an incoming crossing, reflection, and outgoing crossing. The two reductions of three insertions sweep the same braid in the doubled half-plane. Algebraically, one inserts (27.2) three times and uses Yang–Baxter to pass the middle R -matrix through the outer pair. The remaining factors cancel in inverse pairs. $\square$

The RTT algebra governs bulk line exchange. The reflection algebra governs boundary line exchange. The ZF algebra governs charged fields in a fixed representation. Their common coefficient is the same R -matrix, while their products live in three different endomorphism objects.

Quantum vertex realization

28.1 Two expansion regions

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T , state-field map Y , and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 28.1 (Rational quantum vertex datum). A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1) Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1) \quad (28.1)$$

for a sufficiently large N , and weak associativity in every triple of states.

Theorem 28.2 (Exchange coherence). *Suppose S is unitary,*

$$S^{21}(-z)S(z) = 1,$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (28.1) define a braid-group action on every n -point field product. Weak associativity makes the action compatible with iterated operator products.

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. $\square$

When S is the graded flip, the datum is an ordinary local vertex algebra. A nontrivial S retains the ordered chiral product and its exchange monodromy. This is the formal-disc realization of an associative chiral algebra with braiding data in the sense of quantum vertex algebra theory [Li05, Li09].

28.2 The Yangian field

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S -locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u) \left(1 - \frac{\hbar P}{u} \right), \quad f(u)f(-u) = \left(1 - \frac{\hbar^2}{u^2} \right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

Shuffle realization and the positive current algebra

29.1 Pairwise kernel

Let $K(z)$ be a rational function and put

$$\text{Sh}_n = \mathbb{C}(z_1, \dots, z_n)^{\mathfrak{S}_n}.$$

For $f \in \text{Sh}_m$ and $g \in \text{Sh}_n$ define

$$(f * g)(z_1, \dots, z_{m+n}) = \text{Sym} \left(f(z_1, \dots, z_m) g(z_{m+1}, \dots, z_{m+n}) \prod_{i \leq m < j} K(z_i - z_j) \right). \quad (29.1)$$

Theorem 29.1 (Associativity of the shuffle product). *The product (29.1) is associative on every subspace closed under the displayed symmetrization and pole conditions.*

Proof. In a triple product, every ordered pair of variables belonging to different blocks contributes exactly one factor $K(z_i - z_j)$. The product of all pairwise factors is independent of the parenthesization. Final symmetrization gives the same function. $\square$

For the tripled Jordan quiver, one uses

$$K(z) = \prod_{a=1}^3 \frac{z + \epsilon_a}{z}, \quad \epsilon_1 + \epsilon_2 + \epsilon_3 = 0.$$

Wheel conditions remove the forbidden triple residues. The resulting shuffle algebra is the positive current algebra that appears in the critical CoHA of $\mathbb{C}^3$ and in the positive affine Yangian of $\mathfrak{gl}_1$ [SV23, Tsy17].

Calculation 29.2 (Degree one against degree one). For $f(z_1) = z_1^r$ and $g(z_1) = z_1^s$,

$$f * g = z_1^r z_2^s K(z_1 - z_2) + z_2^r z_1^s K(z_2 - z_1).$$

The antisymmetric part records the first current commutator. Its residues at $z_1 - z_2 = -\epsilon_a$ are the three elementary extension channels of the tripled quiver.

Proof. For two variables the symmetrization in the shuffle definition has two terms, which gives the displayed formula. Antisymmetrization exchanges the two chambers. The three numerator factors of the tripled-quiver kernel locate the three elementary shifted extension channels. $\square$

29.2 Coproduct and double

Choose a chamber expansion of $K(z)$ at infinity. Splitting the variables into two ordered groups and expanding every cross-kernel in that chamber defines a completed coproduct. A residue pairing between positive and negative shuffle halves gives a continuous Hopf pairing. The completed Drinfeld double is then a quasitriangular current algebra. Product, coproduct, and pairing are separate operations; each is visible in the shuffle formula that defines it.

The ladder in one diagram

The nonlocal structures form the chain

$$\begin{aligned} \text{ordered coefficient} &\longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S), \\ (V, Y, S) &\longrightarrow \text{Sh}^+ \longrightarrow D(\text{Sh}^+) . \end{aligned}$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S -locality and weak associativity
Sh^+	extension shuffle	pairwise-kernel associativity
$D(\text{Sh}^+)$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

BOOK VI

Field theory and master equations

A finite cotangent BF model

Let L be a finite-dimensional dg Lie algebra. Its cotangent BV space is

$$\mathcal{E}_L = L[1] \oplus L^*[-2],$$

with the canonical degree -1 symplectic pairing. Write $A \in L[1]$ and $B \in L^*[-2]$. The cotangent action is

$$S_L(A, B) = \langle B, d_L A + \frac{1}{2}[A, A] \rangle.$$

Theorem 31.1 (Cotangent master equation). *The Hamiltonian vector field of S_L is square-zero, and consequently $\{S_L, S_L\} = 0$.*

Proof. The A component is the Maurer–Cartan vector field $d_L A + \frac{1}{2}[A, A]$; the B component is its cotangent lift. The square of the first component is the sum of d_L^2 , the derivation identity for d_L , and the Jacobi identity. The cotangent lift preserves commutators of vector fields, so the full Hamiltonian vector field is square-zero. $\square$

Proposition 31.2 (Coordinate form of the modular term). *Let L be an ordinary finite-dimensional Lie algebra with basis e_a and structure constants $[e_a, e_b] = f_{ab}^c e_c$. Let c^a be the odd coordinates on $L[1]$ and b_a their BV-conjugate coordinates. With*

$$S = \frac{1}{2} b_c f_{ab}^c c^a c^b, \quad \Delta = - \sum_a \frac{\partial}{\partial c^a} \frac{\partial}{\partial b_a},$$

one has

$$\Delta S = \sum_b \text{Tr}(\text{ad}_{e_b}) c^b.$$

Thus the one-loop class of the finite cotangent theory is exactly the modular character.

Proof. Differentiation first in b_a gives $\frac{1}{2}f_{ij}^a c^i c^j$. Odd differentiation in c^a gives $f_{aj}^a c^j$, and the chosen sign in Δ converts this to $f_{ja}^a c^j = \text{Tr}(\text{ad}_{e_j})c^j$. This is the displayed formula. $\square$

A finite-dimensional dg Lie algebra is unimodular when the supertrace of every adjoint endomorphism vanishes. Every finite-dimensional nilpotent Lie algebra in degree zero is unimodular because each ad_x is nilpotent.

Theorem 31.3 (Finite-dimensional quantum master equation). *Let L be unimodular and let Δ be the BV Laplacian determined by the translation-invariant Berezinian on $\mathcal{E}_L$. Then*

$$\Delta S_L = 0, \quad \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L = 0,$$

and equivalently

$$\hbar \Delta(e^{S_L/\hbar}) = 0.$$

The quantum observable differential

$$\delta_{S_L} = \{S_L, -\} + \hbar \Delta$$

satisfies $\delta_{S_L}^2 = 0$.

Proof. The classical master equation gives $\{S_L, S_L\} = 0$. The scalar ΔS_L is the divergence of the Chevalley vector field on $L[1]$. Its linear coefficient is the modular character $x \mapsto \text{str}(\text{ad}_x)$, which vanishes by unimodularity. The BV exponential identity

$$\hbar^2 e^{-S_L/\hbar} \Delta(e^{S_L/\hbar}) = \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L$$

proves the exponential form. The standard BV commutator calculation then gives

$$\delta_{S_L}^2 = \left\{ \frac{1}{2}\{S_L, S_L\} + \hbar \Delta S_L, - \right\} = 0.$$

$\square$

Now take the local dg Lie algebra

$$\mathcal{L}_X = \Omega^{0,*}(X) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathbf{I}$$

for a finite-dimensional nilpotent $\mathbf{I}$, and form its shifted cotangent theory with compactly supported density dual. The preceding finite theorem governs every pairing-preserving finite-dimensional deformation retract. The local classical observables are Chevalley cochains and are holomorphic on X and locally constant on $\mathbb{R}$.

Choose the standard interval polarization in which the $L[1]$ variables are fixed at one end and the cotangent variables at the other. Canonical quantization of the boundary linear Poisson algebra gives the associative zero-mode algebra $U(\mathbf{I})$. Gluing

the two augmentation boundary states across an interval gives the derived tensor product

$$\mathbf{1} \otimes_{U(\mathbf{I})}^{\mathbb{L}} \mathbf{1} \simeq \mathrm{Bar}(U(\mathbf{I})) \simeq C_*(\mathbf{I}),$$

the Chevalley–Eilenberg chains. A solution of the renormalized quantum master equation in the local BV deformation complex produces the corresponding infinite-dimensional quantum factorization algebra.

BRST and ordered reduction

Let (A, Q) be a dg ordered algebra, with Q a derivation. Extend Q to the bar factorwise. Write b for the multiplication bar differential.

Proposition 32.1 (BRST–bar bicomplex). *With the suspension sign convention,*

$$Q_{\text{Bar}}b + bQ_{\text{Bar}} = 0.$$

Hence $D = Q_{\text{Bar}} + b$ satisfies $D^2 = 0$.

Proof. Apply Q to a bar face that multiplies adjacent entries. The derivation formula splits the result into the two terms obtained by applying Q before that face. The totalization sign makes their sum the anticommutator. $\square$

The filtration by bar length gives

$$E_1 \cong \text{Bar}(H^*(A, Q)) \implies H^*(\text{Bar}(A), D)$$

under boundedness and completeness. Quantum Drinfeld–Sokolov reduction is a current-algebra realization of this bicomplex.

Stokes and algebraic identities

Let $C_*(K)$ be the oriented cellular chains of a regular manifold with corners. Suppose each codimension-one face F is assigned a degree-one operator ρ_F on a coefficient complex E , and the assignments are compatible with restrictions to corners.

Theorem 33.1 (Stokes-to-master). *Assume that each ρ_F is a chain map of total degree one, that its restriction to a corner depends on the induced contraction, and that disjoint contractions act with the Koszul sign prescribed by the oriented normal bundle. Then the component of*

$$D^2, \quad D = d_E + d_K + \sum_F \rho_F,$$

supported on a codimension-two corner S is the oriented sum

$$\sum_{F \supset S} \epsilon(F, S) \rho_{F/S} \rho_F.$$

Thus D is a differential precisely when these corner sums vanish.

Proof. Expand D^2 and group terms by support strata. The chain-map condition cancels the mixed internal terms. A term supported on S is obtained by crossing two incident faces. The two orders carry the boundary orientations of the two sides of the oriented normal square. Their signed sum is the displayed expression. □

Associahedra give Stasheff identities. Fulton–MacPherson collision strata give chiral and graph identities after a coefficient map is supplied. Stable-graph boundaries give BV and modular master equations.

Two edge colors and two loop operators

Let $\mathbf{G}^{\perp, \text{ch}}$ be the completed vector space spanned by stable graphs with transverse and chiral edges. Contracting an edge that joins distinct vertices gives brackets $[-, -]_{\perp}$ and $[-, -]_{\text{ch}}$. Contracting a loop gives operators $\Delta_{\perp}$ and Δ_{ch} . Product orientation gives

$$\Delta_{\perp} \Delta_{\text{ch}} + \Delta_{\text{ch}} \Delta_{\perp} = 0.$$

Theorem 34.1 (Bicolored master equation). *Let $\mathbf{P}^{\perp, \text{ch}}$ be a two-colored modular operad whose edge contractions carry the product determinant orientation, and let $\mathfrak{g}_{\mathbf{P}}$ be the completed convolution Lie algebra of its Feynman transform with an endomorphism modular operad. Algebra structures on the transform are Maurer–Cartan elements $\Theta \in \mathfrak{g}_{\mathbf{P}}^1$ satisfying*

$$d\Theta + \frac{1}{2}[\Theta, \Theta]_{\perp} + \frac{1}{2}[\Theta, \Theta]_{\text{ch}} + \hbar_{\perp} \Delta_{\perp} \Theta + \hbar_{\text{ch}} \Delta_{\text{ch}} \Theta = 0.$$

Proof. The Feynman-transform differential contracts one edge. Separating edges give the two brackets and loop edges give the two BV operators. The product orientation makes mixed contractions anticommute. The condition that the structure map commute with the transform differential is exactly the displayed Maurer–Cartan equation [GK98]. $\square$

Typed anomalies

An anomaly is a cohomology class in the deformation complex that governs the attempted construction.

construction	complex	class
central extension	local Lie algebra cochains	cocycle
BRST	BRST deformation complex	nilpotence
quantization		obstruction
BV quantization	local BV cochains	quantum master anomaly
conformal-block family	Atiyah algebra of its determinant line	projective curvature
curved Koszul duality	curved convolution Lie algebra	curvature m_0

A comparison theorem consists of a morphism between two complexes and an equality of the transported classes. This formulation keeps the geometric origin of every scalar visible.

BOOK VII

**Calabi–Yau and automorphic
interfaces**

Calabi–Yau categories

Let C be a smooth linear category with a d -Calabi–Yau structure. Two universal consequences are available.

Theorem 36.1 (Cyclic and shifted outputs). *The Hochschild cochains $CC^*(C)$ form a framed E_2 algebra. The derived moduli stack of objects carries a $(2 - d)$ -shifted symplectic structure whenever the representability hypotheses hold.*

Proof. The first assertion is the cyclic Deligne theorem [BR23]. The second is the PTVV–Brav–Dyckerhoff construction: the Calabi–Yau trace pairs the tangent complex $\mathrm{RHom}(E, E)[1]$ with itself in degree $2 - d$ [PTVV13, BD21]. $\square$

A geometric BV theory or an open factorization category supplies an ordered chiral algebra through the recognition theorem of Chapter 3. This is the bridge from Calabi–Yau geometry to the present volume.

Hall and BPS algebras

A Hall multiplication starts from the extension correspondence

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha,\beta}^{\text{ext}} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

With a coefficient theory for which p^* and q_* are defined,

$$a * b = q_* p^*(a \boxtimes b).$$

Associativity follows from the stack of two-step filtrations.

For 2-Calabi–Yau categories with good moduli spaces, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra [DHS23]. For quivers and surface resolutions, preprojective and surface CoHAs recover positive Yangian halves in established cases [SV23, DPS⁺26].

The ordered chiral structure enters when Hall multiplication is promoted to a factorization coefficient over a curve or defect parameter. A coproduct, pairing, and completion then produce the corresponding double.

The Igusa denominator as a bar character

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borchers–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive nilpotent part. Root spaces are finite-dimensional and the positive cone is locally finite. Fix a proper integral height h that is positive on the effective root cone, and set

$$F^{>H}\mathfrak{n}_+ = \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\Delta, \alpha}, \quad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+ / F^{>H}\mathfrak{n}_+.$$

Height additivity makes $F^{>H}\mathfrak{n}_+$ an ideal; properness makes $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. Every bar, Chevalley, enveloping, and character formula in this chapter is interpreted coefficientwise in the separated height completion.

Theorem 38.1 (Bar–Chevalley identification). *For each height H there is a natural quasi-isomorphism of conilpotent coalgebras*

$$\text{Bar}(U(\mathfrak{n}_+^{(H)})) \simeq C_*(\mathfrak{n}_+^{(H)}).$$

The quotient maps are compatible with these quasi-isomorphisms, and therefore induce a quasi-isomorphism in the separated height-complete category

$$\text{Bar}_h^\wedge(U(\mathfrak{n}_+)) := \varprojlim_H \text{Bar}(U(\mathfrak{n}_+^{(H)})) \simeq C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

The coefficientwise Euler supercharacter of the completed Chevalley complex is

$$\chi C_{*,h}^\wedge(\mathfrak{n}_+) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta, \alpha}}.$$

Proof. For every H , the PBW filtration on $U(\mathfrak{n}_+^{(H)})$ identifies the associated graded with $\text{Sym}(\mathfrak{n}_+^{(H)})$. The bar complex of the enveloping algebra and the Chevalley complex have the same associated-graded Koszul resolution. The standard antisymmetrization map is a filtered quasi-isomorphism. Naturality for the quotient maps gives the inverse-limit comparison; proper height makes every homogeneous coefficient stabilize. For an even root vector the exterior Chevalley factor contributes $1 - e^{-\alpha}$; for an odd root vector it contributes its inverse. Multiplication over the root spaces gives the product. $\square$

With the positive-root convention $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$, the denominator theorem reads

$$e^{-\rho} \chi C_{*,h}^{\wedge}(\mathfrak{n}_+) = D_5(2\Omega) = 2^{-6}\Delta_5(2\Omega)$$

[GN97, GN98b]. This identity is the exact meeting point of ordered bar theory and automorphic products.

Finite-height currents and an exact physical model

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\text{Cur}(\mathfrak{n}_+^{(H)}) = \mathbb{K}[\partial] \otimes \mathfrak{n}_+^{(H)}, \quad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H . The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta(a(-1)\mathbf{1}) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \quad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 39.1 (Height-complete current realization). *Let $\mathfrak{n} = \bigoplus_{\alpha > 0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} / \bigoplus_{h(\alpha) > H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\text{ch}}(\text{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar-Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. $\square$

Apply the finite BF model of Chapter 19 to every $\mathfrak{n}_+^{(H)}$. The inverse system gives a height-complete holomorphic–topological classical field theory with completed boundary algebra and interval complex

$$U_h^\wedge(\mathfrak{n}_+) := \varprojlim_H U(\mathfrak{n}_+^{(H)}), \quad C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

This gives an exact algebraic realization, a local classical holomorphic–topological realization, and an exact quantum realization of the finite-dimensional zero-mode sectors of the positive enveloping algebra and its denominator character. The compact $K3 \times E$ identification is the geometric comparison studied in the companion volumes.

Character, bracket, parity, and orientation

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 40.1 (Three reconstruction fibres). *The following fibres contain distinct structured objects with the same scalar image.*

1. *The PBW-character fibre of a graded vector space with basis x, y, z in degrees $a, b, a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*
2. *The superdimension-zero fibre contains both the zero super vector space and $\mathbb{k}^{1|1}$.*
3. *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ .*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 40.2 (Minimal PBW recognition). *Let $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous*

pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. $\square$

Exact dependency theorem

Theorem 41.1 (Ordered-chiral closure). *Fix:*

1. *a six-functor Ran category and its factorization subcategory;*
2. *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate;*
3. *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

Then the three bars, their complete cobars, their module/comodule Koszul transforms, the Hochschild centre, the formal-disc realization, the ribbon theory, and the finite-height current realization are defined functorially and satisfy the identities proved in this volume. Every passage between the pointwise and chiral-convolution monoidal products is carried by the displayed mixed distributive datum.

Proof. Each output is one of the functors constructed in the preceding chapters. The source and target monoidal categories are displayed in every definition. The comparison theorems invoke the explicitly supplied distributive, PBW, completion, duality, or trace datum. Composition of these functors gives the asserted functorial system. $\square$

Synthesis

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$. Associative chirality is an E_1 structure for $\star^{\text{ch}}$. Chiral Lie theory is a Lie structure for $\star^{\text{ch}}$. Their bars are $\text{Bar}^\perp$, $\text{Bar}_{\text{Ass}}^{\text{ch}}$, and $\text{Bar}_{\text{Lie}}^{\text{ch}}$. The aligned projector supplies the exact sector on which a mixed distributive law is defined. Every comparison in the theorem system factors through one of these typed maps.

The constructed operations form the two functorial rows

$$\begin{aligned} \text{HT boundary theory} &\xrightarrow{\text{Obs}_\partial} \text{Alg}_{E_1}(\text{Fact}_D(X)) \xrightarrow{\text{Bar}^{\text{ord}}} \text{Coalg}_{\text{conil}, \text{cpl}}(\text{Fact}_D(X)), \\ \text{Alg}_{E_1}(\text{Fact}_D(X)) &\xrightarrow{Z_{\text{ord}}} \text{Alg}_{E_2}(\text{Fact}_D(X)). \end{aligned}$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

BOOK **VIII**

**Technical appendices to the
preceding book**

Sign calculation for the length-three bar

For homogeneous $a, b, c \in \bar{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}[(ab)c - a(bc)],$$

which vanishes by associativity.

Rational R -matrix verification

Let P_{ij} permute tensor factors i and j . Expanding

$$R_{12}(u)R_{13}(u+v)R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

Concordance with the July 2026 Volume I draft

The present volume preserves the draft's five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative example ladder. It sharpens four interfaces:

1. a factorization carrier is paired explicitly with its operation;
2. the Ran category is fixed before its tensor products are used;
3. the block-ordered operad is introduced after a representing chiral operad is chosen;
4. ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

These refinements make every theorem a statement in a fixed category and every physical interpretation the image of a displayed comparison map.

Book entrance: Mixed Holomorphic–Topological Deligne Theory

*Hamiltonian jets, shifted cotangent fields, universal branes, Duflo quantization,
and the all-arity open–closed map.*

BOOK **IX**

Finite Hamiltonian geometry

The formal symplectic plane

46.1 Hamiltonians and the Poisson bracket

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = dz_1 \wedge dz_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[[z_1, z_2]], \quad \{f, g\} = \partial_{z_1} f \partial_{z_2} g - \partial_{z_2} f \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. The shifted order filtration $F^p \mathfrak{h} = \mathfrak{m}^{p+2}$ is complete and obeys

$$\{F^p \mathfrak{h}, F^q \mathfrak{h}\} \subset F^{p+q} \mathfrak{h}.$$

The affine Hamiltonian algebra $\mathfrak{h}^{\text{aff}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$ also contains translations. Its linear sector is adjoined separately when a moving brane or center-of-mass coordinate is required.

Write $R_r = \text{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r . The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^d R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 46.1 (Finite Hamiltonian jet algebra). *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$ -module $\text{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^d (r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω ; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. $\square$

46.2 Unimodularity

Theorem 46.2 (Unimodularity of every finite jet). *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\text{Tr}_{\mathfrak{h}_{\leq d}}(\text{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^d R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$ -module R_r has trace zero. Linearity gives the result. $\square$

Corollary 46.3 (Canonical finite BV measure). *The linear odd symplectic space $T^*[-1]\mathfrak{h}_{\leq d}[1]$ has a translation-invariant Berezinian for which the cotangent BF Hamiltonian satisfies the quantum master equation.*

Proof. Choose dual linear coordinates on the shifted cotangent space and use their constant Berezinian. The classical term vanishes by Jacobi, while the BV divergence is the adjoint trace. Unimodularity makes that trace zero. $\square$

The shifted cotangent theory

47.1 The Lie algebra and its functions

Put

$$\mathfrak{g}_d = T^*[-1]\mathfrak{h}_{\leq d} = \mathfrak{h}_{\leq d} \ltimes \mathfrak{h}_{\leq d}^\vee[-1],$$

where $\mathfrak{h}_{\leq d}$ acts on the dual by the coadjoint action and the shifted dual is abelian.

Proposition 47.1 (Chevalley–Poisson identification). *There is a canonical isomorphism of commutative dg algebras*

$$C_{\text{CE}}^*(\mathfrak{g}_d) \cong C^*(\mathfrak{h}_{\leq d}, \text{Sym}(\mathfrak{h}_{\leq d})).$$

The differential on the right is the Chevalley differential for the adjoint action on $\text{Sym}(\mathfrak{h}_{\leq d})$. Under the identification $\text{Sym}(\mathfrak{h}_{\leq d}) = \mathcal{O}(\mathfrak{h}_{\leq d}^)$, it is the linear Poisson differential.*

Proof. The dual of $\mathfrak{h}_{\leq d}^\vee[-1]$ shifted by $[-1]$ is $\mathfrak{h}_{\leq d}$ in degree zero. Chevalley cochains therefore contain polynomial functions on $\mathfrak{h}_{\leq d}^*$ and exterior ghosts for $\mathfrak{h}_{\leq d}$. The semidirect bracket produces the coadjoint action, which is the Hamiltonian action of the linear Poisson tensor. $\square$

47.2 The BV action

Choose dual bases e_a and e^a . Let c^a be the ghost coordinates and b_a the cotangent coordinates. The finite action is

$$S_d = \frac{1}{2} b_c f^c_{ab} c^a c^b.$$

The canonical BV bracket satisfies

$$\frac{1}{2}\{S_d, S_d\} = 0$$

by Jacobi.

Theorem 47.2 (Exact finite quantum master equation). *With the translation-invariant BV Laplacian,*

$$\Delta S_d = \sum_b \text{Tr}(\text{ad } e_b) c^b = 0.$$

Consequently

$$\frac{1}{2}\{S_d, S_d\} + \hbar \Delta S_d = 0$$

for every d .

Proof. Contracting one field and one antifield in the cubic Hamiltonian gives the divergence of the adjoint vector field. The preceding unimodularity theorem makes every coefficient vanish. $\square$

The universal boundary brane

48.1 Polarization

Place the cotangent theory on a half-plane. The standard Lagrangian boundary condition fixes the cotangent field and leaves the $\mathfrak{h}_{\leq d}$ field free. Boundary observables form the Rees enveloping algebra

$$U_{\hbar}(\mathfrak{h}_{\leq d}) = T(\mathfrak{h}_{\leq d})[[\hbar]] / \langle xy - yx - \hbar[x, y] \rangle.$$

The Poincaré–Birkhoff–Witt symmetrization gives a filtered vector-space isomorphism

$$\text{sym}_{\hbar} : \text{Sym}(\mathfrak{h}_{\leq d})[[\hbar]] \xrightarrow{\sim} U_{\hbar}(\mathfrak{h}_{\leq d}).$$

Theorem 48.1 (Boundary quantization). *The factorization algebra of boundary observables of the finite cotangent theory is locally constant and its value on an interval is $U_{\hbar}(\mathfrak{h}_{\leq d})$. Fusion of adjacent intervals is multiplication in the enveloping algebra.*

Proof. Canonical quantization turns the linear boundary phase space $\mathfrak{h}_{\leq d}^*$ into the Gutt star algebra. For a linear Poisson bracket the Gutt product is transported by PBW to $U_{\hbar}(\mathfrak{h}_{\leq d})$. Excision identifies ordered interval fusion with multiplication. $\square$

BOOK **X**

The exact open–closed equivalence

Hochschild cochains of the universal brane

49.1 The Cartan–Eilenberg model

For every finite-dimensional Lie algebra $\mathfrak{l}$ there is a natural quasi-isomorphism

$$C_{\text{HH}}^{\bullet}(U\mathfrak{l}, U\mathfrak{l}) \simeq C^*(\mathfrak{l}, U\mathfrak{l}_{\text{ad}}).$$

The same statement holds for the Rees algebra after $\hbar$ -adic completion.

Proposition 49.1 (Filtered Hochschild model). *There is a filtered quasi-isomorphism*

$$C_{\text{HH}}^{\bullet}(U_{\hbar}(\mathfrak{h}_{\leq d}), U_{\hbar}(\mathfrak{h}_{\leq d})) \simeq C^*(\mathfrak{h}_{\leq d}, U_{\hbar}(\mathfrak{h}_{\leq d})_{\text{ad}}).$$

Its associated graded is

$$C^*(\mathfrak{h}_{\leq d}, \text{Sym}(\mathfrak{h}_{\leq d})).$$

Proof. Resolve the diagonal bimodule by the Chevalley–Eilenberg bimodule resolution

$$U_{\hbar}(\mathfrak{l}) \otimes \Lambda^{\bullet} \mathfrak{l} \otimes U_{\hbar}(\mathfrak{l}).$$

Applying bimodule homomorphisms into $U_{\hbar}(\mathfrak{l})$ gives the Chevalley complex with adjoint coefficients. PBW identifies the associated graded coefficient module with $\text{Sym } \mathfrak{l}$. $\square$

The Kontsevich–Duflo map

50.1 Linear Poisson formality

Let π_d be the linear Poisson bivector on $\mathfrak{h}_{\leq d}^*$. Kontsevich’s stable formality morphism, twisted by π_d , gives a filtered L_∞ quasi-isomorphism from polyvectors with differential $[\pi_d, -]$ to Hochschild cochains of the star algebra. Tamarkin’s refinement transports the full homotopy Gerstenhaber structure.

Theorem 50.1 (Finite mixed-HT Deligne equivalence). *For every $d \geq 2$ there is a filtered homotopy- E_2 quasi-isomorphism*

$$\Phi_d : C_{CE}^*(T^*[-1]\mathfrak{h}_{\leq d})[[\hbar]] \xrightarrow{\sim} C_{HH}^\bullet(U_{\hbar}(\mathfrak{h}_{\leq d}), U_{\hbar}(\mathfrak{h}_{\leq d})).$$

Its classical term is the Hochschild–Kostant–Rosenberg map. On invariant degree-zero functions it is the Duflo map

$$\text{Duf} = \text{sym}_{\hbar} \circ j^{1/2}(\hbar\partial), \quad j^{1/2}(x) = \det_{\mathfrak{h}_{\leq d}}^{1/2} \left(\frac{\sinh(\text{ad}_x/2)}{\text{ad}_x/2} \right).$$

Proof. Kontsevich formality identifies the dg Lie algebra of polyvectors on $\mathfrak{h}_{\leq d}^*$, twisted by the Maurer–Cartan element $\hbar\pi_d$, with the Hochschild dg Lie algebra of the corresponding star product. The star algebra of a linear Poisson structure is identified with the Rees enveloping algebra by PBW. Tamarkin formality lifts the equivalence from dg Lie algebras to homotopy E_2 algebras. The wheel graphs in the first Taylor coefficient sum to $j^{1/2}$, which gives Duflo on invariants. $\square$

50.2 The boundary action

The brace operad acts on the pair

$$(C_{HH}^\bullet(A, A), A)$$

for every associative algebra A . The closed color is the derived centre and the open color is the algebra itself.

Corollary 50.2 (All-arity open–closed structure). *Fix a stable formality morphism and a compatible Swiss-cheese refinement. Transport through Φ_d gives the pair*

$$(C_{\text{CE}}^*(T^*[-1]\mathfrak{h}_{\leq d})[[\hbar]], U_{\hbar}(\mathfrak{h}_{\leq d}))$$

a homotopy Swiss-cheese action associated with that refinement. Its binary closed bracket is the Poisson bracket, its open product is the enveloping product, and its first mixed operation is the quantized Hamiltonian action.

Proof. The brace operad acts on Hochschild cochains together with the algebra on which they act. The chosen homotopy- E_2 formality map transports the closed colour and every mixed brace to the Chevalley–Poisson model. The binary and first mixed operations are its first Taylor components. $\square$

The formal inverse limit

51.1 Weight completion

Let $F^{>d}\mathfrak{h}$ be the Hamiltonians of polynomial degree greater than d . Then

$$\mathfrak{h} = \varprojlim_d \mathfrak{h}_{\leq d}.$$

Define

$$\widehat{U}_{\mathfrak{h}}(\mathfrak{h}) = \varprojlim_d U_{\mathfrak{h}}(\mathfrak{h}_{\leq d})$$

and complete every cochain complex in polynomial weight.

Theorem 51.1 (Continuous formal-Darboux equivalence). *A stable formality morphism can be chosen compatibly with the quotient maps $\mathfrak{h}_{\leq d+1} \rightarrow \mathfrak{h}_{\leq d}$. It induces a continuous homotopy- $\mathbf{E}_2$ quasi-isomorphism*

$$\Phi_{\infty} : C_{\text{CE,cont}}^*(T^*[-1]\mathfrak{h})[[\hbar]] \xrightarrow{\sim} C_{\text{HH,cont}}^{\bullet}(\widehat{U}_{\mathfrak{h}}(\mathfrak{h}), \widehat{U}_{\mathfrak{h}}(\mathfrak{h})).$$

A compatible Swiss-cheese refinement gives the full formal mixed holomorphic–topological open–closed structure of the pointed Hamiltonian BF sector. Its cohomological Gerstenhaber structure is independent of the chosen chain-level refinement.

Proof. The Taylor coefficients of a stable formality morphism are universal finite sums of graphs. In any fixed polynomial weight, finitely many jets and graphs contribute. The maps therefore commute with the jet projections. The finite complexes form Mittag–Leffler towers in each weight, so inverse limit preserves the finite-level quasi-isomorphisms. Brace operations are defined by finite insertion in every weight and pass to the same limit. $\square$

Principle 51.2 (Universal brane principle). The boundary algebra $\widehat{U}_{\mathfrak{h}}(\mathfrak{h})$ is the universal formal Hamiltonian brane. Finite matrix branes are representation-theoretic images of this algebra. The bulk is its complete derived centre.

BOOK **XI**

The finite-jet laboratory

The first three Hamiltonian jets

52.1 Quadratic Hamiltonians

Set

$$e = \frac{1}{2}z_1^2, \quad f = -\frac{1}{2}z_2^2, \quad h = -z_1z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \quad \{h, f\} = -2f, \quad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\leq 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. The finite cotangent theory for $d = 2$ is the linear BV theory of $T^*[-1]\mathfrak{sl}_2$ and its boundary algebra is $U_{\hbar}(\mathfrak{sl}_2)$.

52.2 Cubic Hamiltonians

Let

$$v_j = z_1^{3-j}z_2^j, \quad 0 \leq j \leq 3.$$

Then

$$\{h, v_j\} = (3 - 2j)v_j, \quad \{e, v_j\} = jv_{j-1}, \quad \{f, v_j\} = (3 - j)v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 52.1 (The cubic jet algebra). *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\leq 3} \cong \mathfrak{sl}_2 \ltimes \text{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. $\square$

52.3 The first nonlinear layer

Write $w_k = z_1^{4-k} z_2^k$, $0 \leq k \leq 4$. In $\mathfrak{h}_{\leq 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j-i)w_{i+j-1}, \quad (52.1)$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 52.2 (Two brackets).

$$\{z_1^3, z_2^3\} = 9z_1^2 z_2^2 = 9w_2, \quad \{z_1^2 z_2, z_1 z_2^2\} = 3z_1^2 z_2^2 = 3w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio 3 : 1.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1 z_2 \cdot 2z_1 z_2 - z_1^2 \cdot z_2^2 = 3z_1^2 z_2^2$. $\square$

Theorem 52.3 (The quartic jet presentation). *The Lie algebra $\mathfrak{h}_{\leq 4}$ is generated by $e, f, h, v_0, v_1, v_2, v_3$. Its relations are the $\mathfrak{sl}_2$ relations, the irreducible action on the v_j , formula (52.1), and the induced $\mathfrak{sl}_2$ action on R_4 . The quartic subspace is abelian in the quotient.*

Proof. Quadratic and cubic generators produce every quartic monomial through (52.1). Brackets involving a quartic and a cubic have degree five, and brackets of two quartics have degree six, so both vanish after projection. Jacobi follows from the polynomial Poisson bracket before projection. $\square$

Moyal quantization in coordinates

53.1 The bidifferential formula

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2} \right)^n \sum_{r=0}^n (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f) (\partial_1^r \partial_2^{n-r} g). \quad (53.1)$$

Every polynomial pair gives a finite sum.

Theorem 53.1 (Quadratic exactness). *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar \{f, g\}.$$

Thus the quadratic Moyal algebra is the Rees enveloping algebra of $\mathfrak{sl}_2$ together with the Heisenberg translations when linear Hamiltonians are retained.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. $\square$

Calculation 53.2 (Cubic correction). For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (53.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3 (3!) = 3\hbar^3/2$. $\square$

53.2 Weyl generators

Let q act by multiplication and let $p = \hbar \partial_q$. Then

$$[p, q] = \hbar.$$

Weyl ordering identifies polynomial symbols with differential operators. The Moyal formula is the pullback of operator composition under this identification.

Chevalley coordinates and the Poisson differential

Choose a basis e_a of $\mathfrak{h}_{\leq d}$ with structure constants $[e_a, e_b] = f^c_{ab}e_c$. Let c^a be the odd Chevalley coordinate and let x_a be the even coordinate on $\mathfrak{h}_{\leq d}^*$. Then

$$d_{\text{CE}}c^c = -\frac{1}{2}f^c_{ab}c^ac^b, \quad d_{\text{CE}}x_a = f^c_{ab}c^bx_c. \quad (54.1)$$

The first equation is the Lie bracket. The second is the coadjoint action.

Proposition 54.1 (Hamiltonian interpretation). *For a polynomial $F(x)$,*

$$d_{\text{CE}}F = c^a\{x_a, F\}_{\text{lin}}.$$

Consequently $d_{\text{CE}}^2 = 0$ is the Jacobi identity for the linear Poisson tensor.

Proof. The linear Poisson bracket is $\{x_a, x_b\} = f^c_{ab}x_c$. The Leibniz rule gives (54.1) on every polynomial. The square is one half of the Schouten bracket of the Poisson tensor with itself. $\square$

54.1 The $\mathfrak{sl}_2$ invariant

For the quadratic jet, identify the coordinate functions with e, f, h . The invariant

$$q = 2ef + \frac{1}{2}h^2$$

is d_{CE} -closed. Its Duflo image is $\Omega + 1/2$. The local open–closed map therefore sends the classical radius of the coadjoint orbit to the quantum Casimir with its exact zero-point energy.

Residue duality and the formal current algebra

55.1 The continuous dual

Let

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \quad a, b \geq 0.$$

The Grothendieck residue pairing is

$$\text{Res}_0 \left(z_1^i z_2^j \rho_{ab} \right) = \delta_{ia} \delta_{jb}.$$

The continuous dual of $\mathbb{C}[[z_1, z_2]]$ is the direct sum of the ρ_{ab} . The continuous dual of the pointed algebra $\mathfrak{h} = \mathfrak{m}^2$ is the span of the modes with $a + b \geq 2$. The two modes of total degree one pair with the translation sector of $\mathfrak{h}^{\text{aff}}$.

Proposition 55.1 (Coadjoint current formula). *For $f = z_1^i z_2^j$ and ρ_{ab} ,*

$$\text{ad}_f^*(\rho_{ab})(g) = -\rho_{ab}(\{f, g\}).$$

In the residue basis this is a finite linear combination of modes of bidegree

$$(a - i + 1, b - j + 1).$$

Proof. The definition is the coadjoint action. The Poisson bracket lowers the total degree of fg by two, so residue conservation fixes the displayed bidegree. The Poisson formula contains two derivative terms. □

The shifted cotangent current algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}_{\text{cont}}^{\vee}[-1]$$

is therefore a complete topological Lie algebra with a continuous invariant pairing. Its finite jet quotients are precisely the $\mathfrak{g}_d$ used above.

All-arity open–closed operations

56.1 Hochschild braces

For Hochschild cochains $F \in C^p(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F . The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem 56.1 (Brace realization of the universal centre). *For $A = U_{\hbar}(\mathfrak{h}_{\leq d})$, choose the Tamarkin–Kontsevich homotopy–Gerstenhaber refinement of Φ_d . It transports the homotopy Schouten operations of polyvector fields to the Hochschild brace algebra. The binary operation is the Gerstenhaber bracket, and the mixed operation with one boundary word is the quantized Hamiltonian action.*

Proof. Kontsevich formality is an algebra over a cofibrant resolution of the Gerstenhaber operad. Tamarkin’s formality identifies that resolution with the brace operad. Twisting by the linear Poisson Maurer–Cartan element produces the differential and preserves every brace operation. $\square$

56.2 The first mixed identities

Let F be a closed cochain and a, b boundary operators. The first Swiss-cheese identities are

$$F \cdot (ab) = (F_{(1)} \cdot a)(F_{(2)} \cdot b),$$

for the coproduct encoded by the brace diagonal, and

$$[F, G] \cdot a = F \cdot (G \cdot a) - (-1)^{|F||G|} G \cdot (F \cdot a).$$

For linear Hamiltonians these are the Leibniz rule and the commutator identity of derivations. Higher braces provide their coherent homotopies.

Corollary 56.2 (All-arity formal-Darboux theorem). *The inverse limit Φ_∞ equips the pair*

$$\left(C_{\text{CE}}^*(T^*[-1]\mathfrak{h}), \widehat{U}_\hbar(\mathfrak{h}) \right)$$

with a complete homotopy Swiss-cheese algebra. Its closed color is the full derived centre of the universal boundary algebra.

Proof. Every fixed polynomial weight receives finitely many brace insertions, so the finite-level Swiss-cheese operations commute with the jet projections. Their separated inverse limit gives the complete action. Morita theory identifies Hochschild cochains of the boundary algebra with its derived centre. $\square$

Finite Chan–Paton ranks

57.1 Rank one

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

57.2 Rank two and Cayley–Hamilton reduction

For a 2×2 matrix X ,

$$X^2 - (\text{Tr } X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X -degree at most one after adjoining $\text{Tr } X$ and $\det X$. The same statement holds for Y . On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 57.1 (First finite-rank trace relation). *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \quad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20}p_{02} = -(x_1y_2 - x_2y_1)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. $\square$

57.3 Stable passage

The transition from rank N to rank $N + 1$ adds a zero eigenvalue. Each fixed weighted degree stabilizes once N exceeds that degree. The inverse limit is therefore the completed multisymmetric algebra. The stable trace is an exact quotient of the universal centre on the polynomial trace sector.

BOOK **XII**

**Global holomorphic symplectic
descent**

Fedosov descent

58.1 The Weyl algebroid

Let (S, ω) be a holomorphic symplectic surface. The bundle of formal Weyl algebras has fibre the completed Weyl algebra of a formal symplectic plane. A holomorphic symplectic connection and a Fedosov one-form define a flat connection

$$D_F = \nabla + \frac{1}{\hbar}[\gamma, -]$$

whose flat sections form a deformation-quantization algebroid $\mathcal{W}_{S, \hbar}$.

Theorem 58.1 (Globalized formality on a holomorphic symplectic surface). *Fix a stable formality morphism and a compatible homotopy- E_2 refinement. There is a sheafwise homotopy- E_2 quasi-isomorphism*

$$\Phi_S : (\mathrm{PV}_S[[\hbar]], [\hbar\pi, -]) \xrightarrow{\sim} C_{\mathrm{HH}}^\bullet(\mathcal{W}_{S, \hbar}, \mathcal{W}_{S, \hbar}),$$

where $\pi = \omega^{-1}$. A change of holomorphic symplectic connection or Fedosov one-form gives a gauge-equivalent resolution and a homotopic quasi-isomorphism. Holomorphic symplectomorphisms act equivariantly on the resulting homotopy class.

Proof. Fedosov resolutions replace both sheaves by acyclic sheaves of formal fibrewise polyvectors and polydifferential operators. The fixed covariant formality morphism acts fibrewise and intertwines the Fedosov differentials. A change of connection or Fedosov one-form is a gauge transformation in the pronilpotent resolution and therefore gives a homotopy. $\square$

Corollary 58.2 (Global mixed-HT centre). *The deformation-quantization category $\mathrm{Perf}(\mathcal{W}_{S, \hbar})$ has Morita centre*

$$Z_{\mathrm{Mor}}(\mathrm{Perf}(\mathcal{W}_{S, \hbar})) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[[\hbar]], [\hbar\pi, -]).$$

For a K3 surface this gives a global open–closed equivalence for the Hamiltonian BF sector.

Proof. The Morita centre of perfect modules over an algebroid is the derived global section complex of its Hochschild cochains. Apply the globalized formality equivalence and then derived global sections. A K3 surface is holomorphic symplectic, so the stated Hamiltonian sector is the resulting Poisson-BV complex. $\square$

Factorization and descent

59.1 The algebraic mixed-HT realization

The sheaf

$$\mathcal{P}_S = (\mathrm{PV}_S[[\hbar]], [\hbar\pi, -])$$

is a sheaf of complete homotopy- E_2 algebras. Dunn additivity gives its locally constant factorization realization in two oriented topological directions. The holomorphic sheaf variable retains the dependence on S .

Theorem 59.1 (Algebraic factorization realization). *The homotopy- E_2 sheaf $\mathcal{P}_S$ determines a complete mixed holomorphic–topological factorization algebra*

$$\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}} \quad \text{on} \quad \mathbb{R}_{\mathrm{top}}^2 \times S.$$

Its universal boundary category is $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$, and the bulk–boundary morphism is a local equivalence

$$\mathcal{A}_{S,\mathrm{bulk}}^{\mathrm{alg}} \xrightarrow{\sim} Z_{\mathrm{Mor}}^{\mathrm{der}}(\mathrm{Perf}(\mathcal{W}_{S,\hbar})).$$

On every formal Darboux chart the map is the completed Kontsevich–Duflo equivalence.

Proof. The factorization envelope of a complete homotopy- E_2 algebra is locally constant on oriented two-disks. Sheafwise application to $\mathcal{P}_S$ gives the first assertion. The Fedosov algebroid is the boundary quantization of the same linear Poisson data. Globalized formality identifies its Hochschild cochains with $\mathcal{P}_S$, and Morita invariance identifies Hochschild cochains with the centre of the perfect-module category. The construction is local on S and therefore respects factorization descent. $\square$

59.2 BV realization

The cyclic Lie sheaf of formal Hamiltonians has the shifted-cotangent action

$$S_{\text{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \left\langle B, (d + \bar{\partial})A + \frac{1}{2}[A, A] \right\rangle.$$

Its classical observables map to $\mathcal{A}_{S, \text{bulk}}^{\text{alg}}$ by the Chevalley–Poisson identification. A compatible gauge fixing, extension of graph weights to compactified configurations, and solution of the quantum master equation produce a perturbative factorization realization of the same algebraic bulk–boundary map. The finite jet zero modes carry the exact quantum measure constructed above.

BOOK **XIII**

Matrix branes and trace geometry

The derived commuting stack

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \quad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$A_N = C_{\text{CE}}^*(\mathfrak{gl}_N, \text{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])).$$

Proposition 60.1 (Representation of the universal brane). *Substitution of matrices defines a Lie morphism*

$$\rho_N : \mathfrak{h} \longrightarrow \text{Der}(A_N), \quad f \longmapsto \delta_f,$$

which integrates to an algebra morphism

$$\widehat{U}_{\hbar}(\mathfrak{h}) \longrightarrow \text{End}(A_N)[[\hbar]], \quad f \longmapsto \hbar \delta_f.$$

Thus the N -brane stack is a representation of the universal boundary algebra.

Proof. For a polynomial Hamiltonian f , cyclic differentiation gives the matrix vector field

$$\delta_f(X) = -\partial_Y^{\text{cyc}} f(X, Y), \quad \delta_f(Y) = \partial_X^{\text{cyc}} f(X, Y).$$

The cyclic Euler identity implies $\delta_f([X, Y]) = 0$. Hence δ_f commutes with the Koszul differential $Q\psi = [X, Y]$. The same formulas are equivariant for simultaneous conjugation, so δ_f descends to a derivation of the Chevalley–Eilenberg quotient A_N . The trace-moment-map calculation gives

$$[\delta_f, \delta_g] = \delta_{\{f, g\}},$$

first for polynomial Hamiltonians and then continuously for the formal completion. This proves that ρ_N is a continuous Lie morphism. The assignment $f \mapsto \hbar\delta_f$ satisfies

$$[\hbar\delta_f, \hbar\delta_g] = \hbar^2\delta_{\{f,g\}} = \hbar(\hbar\delta_{\{f,g\}}),$$

which is the Rees relation. The universal property of the completed Rees enveloping algebra gives the displayed morphism. It is the universal boundary action evaluated in the N -dimensional Chan–Paton representation. $\square$

Necklace Hamiltonians

Let

$$F = \mathbb{C}\langle x, y \rangle, \quad F_{\text{cyc}} = F/[F, F].$$

For a cyclic word f , define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\delta_f(x) = -\partial_y^{\text{cyc}} f, \quad \delta_f(y) = \partial_x^{\text{cyc}} f.$$

Theorem 61.1 (Necklace Lie algebra). *The bracket*

$$\{f, g\}_{\text{neck}} = [\partial_x^{\text{cyc}} f \partial_y^{\text{cyc}} g - \partial_y^{\text{cyc}} f \partial_x^{\text{cyc}} g]_{\text{cyc}}$$

makes $F_{\text{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \delta_f$ is a Lie morphism. It preserves the moment map:

$$\delta_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\text{cyc}} f] + [y, \partial_y^{\text{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\delta_f, \delta_g] = \delta_{\{f, g\}_{\text{neck}}}$. $\square$

Theorem 61.2 (Trace moment map). *For*

$$J_N(f) = \text{Tr } f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N(\{f, g\}_{\text{neck}}),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^2$ carries $\text{Tr}(dX \wedge dY)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. $\square$

Stable traces

On the semisimple commuting locus write the joint eigenvalues as (x_i, y_i) . Define

$$p_{ab}^{(N)} = \sum_{i=1}^N x_i^a y_i^b.$$

Theorem 62.1 (Stable multisymmetric Poisson algebra). *The inverse stable trace algebra is the completed polynomial algebra*

$$\mathcal{P}_\infty = \mathbb{C}[[p_{ab} \mid a, b \geq 0, a + b > 0]].$$

Its Poisson bracket is

$$\{p_{ab}, p_{cd}\} = (ad - bc)p_{a+c-1, b+d-1}.$$

Proof. The stable fundamental theorem for multisymmetric functions gives polynomial generators p_{ab} . The bracket follows from $\{x_i, y_j\} = \delta_{ij}$ and the Leibniz rule. $\square$

Remark 62.2. The stable trace algebra is a quotient of the universal closed algebra after representation and scalarization. The universal centre retains Hochschild classes, stabilizer directions, and higher central operations that vanish under the trace.

The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \quad v = y_1 - y_2, \quad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 63.1 (The A_1 Kirillov–Kostant cone). *With $\{u, v\} = 2$,*

$$\{H, E\} = 4E, \quad \{H, F\} = -4F, \quad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ Lie–Poisson bracket on the nilpotent cone.

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

BOOK **XIV**

**Wheel corrections and exact
quantum structure**

The Duflo wheel

For $\mathfrak{sl}_2$ take generators

$$[h, e] = 2e, \quad [h, f] = -2f, \quad [e, f] = h.$$

The invariant polynomial and the quadratic Casimir are

$$q = 2ef + \frac{1}{2}h^2, \quad \Omega = ef + fe + \frac{1}{2}h^2.$$

Calculation 64.1 (The zero-point shift). The Duflo map gives

$$\text{Duf}(q) = \Omega + \frac{1}{2}.$$

On the irreducible highest-weight- n module,

$$\left(\Omega + \frac{1}{2}\right)|_{V_n} = \frac{(n+1)^2}{2} \text{id}.$$

Proof. For $\mathfrak{sl}_2$, the second wheel in $j^{1/2}(\partial)$ is the constant shift by $\rho^2 = 1/2$ in the stated normalization. The Casimir eigenvalue is $n(n+2)/2$. $\square$

The shift is the algebraic one-loop term in the open–closed map. Higher wheels assemble into $j^{1/2}$.

The full local theorem

Let

$$\mathcal{A}_{\text{bulk}}^{\text{alg}} = \text{Fact}_{\mathbf{E}_2} \left(C_{\text{CE}, \text{cont}}^* (T^*[-1]\mathfrak{h})[[\hbar]] \right)$$

be the factorization envelope in the two topological directions, with the formal holomorphic coefficient retained at the Darboux point.

Theorem 65.1 (Universal mixed holomorphic–topological Deligne theorem). *Let*

$$C_{\partial} = \text{Perf}(\widehat{U}_{\hbar}(\mathfrak{h})).$$

The formality morphism induces a quasi-isomorphism of complete homotopy- $\mathbf{E}_2$ factorization algebras

$$\mathcal{A}_{\text{bulk}}^{\text{alg}} \xrightarrow{\sim} Z_{\text{Mor}}^{\text{der}}(C_{\partial}).$$

At the fixed formal Darboux insertion point this map is Φ_{∞} . Its restriction to every finite matrix brane is the Rees-scaled necklace representation followed by the Hochschild action.

Proof. The continuous formality theorem gives the complete homotopy- $\mathbf{E}_2$ equivalence at the Darboux point. Factorization envelopes preserve equivalences between complete $\mathbf{E}_2$ algebras. Morita invariance identifies the centre of the perfect-module category with continuous Hochschild cochains. The matrix representation $f \mapsto \hbar \delta_f$ is a module map, so restriction is functorial. $\square$

Corollary 65.2 (Global K3 Hamiltonian sector). *Let S be a K3 surface with holomorphic symplectic form. The Fedosov category $\text{Perf}(\mathcal{W}_{S, \hbar})$ has derived centre equivalent to the global algebraic Hamiltonian bulk $\mathcal{A}_{S, \text{bulk}}^{\text{alg}}$. The equivalence carries the Duflo wheel, cyclic chains, and the selected homotopy Swiss-cheese action.*

Proof. Apply globalized formality and the algebraic factorization realization. Fedosov descent transports the fixed graph operations, including wheel and brace terms, while Morita invariance transports cyclic chains and the centre action. $\square$

The harmonic square

The four complexes

$$C_{\text{CE}}^*(T^*[-1]\mathfrak{h}), \quad C_{\text{HH}}^*(\widehat{U}\mathfrak{h}, \widehat{U}\mathfrak{h}), \quad O(\text{Rep}_\infty B)^{GL_\infty}, \quad H_*^{\text{CE}}(\mathfrak{gl}_\infty B)$$

are connected by explicit maps. The first two are equivalent by formality. The stable trace map reaches the third. Loday–Quillen–Tsygan identifies the primitives of the fourth with shifted cyclic homology. Their common input is the cyclic Hamiltonian class, and each construction retains a different amount of structure.

Theorem 66.1 (Four realizations of cyclic primitives). *For a unital algebra B and a Hamiltonian Lie map $\mathfrak{h} \rightarrow HC_0(B)$, the following maps are functorial:*

$$\begin{aligned} C_{\text{CE}}^*(T^*[-1]\mathfrak{h}) &\xrightarrow{\Phi_\infty} C_{\text{HH}}^*(\widehat{U}\mathfrak{h}, \widehat{U}\mathfrak{h}), \\ \mathfrak{h} &\longrightarrow HC_0(B) \longrightarrow O(\text{Rep}_N B)^{GL_N}, \\ \widehat{\text{Sym}} HC_0(B) &\longrightarrow O(\text{Rep}_\infty B)^{GL_\infty}, \\ \text{Sym}(HC_{*-1}(B)) &\xrightarrow{\sim} H_*^{\text{CE}}(\mathfrak{gl}_\infty B). \end{aligned}$$

The first map preserves the homotopy Gerstenhaber structure, the second preserves the necklace Poisson bracket, the third preserves multiplication, and the fourth preserves the Hopf structure.

Proof. The first arrow is the inverse limit of the finite-jet Kontsevich–Duflo formality maps. The graph formulas preserve braces and the Schouten bracket up to their canonical coherent homotopies, and compatibility with jet quotients passes this structure to the complete limit.

A cyclic class $[b] \in HC_0(B) = B/[B, B]$ defines the invariant function

$$\text{Tr}_N([b])(\rho) = \text{Tr}(\rho(b))$$

on the representation scheme. Cyclicity makes this function independent of the representative of $[b]$. The double-Poisson or Hamiltonian hypothesis on $\mathfrak{h} \rightarrow HC_0(B)$ converts the necklace bracket into the Poisson bracket of trace functions, which proves the second assertion.

Products of single traces define compatible multisymmetric functions as N grows. The universal property of the completed symmetric algebra therefore gives the third map. Its multiplicativity is built into the passage from a cyclic primitive to its multi-trace exponential assembly.

The stable Lie algebra $\mathrm{gl}_\infty B$ has block-sum coproduct. The Loday–Quillen–Tsygan theorem identifies its primitive homology with $HC_{*-1}(B)$ and identifies the full stable homology with the free graded-commutative Hopf algebra on those primitives. This gives the fourth isomorphism and its Hopf compatibility. Naturality of formality, trace evaluation, stabilization, and the Loday–Quillen–Tsygan map proves functoriality of the four constructions. $\square$

Synthesis

The finite jets supply exact BV theories. Their boundary quantizations supply universal branes. Linear Poisson formality identifies bulk and centre. Universal graph formulas pass to the formal limit. Fedosov descent globalizes the equivalence. Matrix substitution produces finite Chan–Paton branes, necklace Hamiltonians, and stable traces. The Duflo wheels measure the quantum difference between symmetric and enveloping coordinates.

The complete chain is

$$\begin{array}{ccc}
 C_{\text{CE}}^*(T^*[-1]\mathfrak{h}) & \xrightarrow[\sim]{\Phi_\infty} & C_{\text{HH}}^*(\widehat{U}_\hbar\mathfrak{h}, \widehat{U}_\hbar\mathfrak{h}) \\
 \text{representation} \downarrow & & \downarrow \text{module action} \\
 \mathcal{O}(\text{Rep}_N)^{GL_N} & \xrightarrow[\text{central action}]{} & C_{\text{HH}}^*(A_N, A_N).
 \end{array}$$

The upper arrow is the universal open–closed equivalence. The lower row is its finite-brane image. This order retains the centre before taking traces and retains the field theory before taking its character.

BOOK **XV**

**Technical appendices to the
preceding book**

Signs in the shifted cotangent complex

Let e_a have degree zero and let $e^a[-1]$ have degree -1 . The Chevalley generators dual to them have degrees 1 and 0. Thus

$$C_{\text{CE}}^*(T^*[-1]\mathbf{I}) = \text{Sym}(\mathbf{I}^\vee[-1] \oplus \mathbf{I}).$$

The degree-zero polynomial factor is $\text{Sym } \mathbf{I} = \mathcal{O}(\mathbf{I}^*)$. The differential on a polynomial p is

$$(d_{\text{CE}}p)(x) = \text{ad}_x p,$$

which is the linear Poisson differential.

A direct Jacobi check for Hamiltonian jets

For monomials

$$f = z_1^a z_2^b, \quad g = z_1^c z_2^d,$$

one has

$$\{f, g\} = (ad - bc) z_1^{a+c-1} z_2^{b+d-1}.$$

Substitution of three monomials gives a scalar multiple of

$$\det((a, b), (c, d)) \det((a + c - 1, b + d - 1), (e, f)) + \text{cyclic}.$$

Writing the three coefficients in terms of the 2×2 determinants of the exponent vectors shows pairwise cancellation. The coordinate-free proof is $[\pi, \pi]_{\text{Sch}} = 0$ for $\pi = \partial_{z_1} \wedge \partial_{z_2}$.

Bibliographic note

The finite and global formality maps used above are the linear and covariant forms of Kontsevich formality and its Fedosov globalization [Kon03, Dol05, CDH07]. The Hochschild model of enveloping algebras is the Cartan–Eilenberg comparison. The all-arity centre action is the Deligne–Tamarkin brace action [MS02, KS00]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

Book entrance: Calabi–Yau Quantum Groups

Calabi–Yau categories, geometric BV theories, Hall correspondences, quantum doubles, K3 symmetries, and compact-background comparison.

BOOK **XVI**

**Categorical and field-theoretic
foundations**

Calabi–Yau categories

71.1 Smoothness, properness, and the Calabi–Yau class

Let C be a small stable $\mathbb{k}$ -linear category. Smoothness means that the diagonal bimodule is perfect. Properness means that the morphism complexes are perfect over $\mathbb{k}$. A d -Calabi–Yau structure is a nondegenerate negative-cyclic class

$$\theta_{\text{CY}} \in HC_d^-(C)$$

whose underlying Hochschild trace identifies the diagonal bimodule with its dual shifted by d .

For $E \in C$, the tangent complex of the derived moduli stack of objects is

$$T_E \mathcal{M}_C \simeq \text{RHom}_C(E, E)[1].$$

The Calabi–Yau trace pairs two tangent vectors by composition and trace.

Theorem 71.1 (The two universal categorical outputs). *Let C be a smooth proper d -Calabi–Yau category.*

1. *The Hochschild cochains $CC^*(C)$ carry a framed E_2 -algebra structure.*
2. *Whenever the moduli stack of objects is representable in the PTVV setting, $\mathcal{M}_C$ carries a $(2 - d)$ -shifted symplectic form.*

Proof. The cyclic Deligne theorem upgrades the brace E_2 structure on Hochschild cochains by the circle action supplied by the cyclic trace [BR23]. The shifted form at E is

$$\text{RHom}(E, E)[1]^{\otimes 2} \xrightarrow{\circ} \text{RHom}(E, E)[2] \xrightarrow{\text{Tr}} \mathbb{k}[2 - d].$$

Nondegeneracy is the Calabi–Yau duality isomorphism, and closedness is the negative-cyclic lift [PTVV13, BD21]. $\square$

The integer d controls the shift of the moduli form. The Hochschild operation remains framed E_2 . These structures meet through deformation theory while retaining their separate operadic degrees.

Local BV theories and factorization observables

72.1 Cyclic elliptic local Lie algebras

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\text{Obs}^{\text{cl}}(U) = C_{\text{CE}}^*(\mathcal{L}_c(U)).$$

Disjoint union of supports gives the prefactorization product.

Theorem 72.1 (Factorization observables). *An elliptic classical BV theory produces a factorization algebra Obs^{cl} . A renormalized quantization satisfying the quantum master equation produces a factorization algebra Obs^{q} over $\mathbb{k}[[\hbar]]$.*

Proof. The Chevalley cochains of a direct sum are the completed tensor product of the cochains, giving the disjoint-support multiplication. Elliptic descent supplies the Weiss cosheaf property. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [CG17, CG21]. $\square$

72.2 Boundary and defect restriction

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \text{RHom}(b, b).$$

Theorem 72.2 (Ordered boundary algebra). *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \text{Alg}_{E_1}(\text{Fact}_D(C), \otimes^!).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b -modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C . The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. $\square$

Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 73.1 (Curve-level faces). A transverse face is

$$A^\perp \in \mathrm{Alg}_{\mathcal{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$A_{\mathrm{Ass}}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathcal{E}_1}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathrm{Lie}}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

Their bars are $\mathrm{Bar}^\perp$, $\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}$, and $\mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}$.

Theorem 73.2 (Source separation). *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ satisfies*

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(U^{\mathrm{ch}}(\mathfrak{g})) \simeq \mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the enveloping lane of chiral Koszul duality. The mixed statement is the distributive-law theorem of Volume I. $\square$

This theorem replaces every dimension-based dispatch of an $\mathcal{E}_n$ level. The geometry of the source selects the available directions; the operadic level records the operations actually constructed.

Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X , and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega^{0,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\text{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 74.1 (Classical hCS factorization algebra). *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X .*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial}A + \frac{1}{2}[A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. $\square$

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [CL19].

Pushforward to a curve

Let $p : X \rightarrow C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathcal{F}$ on X , define

$$(p_*\mathcal{F})(U) = \mathcal{F}(p^{-1}U).$$

Theorem 75.1 (Factorization pushforward). *The assignment $p_*\mathcal{F}$ is a factorization algebra on C . Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathcal{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathcal{F})(U) \simeq \int_{F_p} \mathcal{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C , Riemann–Hilbert algebraization gives a factorization D -module on C .

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D -module setting. $\square$

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

The derived centre and the physical bulk

For an ordered boundary algebra A , define

$$Z_{\text{ord}}(A) = \text{RHom}_{A \otimes A^{\text{op}}}(A, A).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \text{Obs}_{\text{bulk}}|_{\partial} \longrightarrow Z_{\text{ord}}(A).$$

Definition 76.1 (Open–closed completeness). A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase “bulk equals centre” into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

BOOK **XVII**

Hall and BPS algebras

Extension correspondences

Let $\mathcal{M} = \coprod_{\gamma \in \Gamma} \mathcal{M}_\gamma$ be a moduli stack in an exact or derived category. Let $\mathcal{M}_{\alpha,\beta}^{\text{ext}}$ classify extensions

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The correspondence is

$$\mathcal{M}_\alpha \times \mathcal{M}_\beta \xleftarrow{p} \mathcal{M}_{\alpha,\beta}^{\text{ext}} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Theorem 77.1 (Hall associativity). *Assume a coefficient theory in which p^* and q_* are defined and satisfy base change for the two-step filtration stack. Then*

$$a * b = q_* p^*(a \boxtimes b)$$

defines an associative product.

Proof. The two parenthesizations are the two projections from the stack of filtrations $0 \subset E_1 \subset E_2 \subset E_3$. Base change identifies both composites with pullback to this stack followed by pushforward to E_3 . $\square$

For critical CoHAs, the coefficient theory includes vanishing cycles and a coherent orientation. For symplectic CoHAs, it includes the selected equivariant Borel–Moore theory. Completion is chosen so that each output degree receives a convergent sum.

Orientation data and the threefold Hall product

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeyer [JU21]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [KPS24].

Theorem 78.1 (Threefold Hall algebra from strong orientation). *Let C be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [KPS24]. $\square$

Definition 78.2 (Reduced $K3 \times E$ Hall datum). A reduced $K3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

The geometric E -equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\leq 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X , and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\geq 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ . Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E -equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E,\mathrm{mot}} = \widehat{\bigoplus_{\gamma \in \Gamma_X} K_0^E(\mathrm{St}/\mathcal{M}_\gamma)},$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem 79.1 (Geometric reduced Hall algebra). *The extension correspondence in $\mathrm{Coh}_{\leq 1}(X)$ defines an associative E -equivariant motivic Hall product on $\mathcal{H}_X^{E,\mathrm{mot}}$. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E,\mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X]}[\epsilon]/(\epsilon^2)$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E -orbits.

Proof. The stack of short exact sequences is E -equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E -equivariant motivic Hall theorem of Oberdieck–Shen [OS19]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ . $\square$

Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\text{loc}} = \widehat{\bigoplus_{m(\gamma)=0} K_0^E(\text{St}/\mathcal{M}_\gamma)}, \quad \mathcal{H}_X^{\text{wr}} = \widehat{\bigoplus_{m(\gamma)>0} K_0^E(\text{St}/\mathcal{M}_\gamma)}.$$

A pure one-dimensional support of degree zero projects to a finite subset of E . A support of positive degree projects onto E . The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 80.1 (Geometric local–wrapped decomposition). *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\text{loc}}$ is a subalgebra, $\mathcal{H}_X^{\text{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E,\text{mot}} \cong \mathcal{H}_X^{\text{loc}} \ltimes \mathcal{H}_X^{\text{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E . $\square$

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borchers root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

Cohomological refinement and the Gram character

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeyer orientation therefore produces

$$\mathcal{H}_X^{E,\text{coh}} = \widehat{\bigoplus_{\gamma \in \Gamma_X} H_*^{\text{BM},E}(\mathcal{M}_\gamma, \Phi_\gamma)},$$

with its cohomological Hall product.

Theorem 81.1 (Strong-oriented cohomological lift). *For every strong orientation enhancement on the E -equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E,\text{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [KPS24]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. $\square$

For the dyonic charge lattice

$$\Gamma_{\text{phys}} = \widetilde{H}(S, \mathbb{Z}) \oplus \widetilde{H}(S, \mathbb{Z}),$$

write

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right).$$

The polarized central extension of the companion volume makes the graph $c \mapsto (c, \Pi(c))$ additive. Applying this graph to the reduced integration target defines the Gram-polarized character

$$I_{X,\Pi}^E : \mathcal{H}_X^{E,\text{mot}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\text{phys}}][\epsilon]/(\epsilon^2)}.$$

Theorem 81.2 (Exact geometric character map). *The map $I_{X,\Pi}^E$ is multiplicative. Its logarithm is primitive for the completed Hall coproduct whenever that coproduct is defined in the selected localization. In every reduced Donaldson–Thomas sector covered by the equivariant integration theorem, its coefficients are the reduced invariants of $K3 \times E$. The primitive reduced Gromov–Witten theory supplies the independent automorphic character $-\chi_{10}^{-1}$ in classes primitive in the $K3$ factor.*

Proof. Multiplicativity of I_X^E is the reduced integration theorem. The cocycle identity for the polarization cross-term makes the graph of Π multiplicative. The logarithm of a group-like series is primitive. The enumerative identifications are the reduced Hall calculations of Oberdieck–Shen and the primitive Igusa theorem of Oberdieck–Pixton [OS19, OP18]. $\square$

The geometric Hall source and the universal Igusa avatar now occupy explicit objects. A primitive root framing is a morphism from the simple BPS sheaves of $\mathcal{H}_X^{E,\text{coh}}$ to the simple-root spaces of $\mathfrak{g}_{\Delta_5}$ that preserves the Euler pairing, parity, and extension brackets. The universal presentation theorem then extends this framing through Hall, current, quantum-double, and determinant constructions.

BPS primitives

A connected graded bialgebra has primitive Lie algebra

$$\text{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

In cohomological Donaldson–Thomas theory, the BPS sheaf isolates the primitive part of the Hall algebra.

Theorem 82.1 (Two-Calabi–Yau BPS algebra). *For the geometric two-Calabi–Yau categories covered by the BPS-algebra theorem, the BPS algebra is the positive enveloping algebra of a generalized Kac–Moody Lie algebra. This scope includes semistable sheaves on K3 and other symplectic surfaces under the stated good-moduli hypotheses.*

Proof. The theorem is the Davison–Hennecart–Schlegel Mejia construction [DHS23]. The simple BPS sheaves give the simple-root spaces, extension correspondences give the bracket, and the integrality/PBW theorem identifies the Hall algebra with the positive enveloping algebra. $\square$

This result supplies a global K3 generalized Kac–Moody algebra at the two-Calabi–Yau Hall level. Yangian structures arise from additional equivariant correspondences and coproducts in the surface and quiver models described below.

The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta_5,+}$ be the positive Igusa Borcherds Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \quad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta_5,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 83.1 (Integral protected Lie algebra). The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X\boxtimes\Delta} = \mathfrak{b}_X \widehat{\coprod}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 83.2 (Hall–Borcherds amalgamation). *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_X, \quad r_\Delta : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X\boxtimes\Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X , while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ .

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \rightarrow \mathfrak{h}_\Pi \rightarrow \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. $\square$

Definition 83.3 (Comparison ideal). A primitive comparison $\varphi : \mathfrak{n}_{\Delta_5,+} \rightarrow \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta_5,+} \rangle} \subset \mathfrak{b}_{X \boxtimes \Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X \simeq \Delta, \varphi} = \mathfrak{b}_{X \boxtimes \Delta} / \mathcal{J}_\varphi.$$

Theorem 83.4 (Universal comparison quotient). *A Cartan-preserving primitive comparison is equivalent to a quotient of the integral amalgam that identifies the Borchers generators with geometric BPS primitives. If φ is an isomorphism, the quotient is canonically isomorphic to both Borels. In every case, the kernel $\mathcal{J}_\varphi$ measures the complete discrepancy between the geometric and automorphic primitive theories. PBW carries the quotient to enveloping algebras, Chevalley chains, current envelopes, and Fock characters.*

Proof. A quotient identifying $i_\Delta(x)$ with an element of $\mathfrak{p}_X^+$ defines φ , and the Lie relations in the quotient make it a Lie morphism. Conversely, a Lie morphism makes the generators of $\mathcal{J}_\varphi$ compatible with every bracket, so the quotient exists. An isomorphism φ gives mutually inverse maps through the two retractions. Functoriality of enveloping algebras, Chevalley chains, currents, and symmetric-superalgebra completion gives the final statement. $\square$

This amalgam is the maximal theory that retains both sources before a duality identification is imposed. The geometric Hall source and the automorphic Borchers source are its two canonical retracts. A microscopic duality is a diagonal quotient rather than an equality inferred from a scalar character.

Coproducts, pairings, and doubles

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^+ be a complete graded bialgebra, let H^0 be a Cartan algebra, and let H^- be the opposite or dual half. Let

$$\langle -, - \rangle : H^+ \widehat{\otimes} H^- \longrightarrow \mathbb{k}$$

be a nondegenerate Hopf pairing.

Definition 84.1 (Completed Hall double). The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^+) = H^- \widehat{\otimes} H^0 \widehat{\otimes} H^+$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R -matrix whenever the canonical tensor converges.

Theorem 84.2 (Double construction). *Let H^+ and H^- be paired complete graded bialgebras, let H^0 be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^+ \widehat{\otimes} H^- \longrightarrow \mathbb{k}$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^- \widehat{\otimes} H^0 \widehat{\otimes} H^+$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R -matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [Dri87, ES98]. $\square$

Theorem 84.3 (Quantum Hall–Borcherds amalgam). *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H^0 , with continuous Hopf retractions onto H^0 . Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_X \widehat{\ast}_{H^0} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H^0 and has canonical Hopf retractions to both factors. A mixed universal R -matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R -matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H^0 , so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H^0 . A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. $\square$

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

Ordered chiral bialgebras

Let A be an ordered chiral algebra on a curve C . An ordered chiral bialgebra consists of a coproduct

$$\Delta : A \longrightarrow A \widehat{\otimes} A$$

that is a morphism of ordered chiral algebras, together with counit and the chosen completion. An antipode upgrades it to a Hopf object. A spectral coproduct may take values in Laurent series in a difference parameter.

Theorem 85.1 (Representation fusion). *A completed coproduct defines a tensor product of continuous A -modules by restriction of scalars. A universal R -matrix satisfying quasitriangularity defines a braid operator on that module category.*

Proof. For modules M, N , pull back the $A \widehat{\otimes} A$ action on $M \widehat{\otimes} N$ along Δ . Coassociativity supplies the associator. The relation $R\Delta(a) = \Delta^{\text{op}}(a)R$ makes the flip composed with R an intertwiner, and the Yang–Baxter equation gives the braid relation. $\square$

BOOK **XVIII**

Exact model cases

The A_2 Hall algebra

For the quiver $1 \rightarrow 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2, 1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q + 1$	$q + 1$	$q + 1$
1	$q + 1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [Rin90, Gre95].

The tripled Jordan quiver and $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z , with potential

$$W = \text{Tr } x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\text{Tr } W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem 87.1 (Affine-Yangian positive half). *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of $\mathfrak{gl}_1$. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison’s affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters $\epsilon_1, \epsilon_2, \epsilon_3$ and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [Dav26, DM20, SV23, Tsy17]. $\square$

The vacuum representation produces a $W_{1+\infty}$ -type vertex algebra through an evaluation homomorphism. The CoHA, its double, and the vacuum algebra occupy successive levels: positive algebra, paired quantum group, and distinguished representation.

Surface CoHAs and ADE Yangians

Let $S_\Gamma \rightarrow \mathbb{C}^2/\Gamma$ be the minimal resolution for a finite subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{C})$. The exceptional curves have ADE intersection matrix. Moduli of sheaves supported on these curves carry convolution correspondences.

Theorem 88.1 (Curve-supported surface CoHA). *The completed positive CoHA of one-dimensional sheaves supported on the exceptional curves is identified with a nonstandard positive half of the affine Yangian of the corresponding ADE Lie algebra.*

Proof. The theorem is the curve-supported surface construction of Diaconescu–Porta–Sala–Schiffmann–Vasserot [DPS⁺26]. The simple curve sheaves give the simple generators, Hecke correspondences give the current operators, and the completed PBW comparison gives the affine Yangian presentation. $\square$

This theorem is a local K3 model whenever an ADE configuration occurs on a K3 surface. The global K3 surface adds compact curve classes and monodromy beyond the local root lattice.

Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [BFN18]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

BOOK **XIX**

K3 symmetries

The Mukai lattice

For a K3 surface S , the Mukai lattice is

$$\widetilde{H}(S, \mathbb{Z}) = H^0(S, \mathbb{Z}) \oplus H^2(S, \mathbb{Z}) \oplus H^4(S, \mathbb{Z})$$

with pairing

$$((r, c, s), (r', c', s')) = c \cdot c' - rs' - r's.$$

It is even unimodular of signature $(4, 20)$ and rank 24.

The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 91.1 (K3 Heisenberg action). *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m \delta_{m+n,0} \int_S \alpha \cup \beta \text{ id}.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^(S^{[0]})$.*

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha \beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [Nak97]. $\square$

This action is global, compact, and abelian at the Lie level. It is an exact K3 symmetry and supplies the rank-twenty-four Fock sector of any broader K3 algebra.

The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 92.1 (Orthogonal LLV symmetries). *For a K3 surface S , the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\text{LLV}}(S) \cong \mathfrak{so}(\tilde{H}(S), (-, -)_{\text{Muk}}) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\text{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K3 lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S , the extension has signature $(4, 20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [LL97, Ver96]. $\square$

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

Local ADE Yangians

An ADE configuration of (-2) -curves on S gives a root sublattice of $H^2(S, \mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

The K3 symmetry trichotomy

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4, 20)$	$H^*(S)$; separately $\mathfrak{so}(4, 21)$ on $K3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

The K3 correspondence category

Let S be a projective K3 surface. Let $\mathcal{M}_S$ contain the smooth projective moduli spaces $M_\sigma(v)$ of stable objects for primitive Mukai vectors and generic stability conditions, the Hilbert schemes $S^{[n]}$, and the equivariant curve-supported moduli at every ADE enhancement. Close this collection under finite products.

Definition 95.1 (K3 correspondence category). The category $\text{Corr}_{K3}(S)$ has the spaces of $\mathcal{M}_S$ as objects and

$$\text{Hom}(M, N) = H_*^{\text{BM}}(M \times N)$$

for proper objects, with the corresponding localized equivariant Borel–Moore theory on ADE charts. Composition is convolution:

$$\beta \circ \alpha = (p_{13})_* (p_{12}^* \alpha \cap p_{23}^* \beta) .$$

The tensor product is Cartesian product of moduli spaces and external product of correspondences.

Theorem 95.2 (Associativity and monoidality). *Transverse convolution makes $\text{Corr}_{K3}(S)$ a graded monoidal category. The cohomology functor*

$$\omega_S(M) = H^*(M)$$

and its localized equivariant form on ADE charts is strong monoidal and carries convolution to composition of linear operators.

Proof. The two parenthesizations of three correspondences are the two push–pull descriptions on $M_1 \times M_2 \times M_3 \times M_4$. Proper base change and the projection formula identify them. Cartesian products and external products satisfy the interchange law. The standard action of a kernel on cohomology is compatible with both operations. $\square$

Nakajima incidence varieties are morphisms in this category. Cup product and Lefschetz adjoints are diagonal kernels. ADE Hecke correspondences are equivariant morphisms. Thus the three K3 symmetry families occupy one correspondence category before passage to any enveloping algebra.

The common K3 action algebra

Define the completed state space

$$\mathcal{H}_S = \widehat{\bigoplus_{M \in \mathcal{M}_S} \omega_S(M)},$$

completed by Mukai norm, instanton number, and cohomological degree. Every correspondence acts continuously on $\mathcal{H}_S$.

Definition 96.1 (K3 action algebra). The K3 action algebra is the image

$$\mathcal{A}_{K3}(S) = \text{im} \left(\bigoplus_{M, N} \text{Hom}_{\text{Corr}_{K3}(S)}(M, N) \longrightarrow \text{End}_{\text{cont}}(\mathcal{H}_S) \right),$$

closed in the operator topology.

Theorem 96.2 (Exact K3 symmetry envelope). *The algebra $\mathcal{A}_{K3}(S)$ is separated and complete. It contains the Nakajima Heisenberg, every LLV Lefschetz triple, monodromy transport, and every local ADE Yangian action. The cross-relations are the intersection identities of their kernels. This description is faithful by construction.*

Proof. Each named operator is induced by a correspondence in $\text{Corr}_{K3}(S)$. Convolution gives their products. The image quotient removes precisely the correspondences acting trivially on all selected cohomologies, and closure gives completeness. $\square$

The three symmetries retain their intrinsic forms inside one algebra. Their interaction is calculated by intersections of the corresponding cycles. On a Noether–Lefschetz locus, an LLV operator and an ADE Hecke operator meet in the same correspondence space; the resulting convolution is the geometric cross-relation.

Tannaka reconstruction from K3 correspondences

Let $\mathcal{C}_S^{\text{geom}}$ be the rigid Karoubi envelope of the dualizable correspondence objects, modulo the tensor ideal of kernels acting trivially on cohomology, with Tate and parity shifts adjoined. For a finite Mukai-norm truncation, let $\mathcal{C}_S$ be the exact rigid tensor envelope of its image under cohomology: it is closed inside $\text{sVect}_{\text{fd}}^{\text{gr}}$ under finite sums, tensor products, duals, summands, kernels, and cokernels. The induced functor

$$\omega_S : \mathcal{C}_S \longrightarrow \text{sVect}_{\text{fd}}^{\text{gr}}$$

is faithful, exact, and strong monoidal.

Definition 97.1 (K3 correspondence quantum group). The coordinate bialgebra is the coend

$$O(\mathbb{G}_{K3}) = \int^{M \in \mathcal{C}_S} \omega_S(M)^* \otimes \omega_S(M).$$

Its continuous dual is denoted $U(\mathbb{G}_{K3})$.

Theorem 97.2 (K3 Tannaka theorem). *On every finite Mukai-norm truncation, $\mathcal{C}_S$ is equivalent to the rigid exact tensor subcategory of finite comodules over $O(\mathbb{G}_{K3})$ generated by the universal coactions on the fibres $\omega_S(M)$. Every geometric K3 correspondence induces a comodule morphism. The continuous dual acts naturally on every cohomology fibre and represents tensor-compatible natural endomorphisms of the fibre functor. A braided transport on $\mathcal{C}_S$ produces a coquasitriangular structure on the coend.*

Proof. Exact rigid coend reconstruction applies to the faithful exact strong monoidal functor ω_S . The coend carries the universal coaction on each fibre. The exact tensor envelope used

to define $\mathcal{C}_S$ is precisely the subcategory generated from these comodules by the operations preserved by ω_S , which gives the stated equivalence. The quotient map from geometric correspondences makes each correspondence a comodule morphism. Dualizing the universal coaction represents tensor-compatible natural endomorphisms. A braiding supplies the bilinear form whose hexagon identities are the coquasitriangular axioms. $\square$

The action algebra $\mathcal{A}_{K3}(S)$ and the Tannaka algebra $U(\mathbb{G}_{K3})$ are complementary. The first is generated by geometric kernels. The second consists of tensor symmetries compatible with every kernel. Their commutants record two complementary aspects of the K3 correspondence geometry: geometric operators and tensor-compatible natural symmetries.

BOOK **XX**

**The exact $K3 \times E$ protected
completion**

The protected input

The K3 elliptic genus supplies the Jacobi form

$$Z_{K3} = 2\phi_{0,1}.$$

Its Borchers half determines the weight-five product D_5 , the Gritsenko–Nikulin Borchers superalgebra $\mathfrak{g}_{\Delta_5}$, and its simple-root superdatum. The companion volume constructs the Borchers quiver $Q_{\Delta,H}$ in every proper-height window, its coordinate-framed Hall category, its formal three-Calabi–Yau Ginzburg completion, its colored elliptic Ran factorization algebra, its quantum double, and its virtual root Dirac operator.

Definition 98.1 (Protected avatar). The protected avatar of $K3 \times E$ is

$$\mathfrak{P}(K3 \times E) = \left(\Gamma_{\Delta}, \mathcal{B}_{\Delta}, \mathcal{H}_{1p}, \mathcal{F}_{\Delta,E}^{\text{hyb}}, \widehat{U}_q(\mathfrak{g}_{\Delta_5}), \mathbb{A}_{\mathbb{Z}} \right).$$

Here $\mathcal{B}_{\Delta}$ is the ordinary quiver Hall category and $\mathcal{H}_{1p}$ is the parity-refined Borchers root superspace.

Theorem 98.2 (Existence of the universal protected avatar). *The protected avatar exists and has the following structures:*

1. Γ_{Δ} is a locally finite three-Calabi–Yau dg category;
2. $\mathcal{B}_{\Delta}$ is the quiver Hall category of the underlying even Borchers–Cartan datum, while $\mathcal{H}_{1p}$ carries the parity-refined root spaces of $\mathfrak{g}_{\Delta_5+}$;
3. $\mathcal{F}_{\Delta,E}^{\text{hyb}}$ is a local–wrapped factorization algebra with primitive current algebra $\text{Cur}_E(\mathfrak{g}_{\Delta_5+})$;
4. $\widehat{U}_q(\mathfrak{g}_{\Delta_5})$ is a root-height-complete quasitriangular Hopf superalgebra;

5. $\mathbb{A}_Z$ is an oriented virtual Fredholm family with Pfaffian $D_5(2Z)$ after the Weyl shift.

Proof. The assertions are respectively the Calabi–Yau completion theorem, the quiver Hall theorem, the Borchers root-space construction, the colored-Ran Chevalley theorem, Hong’s quantization theorem, and the root-space Fredholm theorem of the companion volume. $\square$

Definition 98.3 (Integral protected avatar). Let $\mathfrak{p}_X^+$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E -equivariant Hall source. The integral protected avatar is

$$\mathfrak{P}_{\text{int}}(X, \Delta) = \left(\mathcal{H}_X^{E, \text{mot}}, \mathfrak{P}(K3 \times E), \mathfrak{b}_{X \boxtimes \Delta}, r_X, r_\Delta, I_{X, \Pi}^E \right).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}_X^+$ and hence the amalgamated primitive component.

Theorem 98.4 (Integral source–target completion). *The integral protected avatar retains the geometric extension theory, the universal Igusa root theory, and their common polarized charge lattice in one functorial system. The retraction r_X recovers the geometric BPS Borel, while r_Δ recovers the Borchers Borel and its denominator. The character morphism factors through the common Cartan grading. Every primitive comparison $\varphi : \mathfrak{n}_{\Delta_5, +} \rightarrow \mathfrak{p}_X^+$ produces a diagonal quotient of the integral avatar, and every diagonal quotient arises in this way.*

Proof. The geometric Hall algebra and its character were constructed before the universal root avatar. The polarized graph map identifies their Cartan gradings. Hall–Borchers amalgamation gives the common primitive object and its two retractions. The universal comparison-quotient theorem identifies primitive comparisons with diagonal quotients. Each step is natural under morphisms preserving the charge lattice, Hall correspondences, and root presentation. $\square$

The noncommutative Calabi–Yau field theory

Every finite-height Ginzburg algebra $\Gamma_{\Delta, H}$ has a cyclic three-Calabi–Yau pairing. Its open string field is a degree-one element a in the completed endomorphism algebra. The action is

$$S_H(a) = \frac{1}{2} \langle a, m_1 a \rangle + \sum_{n \geq 2} \frac{1}{n+1} \langle a, m_n(a^{\otimes n}) \rangle.$$

Theorem 99.1 (Master equation of the universal Igusa open string). *The cyclic A_∞ identities imply*

$$\{S_H, S_H\} = 0.$$

The fixed root ordering and coordinate framing orient each finite Borchers–Ginzburg representation chart and define its vanishing-cycle Hall theory. The proper-height limit gives a complete noncommutative Calabi–Yau field theory whose ordinary boundary Hall algebra is the quiver Hall presentation, whose protected primitive superspace is $\mathfrak{g}_{\Delta_5+}$, and whose quantum symmetry is $\widehat{U}_q(\mathfrak{g}_{\Delta_5})$.

Proof. The cyclic A_∞ relations are the coefficients of the classical master equation. The three-Calabi–Yau pairing gives the odd symplectic form. The coordinate-framed orientation theorem of the companion volume defines the finite-chart Hall product. Proper height makes every fixed coefficient finite, so the inverse limit preserves the master equation and the Hall operations. $\square$

The mixed holomorphic–topological boundary theory

Apply cotangent BF theory to the finite positive Borchers Lie superalgebras $\mathfrak{g}_{\Delta_5+}^{(H)}$ on $\mathbb{R}_{\text{top}}^2 \times E$. The boundary polarization has algebra $U(\mathfrak{g}_{\Delta_5+}^{(H)})$ and derived centre

$$C_{\text{HH}}^{\bullet} \left(U(\mathfrak{g}_{\Delta_5+}^{(H)}), U(\mathfrak{g}_{\Delta_5+}^{(H)}) \right).$$

The shifted-cotangent bulk is

$$C_{\text{CE}}^*(T^*[-1]\mathfrak{g}_{\Delta_5+}^{(H)}).$$

Theorem 100.1 (Finite-height mixed-HT open–closed equivalence). *In every finite height window, linear Poisson formality gives a homotopy- E_2 equivalence*

$$C_{\text{CE}}^*(T^*[-1]\mathfrak{g}_{\Delta_5+}^{(H)})[[\hbar]] \xrightarrow{\sim} C_{\text{HH}}^{\bullet} \left(U_{\hbar}(\mathfrak{g}_{\Delta_5+}^{(H)}), U_{\hbar}(\mathfrak{g}_{\Delta_5+}^{(H)}) \right).$$

The maps are compatible with height truncation. Their inverse limit identifies the completed algebraic bulk with the completed Hochschild centre, root degree by root degree. Applying the current functor on E and retaining the elliptic-degree colour produces compatible local–wrapped factorization objects; the open–closed map is defined on each current coefficient by the same universal formality graphs.

Proof. Each finite algebra is finite-dimensional and nilpotent. Kontsevich formality for its linear Poisson structure gives the first equivalence, and the brace refinement supplies the homotopy- E_2 structure. Universal graph formulas commute with quotient maps. Proper height makes the inverse limit exact in each root degree. The current functor is applied after this algebraic equivalence; root-degree addition preserves the local–wrapped colour. $\square$

The exact scalar and first-order observables

Set

$$\mathrm{Den}_\Delta(Z) = D_5(2Z), \quad J(Z) = Z/2, \quad J^* \mathrm{Den}_\Delta = D_5.$$

The one-particle root spaces give

$$\mathrm{sch Fock}(\mathfrak{g}_{\Delta_5+}) = \prod_{\alpha>0} (1 - e^{-\alpha})^{-\mathrm{sdim} \mathfrak{g}_{\Delta_5+} \alpha}.$$

The oriented root operator gives

$$e^{-\rho} \mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}_{K3 \times E}^{\mathrm{red}} = -(J^* \mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

Theorem 101.1 (Protected square-root theorem). *Within $\mathfrak{P}(K3 \times E)$, the denominator section Den_Δ is simultaneously:*

1. *the Weyl denominator of $\mathfrak{g}_{\Delta_5}$;*
2. *the Euler character of the bar–Chevalley complex;*
3. *the oriented Fredholm Pfaffian of $\mathbb{A}_Z$;*
4. *the PBW determinant of the parity-refined root superspace $\mathcal{H}_{1p}$;*
5. *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borchers Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $j^* \text{Den}_\Delta = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. $\square$

Physical meaning

The protected avatar is the complete universal algebraic model selected by the Igusa root datum. The Ginzburg category supplies its noncommutative three-Calabi–Yau sector. The quiver Hall category supplies its ordinary extension geometry. The Borchers root superspace supplies its protected one-particle states. The colored Ran object supplies its local and wrapped current algebra on the elliptic curve. Hong’s quantum Borchers algebra supplies its line-operator symmetry. The root Dirac family supplies its first-order determinant, and the Fock construction supplies the scalar square.

A compact string construction with specified simple-root states and extension amplitudes maps to the protected avatar through the presentation universal property. The comparison is determined on primitives; PBW propagates it to multiparticle states, the Hall product, the current algebra, the quantum double, and the determinant.

Universal property of the protected avatar

A presentation-preserving protected theory consists of a three-Calabi–Yau category, an oriented Hall boundary, a colored elliptic factorization algebra, a quasitriangular quantum symmetry, an oriented determinant, and comparison functors from the Borchers generators-and-relations construction to these five objects. The chosen simple-root states satisfy the Gritsenko–Nikulin Cartan, parity, and Serre relations.

Theorem 103.1 (Presentation universal property). *Every presentation-preserving protected theory carries a unique primitive Lie-superalgebra morphism*

$$\mathfrak{n}_{\Delta_5,+} \longrightarrow \text{Prim}(\mathfrak{T})$$

extending the chosen simple-root states. The Hall, Ginzburg, current, quantum-group, and determinant functors extend this primitive morphism to the five selected comparison maps. A primitive isomorphism together with equivalences of the selected functorial realizations gives an equivalence of protected data.

Proof. The Borchers generators and relations give the unique primitive morphism. Each remaining construction is functorial in a presentation map: quiver Hall convolution in the quiver map, Calabi–Yau completion in the dg-algebra map, chiral Chevalley theory in the Lie map, Hong quantization in the Cartan superdatum, and the diagonal Fredholm operator in the root-graded map. PBW promotes the primitive isomorphism to the completed enveloping and Fock objects. $\square$

The Borchers-to-Calabi–Yau completion

Theorem 104.1 (Exact root-datum construction). *The sequence*

$$\begin{aligned} \{\text{finite Borchers simple-root superdatum}\} &\longmapsto \mathcal{Q}_{\Delta,H} \longmapsto \Gamma_{\Delta,H}, \\ \Gamma_{\Delta,H} &\longmapsto \mathcal{B}_{\Delta,H} \longmapsto U_q(\mathfrak{g}_{\Delta_5}^{(H)}). \end{aligned}$$

is functorial under full inclusions of finite root windows. Proper-height passage gives a locally three-Calabi–Yau dg category, its Hall presentation, and a complete quantized Borchers superalgebra. The simple-root datum extracted from $\phi_{0,1}$ gives the universal Igusa protected avatar.

Proof. A full inclusion of root data adds vertices and arrows to the Borchers quiver. Keller completion and the quiver Hall functors preserve exact full subcategories. The generator-and-relation map to Hong’s quantized Borchers superalgebra is compatible with root truncation. Proper height makes each coefficient eventually constant. The Gritsenko–Nikulin theorem identifies the resulting root datum with the Igusa denominator algebra. $\square$

BOOK **XXI**

**Quantum groups, exchange, and
representation theory**

From positive half to full quantum group

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^- , carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \widehat{\otimes} H^0 \widehat{\otimes} H^+$. Its universal R -matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\mathbb{k}$ -linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \rightarrow \mathbf{Vect}_{\text{fd}}$$

be a faithful exact strong monoidal functor. Assume the coend

$$H = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M)$$

exists and that the canonical comparison functor to finite-dimensional H -comodules is comonadic. Tensor product makes H a bialgebra, and rigidity supplies its antipode. A braiding gives a coquasitriangular form; a meromorphic braiding gives its spectral version.

Theorem 106.1 (Rigid Tannaka reconstruction). *Under the preceding finite and comonadic hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathbf{Comod}_{\text{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H .

Proof. Matrix coefficients define the universal H -coaction on every $\omega(M)$. Tensor product and duality give the bialgebra and antipode structures. The Barr–Beck comonadic theorem identifies $\mathcal{L}$ with coalgebras for the induced comonad, and the coend identifies that comonad with $H \otimes -$. The hexagon identities transport to the coquasitriangular identities. $\square$

This reconstruction explains why a line-operator category, a fiber functor, and exchange together produce a quantum group. The ordered boundary algebra alone provides fusion; the plane of exchange supplies braiding.

KZ and rational exchange

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R -matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

The centre and the double

The Hochschild centre $Z_{\text{ord}}(A)$ is the universal E_2 algebra acting centrally on an ordered algebra. The Drinfeld centre $Z(\text{Mod}_A)$ is the braided category of modules equipped with half-braidings. Under dualizability hypotheses these are related by the passage from algebraic Hochschild cochains to categorical endomorphisms of the identity. Derived loop-space descriptions of centres provide the geometric form of this relation [BZFN10].

The centre and the double answer complementary questions. The centre classifies central bulk actions. The double reconstructs a full quantum group from paired positive and negative halves. A field theory can supply a comparison by realizing both as line and bulk operator constructions.

BOOK **XXII**

**A compact six-dimensional
protected source**

Holomorphic cotangent BF theory on $K3 \times E$

Let S be a K3 surface with holomorphic symplectic form σ_S , let E be an elliptic curve with holomorphic form dz , and put

$$X = S \times E, \quad \Omega_X = \sigma_S \wedge dz.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta_5,+}^{(H)}$ of the positive Igusa Borchers algebra, take fields

$$A \in \Omega^{0,\bullet}(X, \mathfrak{n}_{\Delta_5,+}^{(H)})[1], \quad B \in \Omega^{0,\bullet}(X, (\mathfrak{n}_{\Delta_5,+}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A, B) = \int_X \Omega_X \wedge \langle B, \bar{\partial} A + \frac{1}{2}[A, A] \rangle. \quad (109.1)$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta_5,+}^{(H)}$ -valued Dolbeault forms.

Theorem 109.1 (Classical compact-background source). *The action (109.1) satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X .*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta_5,+}^{(H)}$ together with $\bar{\partial}^2 = 0$ and the derivation property of $\bar{\partial}$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. $\square$

Tree exactness and the quantum master equation

Every interaction vertex in (109.1) has one B -leg and two A -legs. The propagator pairs an A -leg with a B -leg.

Lemma 110.1 (Loop bound). *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B -half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. $\square$

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta_5,+}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Theorem 110.2 (Perturbative exactness). *Every one-loop Lie weight of the finite-height theory vanishes. The renormalized quantum master equation holds with zero counterterm, and the perturbative effective action is represented by tree graphs. The statement is compatible with the height maps $H \rightarrow H + 1$.*

Proof. The loop bound leaves tree graphs and one-loop graphs. Their Lie factor is a cyclic trace of a product of adjoint or coadjoint operators. Positive height makes each such product strictly triangular, so its trace is zero. The anomaly and the one-loop effective vertex therefore vanish. Tree graphs satisfy the classical master equation by the usual boundary-stratum argument. Every map preserves root height. $\square$

Corollary 110.3 (Height-complete quantum source). *The coefficientwise inverse limit*

$$\text{Obs}_X^\Delta = \varprojlim_H \text{Obs}_X^{(H)}$$

is a perturbative quantum factorization algebra on $K3 \times E$. Its primitive gauge algebra is the complete positive Igusa Borchers algebra.

Proof. Every fixed root height stabilizes in the finite tower. The factorization maps and quantum differential are compatible with the truncations, so the separated inverse limit preserves the quantum master equation and the primitive Lie algebra. $\square$

Harmonic reduction along K3

The Dolbeault algebra of a K3 surface has the strict harmonic model

$$H^{0,\bullet}(S, \mathcal{O}_S) = \mathbb{C}\mathbf{1} \oplus \mathbb{C}\bar{\sigma}_S[-2], \quad \bar{\sigma}_S^2 = 0.$$

The trace pairs $\mathbf{1}$ with $\bar{\sigma}_S$.

Theorem 111.1 (K3 harmonic reduction). *Factorization pushforward along $p_E : S \times E \rightarrow E$, followed by the K3 harmonic deformation retract, gives a quasi-isomorphism*

$$p_{E,*} \text{Obs}_X^{(H)} \simeq \text{Obs}_E \left(T^*[-1] (\text{Cur}_E(\mathfrak{n}_{\Delta_5,+}^{(H)})) \otimes H^{0,\bullet}(S, \mathcal{O}_S) \right).$$

The primitive degree-zero current sector is $\text{Cur}_E(\mathfrak{n}_{\Delta_5,+}^{(H)})$. The degree-two K3 class is its cotangent partner.

Proof. Kähler Hodge theory gives a deformation retract of the Dolbeault complex onto harmonic forms. The harmonic representatives $\mathbf{1}$ and $\bar{\sigma}_S$ are closed under multiplication, so the retract is a cyclic dg-algebra quasi-isomorphism. Tensoring the local cyclic Lie model with this retract and applying homological transfer gives the displayed cotangent current theory. The product $\bar{\sigma}_S^2$ vanishes by complex dimension, so higher transferred products vanish. $\square$

Corollary 111.2 (Exact curve shadow). *The K3 harmonic reduction of the compact six-dimensional source is the cotangent current factorization algebra of the Igusa positive root algebra on E . On a formal disc, its chiral enveloping mode algebra is $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$, and the associative chiral bar of this envelope is the completed Chevalley complex. A topological collar supplies the corresponding transverse interval algebra by the boundary polarization of Volume II.*

Proof. The harmonic reduction theorem identifies the finite-height curve theories. Chiral PBW identifies the associative chiral bar of the current envelope with chiral Chevalley chains. Proper height passes this statement to the complete limit. Boundary polarization of the cotangent theory gives the transverse enveloping algebra on a collar. $\square$

This construction gives a compact effective field-theoretic source whose gauge algebra is selected by the Igusa root datum. The geometric E -equivariant Hall algebra constructed above gives an independent compact D-brane source. A primitive root framing compares its extension algebra with the effective source.

Protected states and the Igusa trace

Let

$$\mathcal{H}_{\text{Ip}} = \widehat{\bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta_5 \alpha}}$$

with its Borchers parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\text{Fock}} = \widehat{\text{Sym}}(\mathcal{H}_{\text{Ip}, \bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\text{Ip}, \bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-Hz} act by $e^{-\alpha(Z)}$ on the root space of degree α .

Theorem 112.1 (Protected Hamiltonian trace). *The supertrace is*

$$\text{str}_{\mathcal{H}_{\text{Fock}}}(e^{-Hz}) = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{-\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\text{Den}_{\Delta}(Z) = D_5(2Z)$. After the half-coordinate pullback and doubling, the disconnected protected scalar is $-(J^ \text{Den}_{\Delta})^{-2} = -\chi_{10}^{-1}$ in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. $\square$

The compact-source comparison

Let $\mathfrak{P}_\Delta(X)$ denote the compact cotangent theory above, its K3 harmonic shadow, its Hall boundary category from the Borchers quiver, and its root Dirac determinant.

Theorem 113.1 (Compact protected source theorem). *The system $\mathfrak{P}_\Delta(K3 \times E)$ contains, in one functorial construction:*

1. *a quantum factorization algebra on the compact Calabi–Yau threefold;*
2. *a curve-level local–wrapped current algebra on E ;*
3. *an ordinary quiver Hall boundary, a parity-refined root superspace, and a three-Calabi–Yau Ginzburg completion;*
4. *Hong’s height-complete quantized Borchers superalgebra;*
5. *the Fredholm Pfaffian Den_Δ and the scalar trace $-(J^* \text{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$.*

Every arrow among these objects preserves root degree, parity, product, and completion.

Proof. The compact BV theory and its quantum exactness were proved above. Harmonic reduction gives the curve theory. The Borchers quiver and Ginzburg completion give the Hall boundary. Hong’s construction gives the quantum group. The root Hamiltonian gives the determinant. Each construction is functorial in finite root windows, and proper height gives the common limit. □

The theorem realizes the complete universal root-datum protected model over the $K3 \times E$ background. The geometric D-brane theory enters through a monoidal functor from its oriented extension category to the Hall boundary of $\mathfrak{P}_\Delta$. Such a functor is determined by primitive branes and their extension amplitudes.

BOOK **XXIII**

Physics of the comparison maps

Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathbb{I})) \simeq C_*(\mathbb{I}).$$

For $\mathbb{I} = \mathfrak{n}_{\Delta_5,+}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

Six-dimensional hCS on $K3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS} . Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S .

A protected reduction to a chiral algebra consists of a deformation retract of the K3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S \times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

Omega backgrounds

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \quad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R -matrix. Toric threefolds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant K3 fibers to the selected generic fiber.

Anomalies as typed classes

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

BOOK **XXIV**

**Automorphic characters and scalar
shadows**

Borcherds products as characters

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi_{C_*(\mathfrak{n}_+)} = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

The universal Borchers weight formula

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borchers lift has weight

$$\text{wt}(\text{Borch}(\varphi)) = \frac{c(0, 0)}{2}.$$

For the K3 half-index, $c(0, 0) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_5, \Delta_2, \Delta_1, \Delta_{1/2}, \nabla_3, \nabla_2, \nabla_{3/2}, \mathcal{Q}_1$$

with weights

$$5, 2, 1, \frac{1}{2}, 3, 2, \frac{3}{2}, 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [CG11]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

Scalarization hierarchy

The constructions descend through four levels:

factorization or Hall object $\longrightarrow$ primitive Lie algebra
 $\longrightarrow$ PBW character $\longrightarrow$ scalar partition function.

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

BOOK **XXV**

The unified theorem system

The foundational theorem

Definition 122.1 (Complete Calabi–Yau quantum-group datum). A complete datum is a governing tuple $\mathfrak{D}$ together with:

1. PTVV-representable moduli for the smooth proper Calabi–Yau category;
2. a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b ;
3. a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;
4. a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;
5. paired negative and Cartan halves, a completed coproduct, a continuous non-degenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;
6. compatible open–closed and Hall–boundary morphisms;
7. a parity-refined primitive comparison for every automorphic identification.

Theorem 122.2 (Calabi–Yau quantum-group assembly). *Let a complete datum consist of:*

1. *a Calabi–Yau category and a representable moduli-of-objects problem;*
2. *a quantized cyclic elliptic BV theory and a boundary or defect generator;*
3. *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*
4. *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*

5. *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

1. *framed E_2 Hochschild cochains and the shifted-symplectic moduli structure;*
2. *the three curve-level algebraic faces and their three bars;*
3. *the positive Hall/BPS algebra and its primitive Lie algebra;*
4. *the completed quantum double and braided representation category;*
5. *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*
6. *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [BR23, PTVV13, BD21]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [KPS24]. The paired bialgebra data give the double and universal R -matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. $\square$

The local quiver, equivariant surface, K3 correspondence, and Igusa Borchers chapters construct objects in their stated categories. A geometric $K3 \times E$ Hall theory determines a morphism from its primitive and determinant-line data to the universal Igusa objects.

Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2}/\Gamma$	$Y^+(\widehat{\mathfrak{g}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
$K3$	$U(\mathfrak{g}_{\text{BPS}}^+)$	Heisenberg and LLV actions	K3 Hall characters
$K3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_5})$ and the height-complete current/BF model	$-\chi_{10}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row combines exact automorphic, Lie, bar, current, BF, Fredholm, and enumerative constructions; geometric Hall data enter through the primitive comparison morphism.

Closure theorem for the established and universal layers

Theorem 124.1 (Foundational closure). *The following layers are complete mathematical constructions:*

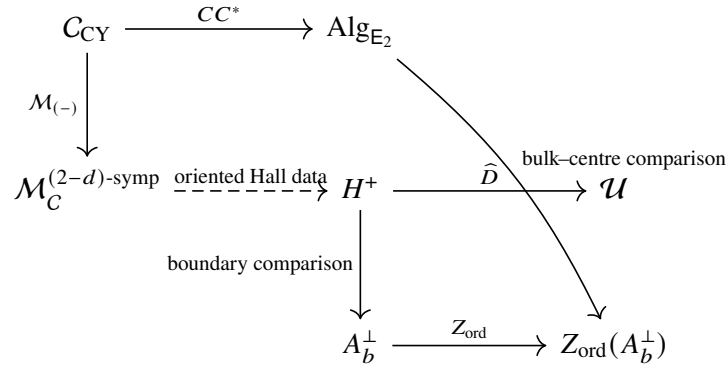
1. *framed Hochschild and shifted-symplectic categorical data;*
2. *perturbative factorization observables from a quantized cyclic elliptic BV theory;*
3. *the three curve-level algebraic faces and their bars;*
4. *Hall products in the established quiver, surface, 2-Calabi–Yau, and strong-oriented 3-Calabi–Yau settings;*
5. *the K3 Heisenberg, LLV, and local ADE Yangian symmetry atlas;*
6. *the Igusa Borchers root algebra, its quantized Borchers superalgebra, proper-height bar/current system, classical local BF theory, exact finite zero-mode BV theory, and analytic determinant;*
7. *the primitive reduced $K3 \times E$ scalar character.*

A compact $K3 \times E$ quantum-group realization is exactly an instance of the compact comparison theorem with a constructed reduced Hall datum.

Proof. Items (1)–(7) are the theorems cited and proved in the corresponding chapters. The last sentence is the definition of a compact realization together with the preceding recognition theorem. $\square$

Common diagram

The relationships are summarized by



Here $A_b^\perp \in \text{Alg}_{E_1}(\text{Fact}_D(C), \otimes^!)$. Every arrow is a construction or a supplied comparison morphism. The diagram separates universal categorical outputs, geometric field-theory outputs, Hall operations, and quantum-group assembly.

Concordance with the original Calabi–Yau volume

The reconstitution preserves the original vision of a bridge among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products. It realizes that vision through the following replacements:

1. framed E_2 Hochschild cochains and $(2 - d)$ -shifted symplectic moduli replace a dimension-dependent Hochschild E_n assertion;
2. a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;
3. pushforward along an actual map $p : X \rightarrow C$ replaces formal cycle-specialization notation;
4. extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;
5. coproduct, Cartan, Hopf pairing, and topology define the double;
6. K3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;
7. the graph of the $K3 \times E$ Gram map becomes multiplicative in the polarized central extension;
8. point-local and wrapped sectors form a hybrid factorization problem;
9. the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;

10. anomaly identities are transported through explicit maps between deformation complexes.

These constructions retain the scale of the original program while placing every operation in its natural category.

BOOK **XXVI**

**Technical appendices to the
preceding book**

Typed data table

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^!$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\text{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull-push product
Bialgebra	product, coproduct, and counit	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R -matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

The three K3 brackets

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\text{Corr}_{K3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

Exact arithmetic checks

The companion verification calculation establishes:

1. the low Fourier rows of $\phi_{0,1}$ through q^3 by exact theta-series division;
2. the vector-valued cocycle identity for the Gram polarization;
3. the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;
4. the A_2 Hall flag counts at ranks zero and one;
5. the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

Book entrance: The Igusa Square Root

The Jacobi seed, Borcherds denominator, root superalgebra, local-wrapped currents, quantum group, Hall model, and Fredholm realization.

BOOK **XXVII**

The automorphic determinant

The protected Jacobi input

130.1 The K3 elliptic genus

Let S be a K3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N} = (4, 4)$ sigma model is the elliptic genus

$$Z_{K3}(\tau, z) = 2\phi_{0,1}(\tau, z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^4 \frac{\vartheta_a(\tau, z)^2}{\vartheta_a(\tau, 0)^2} \in J_{0,1}^w$$

[EZ85]. The coefficient depends on the discriminant $4n - \ell^2$ and on ℓ modulo 2.

Theorem 130.1 (Exact low Fourier coefficients). *The first four q -rows are*

$$\begin{aligned} \phi_{0,1} = & r^{-1} + 10 + r \\ & + q(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2) \\ & + q^2(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3) \\ & + q^3(-64r^{-3} + 808r^{-2} - 2752r^{-1} + 4016 \\ & \quad - 2752r + 808r^2 - 64r^3) + O(q^4). \end{aligned}$$

In particular,

$$f(0, 0) = 10, \quad \frac{f(0, 0)}{2} = 5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t = q^{1/8}$ and $x = r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x . The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 160. $\square$

130.2 The virtual half-index

For every (n, ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\text{sdim } U_{n,\ell} = f(n, \ell).$$

This choice represents the coefficient in $K_0(\text{sVect}_{\text{fd}})$. The determinant construction depends on the superdimension.

Lemma 130.2 (Fock character and Berezinian). *Let $V = \bigoplus_{\gamma} V_{\gamma}$ be a locally finite graded super vector space. Then*

$$\text{sch}(\text{Sym}(V_0) \otimes \bigwedge (V_1)) = \prod_{\gamma} (1 - x^{\gamma})^{-\text{sdim } V_{\gamma}},$$

and

$$\text{sdet}_V(1 - x) = \prod_{\gamma} (1 - x^{\gamma})^{\text{sdim } V_{\gamma}}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^{\gamma})^{-1}$. An odd one-dimensional summand contributes $1 - x^{\gamma}$. Multiplication over a homogeneous basis proves both identities. $\square$

The Borchers product

131.1 The effective chamber

Define

$$\Gamma_{\text{eff}} = \{(n, \ell, m) \in \mathbb{Z}^3 : m > 0, n \geq 0\} \\ \cup \{(n, \ell, 0) : n > 0\} \cup \{(0, \ell, 0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n, \ell, m)$ put

$$V_\gamma = U_{nm, \ell}, \quad \text{sdim } V_\gamma = f(nm, \ell).$$

Definition 131.1 (Monic virtual determinant). The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \text{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\text{eff}}} (1 - q^n r^\ell s^m)^{f(nm, \ell)}.$$

Theorem 131.2 (Borchers–Gritsenko–Nikulin product). *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5 . The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0, 0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [Bor95, Bor98, GN97, GN98b]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. $\square$

131.2 The theta constant normalization

Let $\mathfrak{E}$ be the ten even genus-two characteristics. Define

$$\Delta_5^\theta(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \vartheta[\mathfrak{m}](Z, 0).$$

The cusp leading terms of the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. Thus

$$[q^{1/2}r^{1/2}s^{1/2}]\Delta_5^\theta = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 131.3 (Ten-factor leading-term calculation). *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	scalar	monomial
$(0, 0)$	4	1	1
$(1, 0)$	2	2	$q^{1/8}$
$(0, 1)$	2	2	$s^{1/8}$
$(1, 1)$	2	2	$q^{1/8}r^{1/4}s^{1/8}$.

Multiplication gives

$$1^4 2^2 2^2 2^2 q^{1/2}r^{1/2}s^{1/2} = 64q^{1/2}r^{1/2}s^{1/2}.$$

Proof. Insert each characteristic into the genus-two theta series and retain the unique terms of least order in the lexicographic cusp valuation $(m, n, -\ell)$. For $a = (0, 0)$ the zero lattice vector contributes one. For each nonzero a , the two shortest lattice representatives are exchanged by sign and contribute the scalar two. Their quadratic exponents are the entries in the final column. Summing the exponents and multiplying the six factors of two gives the formula. $\square$

Proposition 131.4 (Boundary factor of the Borcherds product). *The $m = 0$ part of the monic product is*

$$\prod_{n>0, \ell \in \mathbb{Z}} (1 - q^n r^\ell)^{f(0, \ell)} \prod_{\ell < 0} (1 - r^\ell)^{f(0, \ell)} = (1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1})(1 - q^n)^{10}(1 - q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2}r^{1/2}(1 - r^{-1}) \prod_{n \geq 1} (1 - q^n r^{-1})(1 - q^n)^{10}(1 - q^n r) = \eta(\tau)^9 \vartheta_{11}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\vartheta_{11}(\tau, z) = -q^{1/8}r^{-1/2} \prod_{n \geq 1} (1 - q^{n-1}r)(1 - q^n r^{-1})(1 - q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0, -1) = 1$, $f(0, 0) = 10$, and $f(0, 1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ give the second, including its leading sign and monomial. $\square$

Corollary 131.5 (The scalar square). *The monic weight-ten form is*

$$\chi_{10} = D_5^2 = 2^{-12} \Delta_5^{\theta^2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^{\theta} = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. $\square$

Divisor, character, and the meaning of the square root

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = 0\} / \mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 132.1 (Diagonal divisor). *The divisor of D_5 is the $\mathrm{Sp}_4(\mathbb{Z})$ -orbit of H_1 , with multiplicity one. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = D_5^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [Igu62, Igu64, CG11]. $\square$

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

132.1 Automorphic section, orientation line, and Pfaffian operator

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\begin{aligned} &\text{automorphic section} \longrightarrow \text{determinant-line identification} \\ &\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.} \end{aligned}$$

Proposition 132.2 (Logical separation of the four levels). *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. $\square$

The title names the monic automorphic section D_5 , characterized by the exact identity $D_5^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

BOOK **XXVIII**

The denominator algebra

The Lorentzian root lattice

Let

$$L = \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \quad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus 2\mathbb{Z}f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\text{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \quad \alpha(n, \ell, m) = 2nf_2 - \ell f_3 + 2mf_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2(2nm' + 2n'm - \ell\ell'),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

133.1 Real simple roots and Weyl vector

Put

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3.$$

Proposition 133.1 (Type-II Cartan matrix). *The Gram matrix of the real simple roots is*

$$((\delta_i, \delta_j))_{i,j=1}^3 = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. $\square$

The Weyl vector is

$$\rho = f_2 - \frac{1}{2}f_3 + f_{-2}, \quad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$.

The Gritsenko–Nikulin Lie superalgebra

The Borchers product determines a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$ with real simple roots $\delta_1, \delta_2, \delta_3$ and an imaginary correction system determined by $\phi_{0,1}$.

Theorem 134.1 (Root supermultiplicities and denominator). *On the active positive support,*

$$\text{smult } \mathfrak{g}_{\Delta_5} \alpha(n, \ell, m) = f(nm, \ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6} \Delta_5^\theta(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borchers product has exponent $f(nm, \ell)$. The resulting product is the doubled-variable monic Igusa form [GN97, GN98a, GN98b]. $\square$

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

The ordered bar as the denominator complex

Let $\mathfrak{n}_{\Delta_5,+}$ be the positive nilpotent subalgebra of $\mathfrak{g}_{\Delta_5}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H} \mathfrak{n}_{\Delta_5,+} = \bigoplus_{h(\alpha) > H} \mathfrak{g}_{\Delta_5 \alpha}, \quad \mathfrak{n}_{\Delta_5,+}^{(H)} = \mathfrak{n}_{\Delta_5,+} / F^{>H} \mathfrak{n}_{\Delta_5,+}.$$

Height additivity makes $F^{>H} \mathfrak{n}_{\Delta_5,+}$ an ideal. Properness and local finiteness make $\mathfrak{n}_{\Delta_5,+}^{(H)}$ finite-dimensional and nilpotent.

Theorem 135.1 (Height-complete Bar–Chevalley comparison). *For every H , antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$\mathrm{Bar}(U(\mathfrak{n}_{\Delta_5,+}^{(H)})) \simeq C_*(\mathfrak{n}_{\Delta_5,+}^{(H)}).$$

The quotient maps preserve these comparisons and induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^\wedge(U(\mathfrak{n}_{\Delta_5,+})) := \varprojlim_H \mathrm{Bar}(U(\mathfrak{n}_{\Delta_5,+}^{(H)})) \simeq C_{*,h}^\wedge(\mathfrak{n}_{\Delta_5,+}) := \varprojlim_H C_*(\mathfrak{n}_{\Delta_5,+}^{(H)}).$$

Its root-graded Euler supercharacter is

$$\chi_{\mathrm{root}} C_{*,h}^\wedge(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\mathrm{sdim} \mathfrak{g}_{\Delta_5 \alpha}}.$$

Proof. For each H , the antisymmetrization map sends $\mathrm{Sym}(\mathfrak{n}_{\Delta_5,+}^{(H)}[1])$ to the normalized bar of $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$. The PBW filtrations have the same associated-graded Koszul resolution of $\mathrm{Sym}(\mathfrak{n}_{\Delta_5,+}^{(H)})$, so antisymmetrization is a filtered quasi-isomorphism. Naturality gives a

morphism of inverse systems. Every fixed root degree is visible in all sufficiently large height quotients and is unchanged thereafter; hence the inverse-limit map is a quasi-isomorphism coefficientwise. An even root vector becomes odd after the Chevalley shift and contributes $1 - e^{-\alpha}$; an odd root vector becomes even and contributes its reciprocal. Multiplication over the finite-dimensional root spaces gives the product. $\square$

Let $\mathbb{T}_{LII}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \quad e^{-\rho} = Q^{1/2} R^{1/2} S^{1/2}.$$

Corollary 135.2 (Bar denominator identity). *On the formal root torus,*

$$e^{-\rho} \chi_{\text{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\text{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [GN97, GN98b]. Hence

$$e^{-\rho} \chi_{\text{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \text{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. $\square$

The denominator is therefore the Weyl-shifted Euler supercharacter of the height-complete interval bar of the positive enveloping algebra.

Finite-height currents

Retain the finite-height quotients $\mathfrak{n}_{\Delta_5,+}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\mathrm{Cur}(\mathfrak{n}_{\Delta_5,+}^{(H)}) = \mathbb{K}[\partial] \otimes \mathfrak{n}_{\Delta_5,+}^{(H)}, \quad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\mathrm{ch}}\left(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+}^{(H)})\right).$$

Theorem 136.1 (Height-complete current envelope). *The quotient maps induce a coefficientwise inverse limit*

$$\widehat{V}_{\Delta_5} = \varprojlim_H V_H.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\mathrm{sch} \widehat{V}_{\Delta_5} = \prod_{j \geq 1} \prod_{\alpha > 0} (1 - t^j e^{-\alpha})^{-\mathrm{sdim} \mathfrak{g}_{\Delta_5} \alpha}.$$

Moreover,

$$\mathrm{Bar}_{\mathrm{Ass}}^{\mathrm{ch}}(V_H) \simeq \mathrm{Bar}_{\mathrm{Lie}}^{\mathrm{ch}}\left(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+}^{(H)})\right),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes contribute exterior powers, giving the product. Proper height makes every coefficient stationary. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra; the complete separated PBW filtration is coefficientwise finite. $\square$

136.1 Classical local theory and exact quantum zero modes

For every H , the finite-dimensional nilpotent Lie superalgebra $\mathfrak{n}_{\Delta_5,+}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{0,*}(E) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta_5,+}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + d_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 136.2 (Finite-height BF realization). *The action S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra. Harmonic reduction along a compact interval produces the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta_5,+}^{(H)}$. The positive-root height grading makes every even adjoint endomorphism strictly height-increasing and every odd adjoint endomorphism odd; hence the modular supercharacter vanishes and the reduced theory satisfies the quantum master equation.*

Choose the standard complementary Lagrangian boundary polarizations, fixing the $\mathfrak{n}_{\Delta_5,+}^{(H)}[1]$ coordinate at one endpoint and its cotangent coordinate at the other. The algebraic boundary quantization, the interval amplitude between the two augmentation states, and the holomorphic boundary current envelope are then

$$U(\mathfrak{n}_{\Delta_5,+}^{(H)}), \quad \mathbf{1} \otimes_{U(\mathfrak{n}_{\Delta_5,+}^{(H)})}^{\mathbb{L}} \mathbf{1} \simeq C_*(\mathfrak{n}_{\Delta_5,+}^{(H)}), \quad V_H.$$

The height quotient maps preserve the polarizations and yield the inverse systems

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \quad C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}), \quad \widehat{V}_{\Delta_5}.$$

Proof. The classical master equation is the Jacobi identity paired with invariance of the cotangent pairing. Elliptic descent gives the classical factorization algebra. Harmonic reduction along the interval gives a finite-dimensional BV space. If x is even of positive root height, ad_x raises height and is nilpotent on the finite quotient, so its restrictions to the even and odd summands both have trace zero; if x is odd, ad_x is odd and has supertrace zero. Thus the modular character vanishes and the finite BV quantum master equation follows.

The chosen boundary polarization quantizes the linear Kirillov–Kostant boundary bracket to $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$. Gluing the two augmentation states gives the two-sided derived bar, and the PBW antisymmetrization identifies it with $C_*(\mathfrak{n}_{\Delta_5,+}^{(H)})$. Holomorphic descendants generate $U^{\text{ch}}(\text{Cur}_E \mathfrak{n}_{\Delta_5,+}^{(H)}) = V_H$. All maps preserve root height, so they pass coefficientwise to the separated inverse limits. $\square$

BOOK **XXIX**

The compact scalar character

Reduced curve counting on $K3 \times E$

Let S be a nonsingular projective K3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 137.1 (Primitive Igusa cusp-form theorem). *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [OP18]; their normalization is the monic Borcherds product χ_{10} . The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. $\square$

The theorem is a scalar trace identity and fixes every primitive reduced invariant. The Borcherds root algebra supplies the fixed algebraic target whose PBW character is the same product.

Corollary 137.2 (Exact character comparison). *Let $J : \mathbb{H}_2 \rightarrow \mathbb{H}_2$ be the half-coordinate map $J(Z) = Z/2$, and let*

$$\text{Fock}(\mathfrak{n}_{\Delta_5,+}) = \text{Sym}(\mathfrak{n}_{\Delta_5,+ \bar{0}}) \otimes \bigwedge (\mathfrak{n}_{\Delta_5,+ \bar{1}}).$$

Then

$$\mathcal{Z}_{S \times E}^{\text{red}} = -J^* \left(e^{2\rho} \text{sch}(\text{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}) \right) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_5} \alpha}.$$

The denominator theorem gives $\text{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\text{sch Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho} \text{sch}(\text{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}) = \text{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by J sends $\text{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10} . The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. $\square$

The equality is an identity of completed protected characters. The universal root algebra supplies the primitive bracket, and reduced curve counting supplies the geometric trace.

Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \widetilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\text{phys}} = \Lambda \oplus \Lambda, \quad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\text{ind}}.$$

Proposition 138.1 (Polarization identity). *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = (Q_1 \cdot Q_2, Q_1 \cdot P_2 + Q_2 \cdot P_1, P_1 \cdot P_2).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B . $\square$

138.1 The polarized central extension

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\text{ind}} \longrightarrow \tilde{\Gamma}_{\Pi} \longrightarrow \Gamma_{\text{phys}} \longrightarrow 0, \quad \tilde{\Gamma}_{\Pi} = \Gamma_{\text{phys}} \times \Gamma_{\text{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = (c + d, \lambda + \mu + B(c, d)).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_{\Pi} : \Gamma_{\text{phys}} \longrightarrow \tilde{\Gamma}_{\Pi}, \quad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\text{phys}}} A_c$ be a charge-graded associative algebra, and let $\mathbb{k}[\Gamma_{\text{ind}}]$ have monomials u^{λ} . Define the polarized central-extension algebra

$$\mathcal{R}_{\Pi}(A) = A \otimes \mathbb{k}[\Gamma_{\text{ind}}]$$

with product

$$(a_c u^{\lambda}) \star (b_d u^{\mu}) = a_c b_d u^{\lambda + \mu + B(c, d)}.$$

Give $a_c u^{\lambda}$ degree $(c, \lambda) \in \tilde{\Gamma}_{\Pi}$.

Theorem 138.2 (Graph-multiplicative polarization). *The product $\star$ is associative and $\tilde{\Gamma}_{\Pi}$ -graded. The map*

$$\iota_{\Pi} : A \longrightarrow \mathcal{R}_{\Pi}(A), \quad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_{\Pi}(c)$.

Proof. The two parenthesizations differ by

$$B(c, d) + B(c + d, e) - B(d, e) - B(c, d + e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_{\Pi}(a_c) \star \iota_{\Pi}(b_d) = a_c b_d u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. $\square$

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

Point-local and wrapped sectors

Let $p_E : S \times E \rightarrow E$ be projection.

Lemma 139.1 (Positive elliptic degree is wrapped). *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E . Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C , is all of E . $\square$

Projection-finite objects define point-supported sectors on $\text{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 139.2 (The local–wrapped factorization operad). Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$l, \quad w_m \quad (m \geq 1)$$

by the following operation objects.

1. Operations with local output are

$$\text{Hyb}_E(l^{\otimes k}; l) = \text{Ch}_E(k).$$

2. For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\text{Hyb}_E(l^{\otimes k}, w_{m_1}, \dots, w_{m_r}; w_m) = \text{Ch}_E(k) \otimes \text{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 139.3 (Hybrid local–wrapped algebra). A hybrid local–wrapped algebra over E is an algebra over Hyb_E . It consists of a factorization algebra $\mathcal{F}_{\text{loc}}$, a height- or charge-complete family $W = \prod_{m>0} W_m$, ordered products

$$\mu_{m_1, \dots, m_r} : W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1 + \dots + m_r},$$

and factorization-compatible actions of the local operation objects $\text{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E -algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

BOOK **XXX**

**The Borcherds algebra and its
quantum group**

The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m . Set

$$\mathfrak{n}_{\Delta_5, +\text{loc}} = \bigoplus_{\substack{\alpha > 0 \\ m(\alpha) = 0}} \mathfrak{g}_{\Delta_5 \alpha}, \quad \mathfrak{n}_{\Delta_5, +\text{wr}} = \bigoplus_{\substack{\alpha > 0 \\ m(\alpha) > 0}} \mathfrak{g}_{\Delta_5 \alpha}.$$

Proposition 140.1 (Semidirect local–wrapped decomposition). *The space $\mathfrak{n}_{\Delta_5, +\text{loc}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_5, +\text{wr}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_5, +} = \mathfrak{n}_{\Delta_5, +\text{loc}} \ltimes \mathfrak{n}_{\Delta_5, +\text{wr}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. $\square$

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_5} = \widehat{U}_h(\mathfrak{n}_{\Delta_5, +}), \quad \mathbb{V}_{\Delta_5, E} := \widehat{V}_{\Delta_5} = \varprojlim_H U^{\text{ch}}\left(\text{Cur}_E(\mathfrak{n}_{\Delta_5, +}^{(H)})\right).$$

The adjoint action of $\mathfrak{n}_{\Delta_5, +\text{loc}}$ on $\mathfrak{n}_{\Delta_5, +\text{wr}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_5, +\text{wr}})$. Let $\#$ denote the corresponding completed smash product.

Definition 140.2 (Current-generated local–wrapped operad). The colored operad $\text{Hyb}_{E, \Delta}^{\text{cur}}$ is generated by the chiral current operations of $\text{Cur}_E(\mathfrak{n}_{\Delta_5, +\text{loc}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_5, +\text{wr}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 140.3 (Local–wrapped Borchers decomposition). *Let $\mathfrak{n}_{\Delta_5,+_{\text{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_5,+_{\text{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\text{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\text{wr}}},$$

where $\mathfrak{n}_{\Delta_5,+_{\text{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_5,+_{\text{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\text{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\text{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \simeq \text{Tot } C_*(\mathfrak{n}_{\Delta_5,+_{\text{loc}}}; \widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+_{\text{wr}}})) .$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_5,+_h}} = \varprojlim_H \mathfrak{n}_{\Delta_5,+}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_5(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. $\square$

Remark 140.4. The theorem constructs a Borchers root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

The formal Igusa quantum group

Let $\mathfrak{b}_+$ be the positive Borchers Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee, \text{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted $\mathfrak{g}_{\Delta_5}^{\text{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 141.1 (Height-admissible Lie superbialgebra). A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 141.2 (Height admissibility). *The paired Gritsenko–Nikulin Borchers Lie superbialgebra $\mathfrak{g}_{\Delta_5}^{\text{pair}}$ is height-admissible.*

Proof. The Borchers root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. $\square$

Theorem 141.3 (Quantized Borchers superalgebra). *Choose the symmetrizable Borchers–Cartan superdatum and multiplicity index set that present $\mathfrak{g}_{\Delta_5}$. Hong’s construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta_5})$$

over $\mathbb{Q}(q)$, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition

$$U_q(\mathfrak{g}_{\Delta_5}) \cong U_q^-(\mathfrak{g}_{\Delta_5}) \otimes U_q^0(\mathfrak{g}_{\Delta_5}) \otimes U_q^+(\mathfrak{g}_{\Delta_5}).$$

Its PBW root spaces have the same graded ranks as the classical Borchers superalgebra. The canonical pairing of the two Borel halves yields a universal R -matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta_5}) \widehat{\otimes} U_q^-(\mathfrak{g}_{\Delta_5}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_5(2Z)$.

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borchers–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borchers superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R -matrix are established by Hong [Hon99]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. $\square$

Remark 141.4. The quantum group is attached to the Borchers generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

A holomorphic–topological BF realization

For every height bound H , let $\mathfrak{n}_{\Delta_5,+}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{0,*}(E) \widehat{\otimes} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta_5,+}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + d_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 142.1 (Classical local and exact zero-mode realization). *For every H :*

1. S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra;
2. harmonic reduction along a compact interval gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta_5,+}^{(H)}$, whose positive-root grading makes its modular supercharacter zero and hence gives the quantum master equation;
3. after choosing the complementary Lagrangian boundary polarizations, the boundary algebra is $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$, the interval amplitude is $\mathbf{1} \otimes_{U(\mathfrak{n}_{\Delta_5,+}^{(H)})}^{\mathbb{L}} \mathbf{1} \simeq C_*(\mathfrak{n}_{\Delta_5,+}^{(H)})$, and the holomorphic boundary current envelope is V_H .

The transition maps yield a pro-object with boundary algebra $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$, interval complex $C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+})$, and current envelope $\widehat{V}_{\Delta_5}$.

Proof. The classical master equation is the Jacobi identity and invariance of the cotangent pairing. The classical observables satisfy factorization descent. Harmonic reduction is finite-dimensional. An even adjoint operator raises positive-root height and is nilpotent on

the finite quotient; an odd adjoint operator is odd. Both have supertrace zero, so the modular term in the finite BV Laplacian vanishes. The chosen polarization quantizes the boundary Lie–Poisson bracket to the enveloping algebra; the two augmentation states give the two-sided bar, identified with Chevalley chains by PBW. Holomorphic descendants give the current algebra. All structure maps preserve height. $\square$

The theorem gives the precise field-theoretic realization used in this volume: a classical local theory together with an exact quantum protected zero-mode sector.

The determinant and Fredholm square-root operators

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha > 0 \\ h(\alpha) \leq H}} \mathfrak{g}_{\Delta_5 \alpha}$$

and define

$$\mathcal{D}_H|_{\mathfrak{g}_{\Delta_5 \alpha}} = (1 - e^{-\alpha}) \text{id}.$$

Theorem 143.1 (Universal determinant section). *The finite-height Berezinian is*

$$\text{Ber}(\mathcal{D}_H) = \prod_{\substack{\alpha > 0 \\ h(\alpha) \leq H}} (1 - e^{-\alpha})^{\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho} \text{Ber}(\mathcal{D}_\infty) = D_5(2Z).$$

After pullback by $J(Z) = Z/2$,

$$J^*(e^{-\rho} \text{Ber}(\mathcal{D}_\infty)) = D_5(Z), \quad D_5(Z)^2 = \chi_{10}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. $\square$

143.1 Trace-class realization on the root Hilbert superspace

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta_5 \alpha}}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\text{abs}}$ be the subchamber on which

$$\sum_{\alpha > 0} \dim \mathfrak{g}_{\Delta_5 \alpha} \left| e^{-\alpha(Z)} \right| < \infty.$$

For $Z \in \mathfrak{C}_{\text{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta_5 \alpha}} = e^{-\alpha(Z)} \text{id}, \quad D_Z = 1 - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z, \bar{\epsilon}} = \begin{pmatrix} 0 & D_{Z, \bar{\epsilon}} \\ -D_{Z, \bar{\epsilon}}^\dagger & 0 \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = 0$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\text{sPf}_F(\mathbb{A}_Z) = \frac{\text{Pf}_F(\mathbb{A}_{Z, \bar{0}})}{\text{Pf}_F(\mathbb{A}_{Z, \bar{1}})}.$$

Theorem 143.2 (Polarized Fredholm realization). *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta_5 \alpha}} = e^{-\alpha(Z)} \text{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = 1 - K_Z$,*

$$\text{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha > 0} (1 - e^{-\alpha(Z)})^{\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_\Delta \oplus \mathcal{H}_\Delta^$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z, \bar{0}}$ and $\mathbb{A}_{Z, \bar{1}}$ have an oriented Fredholm super-Pfaffian*

$$\text{sPf}_F(\mathbb{A}_Z) = \frac{\text{Pf}_F(\mathbb{A}_{Z, \bar{0}})}{\text{Pf}_F(\mathbb{A}_{Z, \bar{1}})} = \text{sdet}_F(D_Z).$$

Hence

$$e^{-\rho} \text{sdet}_F(D_Z) = D_5(2Z).$$

Proof. The preceding convergence theorem gives the trace-class criterion for the diagonal operator. The Fredholm determinant of $1 - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z . Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. $\square$

Remark 143.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This theorem constructs the square-root operator on the universal Borchers root Hilbert superspace. A geometric realization is a determinant-line intertwiner from a protected geometric Hilbert complex to this universal operator family.

What the automorphic character determines

Theorem 144.1 (Denominator data and reconstruction data). *The Borchers product determines:*

1. the signed root multiplicity $\text{sdim } \mathfrak{g}_{\Delta_5, \alpha}$ in every active root degree;
2. the Weyl vector and denominator character;
3. the divisor and automorphic multiplier of the section.

A Lie bracket is supplied by the Borchers–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. $\square$

Recognition Theorem 144.2 (Complete primitive recognition). *Let $\mathfrak{p}$ be a positive graded Lie superalgebra and let $\Psi : \mathfrak{n}_{\Delta_5, +} \twoheadrightarrow \mathfrak{p}$ be a graded Lie map. The map is an isomorphism if either:*

1. the parity-refined Hilbert series of $U(\mathfrak{n}_{\Delta_5, +})$ and $U(\mathfrak{p})$ agree; or
2. each primitive degree has prescribed pure parity and the signed PBW characters agree.

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. $\square$

BOOK **XXXI**

**The universal Borchers Hall and
three-Calabi–Yau models**

Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borcherds superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{i \in I_\Delta : h(i) \leq H\}.$$

The restriction of the invariant form gives a finite symmetric Borcherds–Cartan matrix

$$\begin{aligned} A_H &= (a_{ij})_{i,j \in I_H}, \\ a_{ii} &= 2 && \text{for real } i, \\ a_{ii} &\leq 0 && \text{for imaginary } i, \\ a_{ij} &\leq 0 && \text{for } i \neq j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 145.1 (The Igusa Borcherds quiver). Order I_H . For $i < j$ place $-a_{ij}$ arrows from i to j . At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $\mathcal{Q}_{\Delta,H}$.

Proposition 145.2 (Euler form). *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H :*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \neq j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij} . At i the value is $2 - 2g_i = a_{ii}$. $\square$

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \quad \delta_2 = 2f_{-2} - f_3, \quad \delta_3 = f_3,$$

with Gram matrix

$$A_{\text{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 146.1 (Real-root Hall seed). *The symmetrized Euler form of Q_{re} is A_{re} . The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re} .*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2 . The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. $\square$

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

146.1 The first Ginzburg differential

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \rightarrow j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \rightarrow i$ of degree -1 and loops t_i of degree -2 . The canonical three-Calabi–Yau completion has

$$da_{ij}^{(r)} = 0, \quad d(a_{ij}^{(r)})^* = 0,$$

and

$$dt_i = \sum_{a:s(a)=i} aa^* - \sum_{a:t(a)=i} a^*a. \quad (146.1)$$

The identity $d^2 = 0$ follows immediately. Formula (146.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 146.2 (Vertex one). At vertex 1,

$$dt_1 = \sum_{r=1}^2 \left(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \right).$$

At vertex 3 the same four terms occur with reverse order and negative sign. Summing the three equations gives zero, as required by the trace of the moment map.

Proof. Vertex 1 is the source of the four displayed arrows and the target of none, so the Ginzburg moment-map formula contains exactly the four positive terms. Vertex 3 is their terminal counterpart. Every arrow–reverse-arrow pair occurs once with each sign in the sum over vertices. $\square$

Hall realization of the positive algebra

Let $\mathcal{A}_H = \text{Rep}_{\mathbb{F}_q}(Q_{\Delta, H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N \rangle} \sum_{[L]} \frac{|\{N' \subset L : N' \cong N, L/N' \cong M\}|}{|\text{Aut } M| |\text{Aut } N|} [L], \quad v^2 = q.$$

Theorem 147.1 (Finite-window Hall theorem). *The composition subalgebra generated by the simple representations of $Q_{\Delta, H}$ is the positive quantum generalized Kac–Moody algebra attached to the underlying even Borcherds–Cartan matrix A_H . Semisimple perverse sheaves on the representation varieties give its canonical basis. The semi-derived Hall algebra of two-periodic complexes realizes the corresponding full ordinary quantum generalized Kac–Moody algebra after Cartan reduction.*

Proof. The positive statement is the quiver-with-loops construction of Kang–Schiffmann. Their perverse-sheaf category gives the canonical basis. Lu’s semi-derived Hall theorem realizes the full quantum generalized Kac–Moody algebra from the same quiver with loops; the Cartan reduction identifies the two torus copies [KS06, Lu21, AL19]. $\square$

147.1 Parity and the quantized superdatum

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. Extend it additively to the root lattice. Hong’s construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta_5}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta_5}^{(H)})$.

Theorem 147.2 (Parallel Hall and supergroup realizations). *The quiver Hall algebra and Hong's quantum supergroup are two functorial realizations of the same finite Borcherds–Cartan window:*

1. *the quiver Hall algebra realizes the underlying ordinary generalized Kac–Moody presentation and its canonical basis;*
2. *Hong's algebra realizes the parity-refined super presentation and its universal R -matrix;*
3. *their root lattices, Cartan forms, height filtrations, and Weyl denominators agree after applying the parity supertrace.*

Proof. The first statement is the finite-window Hall theorem. The second is Hong's quantized Borcherds-superalgebra theorem. Both presentations use A_H and the same simple-root lattice. The parity map changes ordinary dimensions into superdimensions in the PBW character. The denominator theorem identifies this signed PBW product with the Gritsenko–Nikulin product. $\square$

The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its completed three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Equivalently, it is the Ginzburg dg algebra of the doubled quiver with a degree -1 reverse arrow for every arrow and a degree -2 loop at every vertex, with differential determined by the canonical potential.

Theorem 148.1 (Borcherds–Ginzburg tower). *For every H :*

1. $\Gamma_{\Delta,H}$ is a complete three-Calabi–Yau dg algebra;
2. the degree-zero inclusion $kQ_{\Delta,H} \rightarrow \Gamma_{\Delta,H}$ is functorial in full-subquiver inclusions;
3. the Hall composition algebra of $kQ_{\Delta,H}$ is the underlying ordinary positive quantum generalized Kac–Moody algebra;
4. the inclusions $H \leq H'$ form a compatible tower of dg categories and Hall algebras.

Proof. The Calabi–Yau assertion is Keller’s theorem on deformed Calabi–Yau completions, in the Ginzburg model [Kel11, Gin06]. Full-subquiver inclusions extend to reverse arrows, loops, and differentials. The Hall statement is the finite-window Hall theorem. $\square$

Define the proper-height objects

$$\Gamma_{\Delta} = \varinjlim_H \Gamma_{\Delta,H}, \quad \widehat{U}_q(\mathfrak{g}_{\Delta_5}) = \varprojlim_H U_q(\mathfrak{g}_{\Delta_5}^{(H)}).$$

The first is locally three-Calabi–Yau: every finite collection of objects lies in a finite three-Calabi–Yau window. The second is Hong’s root-height-complete quantum symmetry.

Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the canonical-basis construction. Their equivariant hypercohomology groups give finite Hall state spaces in every dimension vector.

Definition 149.1 (Igusa Hall category). For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_H \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{1p,H} = \bigoplus_{\alpha > 0} \mathfrak{g}_{\Delta_5 \alpha}^{(H)},$$

with the parity of the Borcherds superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$.

Theorem 149.2 (Hall and protected-state comparison). *The Hall Grothendieck algebra is the finite-window ordinary positive quantum generalized Kac–Moody algebra. The protected one-particle superspace has signed root multiplicities*

$$\mathrm{sdim} \mathcal{H}_{1p,H,\alpha} = \mathrm{sdim} \mathfrak{g}_{\Delta_5 \alpha}^{(H)}.$$

Its height-complete PBW character is

$$\text{sch Fock}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha > 0} (1 - e^{-\alpha})^{-\text{sdim } \mathfrak{g}_{\Delta_5} \alpha},$$

and its Weyl-shifted inverse is $D_5(2\mathbb{Z})$. The Hall canonical basis and the protected superbasis share the simple-root indexing and the Cartan grading; the parity refinement belongs to the Borchers superdatum.

Proof. The Hall assertion is the finite-window theorem. The protected assertion is PBW for the Borchers Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. $\square$

Coordinate-framed orientation on finite critical charts

Fix H and a dimension vector $\mathbf{d}$. The representation space of the graded doubled quiver underlying $\Gamma_{\Delta,H}$ is affine. Order its vertex, arrow, reverse-arrow, and degree-2 loop coordinates. The corresponding volume line is trivial. The derived moment-map equations form a symmetric three-Calabi–Yau deformation complex whose determinant is the square of this ordered coordinate line.

Theorem 150.1 (Multiplicative coordinate-framed orientation). *A choice of order for the vertex, arrow, reverse-arrow, and loop coordinates trivializes the ambient canonical line and supplies an orientation of every finite-dimensional critical representation chart of $\Gamma_{\Delta,H}$. Under direct sum and the Hall extension correspondence, the orientation line is the tensor product of the two diagonal orientation lines. The construction is associative for three summands and compatible with full-subquiver inclusions.*

Proof. On the extension chart, every arrow matrix is block upper triangular. The tangent complex splits into two diagonal deformation complexes and two off-diagonal complexes exchanged by three-Calabi–Yau duality. The paired off-diagonal part carries the standard determinant square root induced by the three-Calabi–Yau duality pairing. The ordered diagonal coordinate lines therefore multiply. A three-block decomposition gives the same tensor order in both parenthesizations. Full-subquiver inclusion appends an ordered set of coordinates. $\square$

Corollary 150.2 (Oriented critical chart tower). *The finite critical charts carry compatible coordinate-framed vanishing-cycle complexes. Their Grothendieck groups refine the quiver Hall tower by the three-Calabi–Yau critical coefficients.*

Proof. The chosen coordinate framings are compatible with direct sum, extension correspondences, and full-subquiver inclusions in the fixed root order. Vanishing cycles are functorial for these oriented critical charts, so their convolution complexes and Grothendieck groups form the stated tower. $\square$

The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \bigsqcup_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta_5,+} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta_5,+}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta_5,+}[m], \mathfrak{n}_{\Delta_5,+}[n]] \subset \mathfrak{n}_{\Delta_5,+}[m+n].$$

Construction 151.1 (Hybrid factorization algebra). Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}_{\Delta,E}^{\mathrm{hyb}} = C_*^{\mathrm{ch}}(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+})).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree- m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 151.2 (Local–wrapped factorization). $\mathcal{F}_{\Delta,E}^{\mathrm{hyb}}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta_5,+}[0]$. Every degree- m color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m+n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+})$, and constant modes recover $\mathfrak{n}_{\Delta_5,+}$.

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. $\square$

This colored Ran space is the exact carrier of the positive elliptic-degree direction. The wrapped label survives when the point support is empty and interacts with local insertions through the same Hall bracket.

Physical charge and Gram degree

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c, d) = \Pi(c + d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\tilde{\Gamma}_{\Pi} = \Gamma_{\text{phys}} \times \Gamma_{\text{ind}}$$

with multiplication

$$(c, \lambda)(d, \mu) = (c + d, \lambda + \mu + B(c, d)).$$

Theorem 152.1 (Graph homomorphism). *The map*

$$j_{\Pi} : \Gamma_{\text{phys}} \longrightarrow \tilde{\Gamma}_{\Pi}, \quad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys} -graded algebra A , the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\tilde{\Gamma}_{\Pi}$ -grading with cocycle $u^{-B(c,d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c)j_{\Pi}(d) = (c + d, \Pi(c) + \Pi(d) + B(c, d)) = j_{\Pi}(c + d).$$

The algebra identity follows immediately. $\square$

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K3 \times E$ theory enters through a charge functor to $\tilde{\Gamma}_{\Pi}$.

The universal protected field theory

For each H take cotangent BF theory with gauge Lie superalgebra $\mathfrak{n}_{\Delta_5,+}^{(H)}$ on $\mathbb{R}_{\text{top}}^2 \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \widehat{\otimes} \Omega^{0,\bullet}(E) \widehat{\otimes} \left(\mathfrak{n}_{\Delta_5,+}^{(H)}[1] \oplus (\mathfrak{n}_{\Delta_5,+}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (d + \bar{\partial})A + \frac{1}{2}[A, A] \rangle.$$

Theorem 153.1 (Classical local theory and exact quantum harmonic sector). *For every finite height H the shifted-cotangent action defines a classical mixed holomorphic–topological factorization algebra on $\mathbb{R}_{\text{top}}^2 \times E$. Harmonic reduction on a compact product cell gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta_5,+}^{(H)}$, and this reduced theory satisfies the quantum master equation. Complementary Lagrangian boundary polarizations give the boundary enveloping algebra $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$, the two-sided bar complex $C_*(\mathfrak{n}_{\Delta_5,+}^{(H)})$, and the current envelope on E . The proper-height inverse system is coefficientwise finite and defines a complete classical local theory together with an exact quantum harmonic sector.*

Proof. The invariant cotangent pairing and the Jacobi identity give the classical master equation and factorization descent. Harmonic reduction leaves a finite positive nilpotent Lie superalgebra. Every even adjoint endomorphism raises proper height and is strictly upper triangular; every odd adjoint endomorphism has vanishing supertrace. The reduced BV divergence therefore vanishes. Canonical quantization in the selected polarization gives the enveloping algebra, and the augmentation boundary states give the two-sided bar, identified with Chevalley chains by PBW. Each map preserves height, so the inverse system is defined coefficientwise. □

Absolute convergence of the virtual root operator

For each positive root α , let V_α be the minimal parity realization of the signed multiplicity: it is purely even of dimension $\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}$ when this integer is nonnegative and purely odd of dimension $-\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}$ when it is negative. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus_{\alpha>0} V_\alpha}.$$

This virtual root superspace is the canonical linear carrier of the denominator character; the Lie bracket remains carried by $\mathfrak{g}_{\Delta_5}$.

Let $h(\alpha)$ be the proper height. Jacobi-form coefficient estimates give constants $C_0, C_1 > 0$ such that

$$\dim V_\alpha = |\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}| \leq C_0 \exp(C_1 h(\alpha)) \quad (154.1)$$

for every positive root. The number of Gram triples of height H is polynomial in H .

Theorem 154.1 (Nonempty trace-class chamber). *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha>0} \dim V_\alpha \left| e^{-\alpha(tZ_0)} \right| < \infty.$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re \alpha(Z_0) \geq c(Z_0)h(\alpha)$. Combine this with (154.1). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp((C_1 - tc(Z_0))H).$$

The series converges when $tc(Z_0) > C_1$. □

Corollary 154.2 (Trace-class root Hamiltonian). *On the conical tail, the diagonal operator*

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)} \text{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borchers product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. $\square$

The root Dirac operator

Let

$$\mathcal{H}_\Delta = \widehat{\bigoplus_{\alpha>0} \mathfrak{g}_{\Delta_5 \alpha}}$$

with Hilbert norm chosen root space by root space. In the absolute-convergence chamber define

$$K_Z|_{V_\alpha} = e^{-\alpha(Z)} \text{id}, \quad D_Z = 1 - K_Z.$$

Theorem 155.1 (Oriented Fredholm square root). *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\text{sdet}_F(D_Z) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{\text{sdim } \mathfrak{g}_{\Delta_5 \alpha}}.$$

Choose an order of the positive roots, first by height and then lexicographically. This choice orients every finite-height root space. The skew doubling

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^\dagger & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\text{sPf}_F(\mathbb{A}_Z) = \text{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

Proof. Absolute convergence of the Borchers product is the trace-norm summability condition. The determinant formula follows from the spectrum. At finite height the Pfaffian of the oriented skew doubling is the determinant. Trace-norm convergence passes the identity to the limit. The denominator theorem supplies the Weyl shift. $\square$

155.1 Automorphic orientation

The product defines a section of the automorphic line $\mathcal{L}^5 \otimes \nu_5$. On the absolute-convergence chamber, the oriented Fredholm Pfaffian is a trivialization of this line. Analytic continuation uses the automorphy factor of D_5 and therefore glues the chamberwise Fredholm models with transition character ν_5 .

Corollary 155.2 (Orientation character). *The automorphic orientation line of the root Dirac family is the sign local system ν_5 . On every real simple reflection it has value -1 .*

Proof. The Fredholm product equals D_5 on the convergence chamber, and D_5 transforms with automorphy factor ν_5 . This transformation law is the gluing rule for the automorphic orientation line. A real simple root occurs with multiplicity one, so reflection reverses its oriented normal line. $\square$

The universal Igusa protected effective theory

Definition 156.1 (Universal Igusa completion). The protected completion selected by the K3 half-index is the quintuple

$$\mathfrak{R}_\Delta = \left(\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{1p}, \mathcal{F}_{\Delta,E}^{\text{hyb}}, \mathbb{A}_Z \right).$$

Its components are the locally three-Calabi–Yau completion, the ordinary quiver Hall category, the Borchers one-particle Lie superalgebra, the local–wrapped current factorization algebra, and the virtual root Dirac family.

Theorem 156.2 (Igusa protected completion). *The quintuple $\mathfrak{R}_\Delta$ satisfies:*

1. *its one-particle Lie superalgebra is $\mathfrak{g}_{\Delta_5}$;*
2. *its ordinary Hall presentation and Hong’s parity-refined quantum supergroup share the Borchers–Cartan grading and signed PBW denominator;*
3. *its chiral primitive algebra on E is the local–wrapped current algebra of $\mathfrak{g}_{\Delta_5}$;*
4. *its raw multiparticle supercharacter is P^{-1} , where $e^{-\rho}P = \text{Den}_\Delta$; equivalently $e^\rho \text{sch Fock}(\mathfrak{n}_{\Delta_5,+}) = \text{Den}_\Delta^{-1}$;*
5. *the half-coordinate pullback of the doubled Weyl-normalized character is $-\chi_{10}^{-1}$ in the Oberdieck–Pixton normalization;*
6. *its oriented first-order determinant is Den_Δ and its orientation monodromy is ν_5 .*

Proof. The first three statements are the Borchers root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = D_5^2$ and the established reduced scalar theorem give the fifth. The Fredholm theorem gives the sixth. $\square$

This theorem constructs the protected effective theory selected by the Igusa denominator. A geometric type-IIA or topological-string model determines a realization functor into this universal completion by its primitive states and extension amplitudes. Equivalence of protected observables is equivalence of the oriented Hall and factorization objects carried by that functor.

Universal property

A presentation-preserving protected realization consists of an oriented root-graded theory $\mathfrak{T}$, a map from each Igusa simple-root space to its primitive states, and chosen comparison maps from the Hall, Ginzburg, current, quantum-double, and determinant functors applied to that simple-root map.

Theorem 157.1 (Universal presentation map). *Every presentation-preserving protected realization carries a unique Lie-superalgebra morphism*

$$\mathfrak{n}_{\Delta_5,+} \longrightarrow \text{Prim}(\mathfrak{T})$$

extending the simple-root map. Functoriality produces compatible morphisms from the Hall algebra, the Ginzburg completion, the local–wrapped current theory, the completed quantum group, and the virtual root determinant of $\mathfrak{R}_\Delta$ to their selected targets in $\mathfrak{T}$. These morphisms are equivalences when the primitive map and the selected functorial comparison maps are equivalences.

Proof. The Borchers generators and relations give the unique primitive Lie morphism. Hall convolution, Calabi–Yau completion, chiral Chevalley chains, quantized doubling, and the diagonal Fredholm construction are functorial in the root presentation. PBW promotes a primitive equivalence to the completed enveloping and Fock objects. The remaining equivalences are exactly the selected comparison maps in the definition of a presentation-preserving realization. □

The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two modular forms whose divisor is the diagonal with order one [CG11]. Gritsenko–Nikulin and the CHL denominator constructions give the sourced reflective presentations used for the corresponding rows [GN97, GN98a, GN98b, GGK09, Gov11, GS19]. Every row has a minimal virtual product superspace obtained from its Fourier exponents, together with a Fock determinant and a diagonal Fredholm operator. A sourced denominator presentation refines this product model by its Borchers–Cartan bracket, Weyl group, and quantum supergroup. The half-integral rows are formed on their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 158.1 (Eight-row algebraic realization). *Each diagonal-divisor row has a proper-height virtual root superspace, a super-Fock determinant, a colored abelian current theory, and a trace-class Fredholm denominator operator whose character is the corresponding automorphic product. For every row equipped with its reflective denominator presentation, the finite-window Borchers quiver, ordinary Hall category, three-Calabi–Yau completion, and parity-refined quantum supergroup refine this product model and recover the same denominator character.*

Proof. Local finiteness of the product exponents gives the graded superspace and its proper-height completion. The Fock and Fredholm constructions depend only on these signed

multiplicities and reproduce the product coefficientwise. A reflective denominator presentation supplies a symmetrizable Borchers superdatum; the quiver, Hall, Ginzburg, current, and Hong constructions then apply to that datum and preserve its PBW denominator. $\square$

Algebraic and physical closure

Theorem 159.1 (Universal Igusa realization theorem). *The following objects are canonically connected by constructed morphisms:*

$$\begin{array}{ccccc}
 \phi_{0,1} \xrightarrow{\text{Borch}} D_5, & \phi_{0,1} \xrightarrow{\text{root datum}} \mathfrak{g}_{\Delta_5}, & A_{\Delta} \xrightarrow{\text{quiver}} Q_{\Delta,H} \xrightarrow{\text{Hall}} \mathcal{B}_{\Delta}, \\
 \mathcal{B}_{\Delta} \xrightarrow{\Pi_3} \Gamma_{\Delta}, & \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Cur}_E} \mathcal{F}_{\Delta,E}^{\text{hyb}}, & \mathfrak{g}_{\Delta_5} \xrightarrow{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}), \\
 \mathfrak{g}_{\Delta_5} \xrightarrow{\text{root spectrum}} \mathbb{A}_Z, & e^{-\rho} \text{sPf}_F(\mathbb{A}_Z) = \text{Den}_{\Delta}(Z) = D_5(2Z), \\
 \mathcal{Z}_{K3 \times E}^{\text{red}} = -(J^* \text{Den}_{\Delta})^{-2} = -\chi_{10}^{-1}.
 \end{array}$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borchers root bracket, a local-wrapped collision law, a quantum supergroup, an automorphic orientation line, a first-order Fredholm determinant, and the reduced scalar square.

Proof. Each arrow is one of the preceding constructions. Compatibility follows from their common root grading and PBW filtration. The final scalar identity is the primitive reduced curve-counting theorem. $\square$

BOOK **XXXII**

**Technical appendices to the
preceding book**

Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\begin{aligned}\vartheta_2(\tau, z) &= \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1}, \\ \vartheta_3(\tau, z) &= \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n}, \\ \vartheta_4(\tau, z) &= \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.\end{aligned}$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r) .

The root reflection matrices

In the basis (f_2, f_3, f_{-2}) , the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \quad S_i^\top G S_i = G.$$

Concordance with the earlier Igusa draft

The reconstitution retains the half-index determinant, theta normalization, Borcherds denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum. It reorganizes their dependencies as follows:

1. the automorphic square root is completed entirely within the Borcherds product theorem;
2. the BKM bracket comes from the Gritsenko–Nikulin algebra;
3. the bar–Chevalley comparison gives the denominator a homological source;
4. the current and BF constructions give an exact first-order algebraic model;
5. orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;
6. Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;
7. the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

Book entrance: Universal Chiral BV and Einstein Completion

*The complete mixed boundary datum, its centre, deformation algebra, BV
envelope, gravitational completion, and anomaly geometry.*

BOOK **XXXIII**

The full mixed boundary datum

Three products and one interface

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-Mod}(\text{Ran } X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\text{ch}}$ for chiral convolution.

Definition 163.1 (Complete mixed boundary datum). A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, A^{\text{ch}}, A^\perp, K, \iota, \lambda)$$

with the following components.

1. L is a complete chiral Lie algebra.
2. A^{ch} is a complete associative algebra for $\star^{\text{ch}}$.
3. $A^\perp$ is a complete E_1 algebra in $(\text{Fact}_D(X), \otimes^!)$.
4. K is an $(A^{\text{ch}}, A^\perp)$ interface bimodule.
5. $\iota : L \rightarrow (A^{\text{ch}})_{\text{Lie}}$ is a chiral Lie morphism.
6. λ is the aligned mixed distributive law relating the two module actions on K .

Every structure map is continuous for the chosen weight filtrations.

The three native bars are

$$\text{Bar}_{\text{Lie}}^{\text{ch}}(L), \quad \text{Bar}_{\text{Ass}}^{\text{ch}}(A^{\text{ch}}), \quad \text{Bar}^\perp(A^\perp).$$

The interface has a fourth bar. Its degree- r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 163.2 (Interface bar). Let $\overline{A}^{\text{ch}}$ and $\overline{A}^\perp$ be the augmentation ideals. Set

$$\text{Bar}^{\text{mix}}(\mathbb{A}; K) = \bigoplus_{p, q \geq 0} (s\overline{A}^{\text{ch}})^{\otimes p} \otimes sK \otimes (s\overline{A}^\perp)^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K , the right action on K , and the mixed terms supplied by λ .

Theorem 163.3 (Four-colour differential). *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. $\square$

Compatible modules and the mixed centre

A compatible module is a triple $(M^{\text{ch}}, M^{\perp}, \kappa)$ with an A^{ch} action, an $A^{\perp}$ action, and an interface comparison over K . Let $\text{Mod}_{\mathbb{A}}^{\text{mix}}$ be their stable category.

Definition 164.1 (Mixed derived centre). The mixed derived centre is

$$Z_{\text{mix}}(\mathbb{A}) = \text{End}\left(\text{id}_{\text{Mod}_{\mathbb{A}}^{\text{mix}}}\right).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 164.2 (Universal interior action). $Z_{\text{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every coherent interior action on the mixed boundary category factors uniquely through this centre. Morita-equivalent complete mixed data have equivalent centres.

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. An interior operator acts naturally on every compatible module and therefore defines an endomorphism of the identity. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. $\square$

Relative induction from a chiral algebra

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\text{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V , modulo the chiral relations already present in V . Weight completion gives $\widehat{\text{Coll}}(V)$.

Theorem 165.1 (Universal collar). *For every complete transversely ordered chiral algebra B and every chiral morphism $V \rightarrow B$, there is a unique continuous E_1 morphism*

$$\widehat{\text{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. $\square$

The full envelope of V is obtained by taking $L = V_{\text{Lie}}$, $A^{\text{ch}} = U^{\text{ch}}(L)$, $A^{\perp} = \widehat{\text{Coll}}(V)$, the regular interface, and the canonical aligned distributive law.

BOOK **XXXIV**

**The deformation algebra and its
BV envelope**

The centre-extended deformation algebra

Fix a cofibrant coloured operad $\mathcal{P}_{\text{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\text{mix}}^!$ be its bar cooperad. The convolution complex

$$\text{Def}_{\mathbb{A}}^{\text{full}} = \text{Hom}_{\mathbb{S}} \left(\mathcal{P}_{\text{mix}}^!, \text{End}_{\mathbb{A}} \right)$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 166.1 (Complete deformation L_{∞} algebra). *The convolution complex has a canonical complete L_{∞} structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_{∞} identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\text{mix}}$ to $\text{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. $\square$

166.1 Field equations and Bianchi identities

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_\alpha x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 166.2 (Universal Bianchi identity). *For every α ,*

$$d_\alpha \mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_\alpha^2 = 0$.

Proof. Both identities are the exponential form of the L_∞ Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^α and $e^\alpha x$. $\square$

Shifted cotangent transgression

Let $\mathcal{I}$ be a complete elliptic L_∞ algebra on a three-manifold M . Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathcal{I}[1] = \mathcal{I}[1] \oplus \mathcal{I}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 167.1 (Universal BV envelope). *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \quad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. $\square$

Definition 167.2 (Universal chiral BV envelope). For a complete mixed datum $\mathbb{A}$, let $\mathcal{I}_\mathbb{A}^{\text{full}}$ be the analytic localization of $\text{Def}_\mathbb{A}^{\text{full}}$ on $X^{\text{an}} \times \mathbb{R}$. Define

$$\mathbf{BVEnv}_3^{\text{full}}(\mathbb{A}) = T^*[-1] \text{MC}(\mathcal{I}_\mathbb{A}^{\text{full}}).$$

Intrinsic algebraic quantization

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ : the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$\mathcal{O}(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The homological orientation defines the divergence operator

$$\Delta_{\mu} : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 168.1 (Oriented BV complex). *The operator Δ_{μ} has order two and satisfies*

$$\Delta_{\mu}^2 = 0, \quad [d_{\mathfrak{X}}, \Delta_{\mu}] = 0,$$

$$\Delta_{\mu}(fg) = \Delta_{\mu}(f)g + (-1)^{|f|} f \Delta_{\mu}(g) + (-1)^{|f|} \{f, g\}.$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[[\hbar]], d_{\mathfrak{X}} + \hbar \Delta_{\mu} \right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_{μ} is the sum of paired second derivatives. Their graded commutation gives $\Delta_{\mu}^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_{\mu}] = 0$. Expanding the operator on a product gives the bracket as the cross term. $\square$

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_{\mu} \Delta_{\mu} = d_{\mathrm{dR}} \iota_{\mu}.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 168.2 (Intrinsic quantum complex of a mixed datum). For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\mathrm{MC}(\mathbf{I}_{\mathbb{A}}^{\mathrm{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 168.3 (Quantum master equation). *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar}\mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar\Delta_{\mu}S = 0.$$

The equation is equivalent to

$$(d + \hbar\Delta_{\mu})e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. $\square$

Perturbative factorization realization

Choose elliptic gauge-fixing data for $\mathbb{I}_{\mathbb{A}}^{\text{full}}$. The heat kernel gives a propagator on $M \times M \setminus \Delta$. Fulton–MacPherson compactification resolves simultaneous collisions of vertices.

Theorem 169.1 (Graphwise factorization). *Renormalized graph weights that extend to the compactified configuration spaces and satisfy the quantum master equation define a factorization algebra of quantum observables. The boundary restriction of this factorization algebra realizes the original mixed datum $\mathbb{A}$.*

Proof. Disjoint unions of configurations multiply graph weights. Stokes’ theorem on each compactified configuration space expresses the differential of a weight as the sum over boundary contractions. The quantum master equation cancels the codimension-one contractions. The resulting cochains satisfy Weiss descent. Restriction to a collar recovers the generating operations used to define the deformation algebra. $\square$

BOOK XXXV

Boundary duality, exchange, and reduction

The Koszul–Morita square

Let P be a dualizable boundary state for $A^\perp$. Its boundary dual is

$$A_P^\dagger = \mathrm{RHom}_{A^\perp}(P, P).$$

The two-sided bar sends the diagonal bimodule to a diagonal bicomodule.

Theorem 170.1 (Mixed Koszul–Morita transport). *On the positive complete locus, the universal twisting cochains identify complete mixed modules with compatible comodules over the four bar objects. Derived tensor products of interfaces become derived cotensor products. The mixed centre and mixed Hochschild chains are preserved.*

Proof. Apply one-colour bar–cobar duality to each associative colour. The interface bar is a bimodule bar and its filtration is finite in each total weight. The aligned distributive law identifies the two orders of transformation. Eilenberg–Moore comparison sends tensor to cotensor. Endomorphisms of the diagonal and its trace are invariant under the induced Morita equivalence. $\square$

Planar exchange and Tannaka reconstruction

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. A rigid meromorphic line category $\mathcal{L}$ with a faithful fibre functor ω has coend

$$O(\mathbb{G}) = \int^{M \in \mathcal{L}} \omega(M)^* \otimes \omega(M).$$

Theorem 171.1 (Quantum symmetry from exchange). *The coend is a coquasitriangular bialgebra. Its continuous dual acts on every line object. The meromorphic braiding gives the universal R-matrix on completed matrix coefficients. A spectral rational braiding yields an RTT presentation.*

Proof. Tensor product in $\mathcal{L}$ gives the coproduct on matrix coefficients. Evaluation gives the counit. Rigidity gives the antipode on the Hopf envelope. The braiding defines the coquasitriangular pairing, and the hexagon identities give the Yang–Baxter equation. Writing this naturality on a generating fibre produces the RTT relation. $\square$

BRST reduction

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 172.1 (BRST–bar bicomplex). *The total differential*

$$D = Q + b_{\text{ch}} + b_{\perp} + b_{\text{mix}}$$

satisfies $D^2 = 0$. A bounded-below complete filtration gives a convergent spectral sequence

$$E_1 = H_Q(\text{Bar}^{\text{mix}}(\mathbb{A}; K)) \implies H_D(\text{Bar}^{\text{mix}}(\mathbb{A}; K)).$$

When Q -cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_1 differential with the bar differential of the reduced operations. $\square$

BOOK **XXXVI**

Conformal covariance and Einstein completion

Projective connections and the Virasoro tangent complex

Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c . At a projective connection p , its tangent complex is locally

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{0,\bullet}(X, K_X^{\otimes 2}),$$

where

$$\mathcal{D}_{p,c}(\epsilon) = \epsilon \partial p + 2(\partial \epsilon)p + \frac{c}{12} \partial^3 \epsilon.$$

Theorem 173.1 (Projective covariance). *A conformal map $\text{Vir}_c \rightarrow A$ defines a formal family of mixed structures over $\mathcal{P}_c(X)$. Its tangent algebra is the semidirect product of the Virasoro projective complex with the deformation algebra relative to a fixed projective connection.*

Proof. The conformal vector acts on fields by infinitesimal coordinate change. The central term is the affine cocycle in the transformation law of a projective connection. The action therefore gives a map from the projective tangent algebra to derivations of the mixed datum. The tangent algebra of the associated family is the semidirect product. $\square$

Three-dimensional gravity as a flat Cartan theory

Let M be an oriented three-manifold, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have basis J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^c J_c, \quad [J_a, P_b] = \epsilon_{ab}{}^c P_c, \quad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^c J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 174.1 (Cartan–Einstein equivalence). *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \quad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \quad g_{\mu\nu} = \eta_{ab} e_\mu^a e_\nu^b.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form with the off-diagonal invariant pairing gives $e \wedge R + \ell^{-2} e^3$ plus an exact form [AT86, Wit88]. $\square$

Four equivalent linear complexes

Fix a constant-curvature solution (e, ω) . There are four deformation complexes.

1. The flat adjoint complex $\Omega^\bullet(M, \mathfrak{g}_\Lambda)$ with differential $d_{\mathcal{A}}$.
2. The Einstein–Cartan complex on $(\delta e, \delta \omega)$ with linearized torsion and curvature.
3. The metric Einstein complex on $h_{\mu\nu}$ with linearized Einstein operator.
4. The Calabi complex resolving local Killing fields by the Killing, linearized curvature, and Bianchi operators.

Theorem 175.1 (Chain equivalence of gravitational presentations). *An invertible torsion-free dreibein gives chain maps among the four complexes. These maps are quasi-isomorphisms on every geodesically convex open set. Their cohomology describes infinitesimal isometries, physical linearized fields, and Bianchi identities in the four languages.*

Proof. The dreibein identifies internal translations with tangent vectors and internal Lorentz rotations with skew endomorphisms. Splitting $d_{\mathcal{A}}$ in this basis gives the linearized torsion and curvature operators. Eliminating $\delta \omega$ by the linearized torsion equation gives the metric Einstein operator. The symbol sequence is the Calabi symbol sequence and is exact on constant-curvature space. The Poincaré lemma for the flat adjoint connection and exactness of the symbol homotopies give local quasi-isomorphisms. $\square$

Brown–Henneaux and the projective boundary

Write the AdS connection as

$$A^\pm = \omega \pm \ell^{-1} e.$$

The Einstein action is the difference of two $\mathfrak{sl}_2$ Chern–Simons actions with level

$$k = \frac{\ell}{4G}.$$

Theorem 176.1 (Brown–Henneaux central charge). *The classical asymptotic Poisson algebra of each chiral Chern–Simons sector has Virasoro central charge*

$$c = 6k = \frac{3\ell}{2G}.$$

The projective connection carried by the boundary stress tensor is the projective variable of the preceding conformal family.

Proof. Hamiltonian reduction of the boundary $\mathfrak{sl}_2$ current Poisson algebra gives the classical Virasoro bracket. Its central term is $6k$. Substitution of the gravitational level gives the second equality. The coadjoint transformation of the boundary stress tensor is the projective-connection transformation law [BH86]. $\square$

Einstein completion of a conformal pair

Let A_L and A_R be conformal mixed boundary data with equal Brown–Henneaux central charge. Their projective families map to the same $\mathcal{P}_c(X)$. Define their relative field algebra by the homotopy fibre product over the projective tangent algebra.

Definition 177.1 (Einstein BV completion). The Einstein completion is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1] \mathbf{MC}(\mathfrak{g}_{\text{grav}} \ltimes_{\text{Ham}} \mathfrak{m}_{A_L, A_R}).$$

Here $\mathfrak{g}_{\text{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L, A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 177.2 (Einstein completion theorem). *The completion has a canonical BV action. Its pure gravitational sector is three-dimensional Einstein gravity. Its matter sector is the relative deformation theory of the conformal pair. The coupling is the moment map for the projective and diffeomorphism actions. The Bianchi identity of the full system is the L_∞ identity of the semidirect product.*

Proof. The homotopy fibre product matches the projective gauge fields and retains the relative matter directions. The gravitational algebra acts on those directions by conformal transport. The semidirect product is therefore an L_∞ algebra. Shifted cotangent transgression gives the BV action and its master equation. Restriction to the gravitational summand gives the Cartan action. The mixed term is the Hamiltonian pairing of the gauge field with the matter moment map. □

Higher-spin completion

Replace $\mathfrak{sl}_2$ by $\mathfrak{sl}_N$ and choose its principal $\mathfrak{sl}_2$ embedding. The adjoint representation decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^N V_{2s-2}, \quad \sum_{s=2}^N (2s-1) = N^2 - 1.$$

Theorem 178.1 (Principal higher-spin spectrum). *The principal embedding gives one massless gauge field of each integer spin $2, 3, \dots, N$. Classical Drinfeld–Sokolov reduction of the boundary affine Poisson algebra gives the principal W_N algebra. A pair of principal W_N boundary data therefore has an $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ Einstein–Cartan completion with the same spin spectrum.*

Proof. The irreducible $\mathfrak{sl}_2$ module V_{2s-2} corresponds to a frame-like field of spin s . The dimension identity exhausts the adjoint. Principal Drinfeld–Sokolov reduction produces one generating current in each conformal weight $2, \dots, N$. The boundary currents and bulk gauge fields therefore match representation by representation [CFPT10, HR10]. $\square$

BOOK **XXXVII**

**Exact gravitational complexes and
quantum tests**

The four complexes in coordinates

179.1 Flat adjoint complex

At a flat Cartan connection $\mathcal{A}$, the first complex is

$$\Omega^0(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M, \mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M, \mathfrak{g}_\Lambda).$$

Degree zero is the gauge parameter, degree one is the fluctuation, degree two is the linearized curvature, and degree three is the Bianchi identity.

179.2 Einstein–Cartan complex

Write a gauge parameter as (ξ^a, λ^a) and a fluctuation as $(u^a, v^a) = (\delta e^a, \delta \omega^a)$. The first differential is

$$\begin{aligned} u^a &= D_\omega \xi^a + \epsilon^a_{bc} e^b \lambda^c, \\ v^a &= D_\omega \lambda^a + \ell^{-2} \epsilon^a_{bc} e^b \xi^c. \end{aligned}$$

The second differential is

$$\begin{aligned} \delta T^a &= D_\omega u^a + \epsilon^a_{bc} v^b \wedge e^c, \\ \delta F^a &= D_\omega v^a + \ell^{-2} \epsilon^a_{bc} e^b \wedge u^c. \end{aligned}$$

The third differential is the linearized pair of Cartan Bianchi identities.

Proposition 179.1 (Chain identity). *The composite of the first two Einstein–Cartan differentials vanishes on a constant-curvature torsion-free background.*

Proof. Apply D_ω^2 to ξ and λ . Replace the background curvature by $-\ell^{-2}e \wedge e$ and use $D_\omega e = 0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining expressions are Jacobi identities for $\epsilon_{ab}{}^c$. $\square$

179.3 Metric complex

The dreibein identifies the translation parameter with a vector field. Modulo Lorentz rotations,

$$h_{\mu\nu} = 2\eta_{ab}e_{(\mu}^a u_{\nu)}^b.$$

The metric differential begins with the Killing operator

$$K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

The linearized cosmological Einstein operator is

$$(E_\Lambda h)_{\mu\nu} = \frac{1}{2} \left(-\nabla^2 h_{\mu\nu} - \nabla_\mu \nabla_\nu h + \nabla_\mu \nabla^\rho h_{\rho\nu} + \nabla_\nu \nabla^\rho h_{\rho\mu} \right. \\ \left. - g_{\mu\nu} (\nabla^\rho \nabla^\sigma h_{\rho\sigma} - \nabla^2 h) \right) - \Lambda(h_{\mu\nu} - g_{\mu\nu} h).$$

It satisfies $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$.

Theorem 179.2 (Explicit metric reduction). *Solving the linearized torsion equation for $v = \delta\omega$ and substituting into the linearized curvature equation gives $E_\Lambda h = 0$. The kernel of the map from (u, v) to h is the infinitesimal Lorentz action.*

Proof. The torsion equation is algebraic in the antisymmetric part of the covariant derivative of u and determines v . The symmetric part of u is $h/2$. Substitution into $\delta F = 0$ yields the standard variation of the Ricci tensor and the displayed Einstein operator. An antisymmetric $u_{\mu\nu}$ is exactly a Lorentz rotation of the dreibein. $\square$

179.4 Calabi complex

On a constant-curvature manifold, the Calabi complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2 T^* M) \xrightarrow{R'} \Gamma(\mathcal{R}) \xrightarrow{B'} \Gamma(\mathcal{B}),$$

where R' is the linearized Riemann tensor and B' is the linearized algebraic and differential Bianchi operator. In three dimensions the Riemann tensor is equivalent to the Einstein tensor, so the metric and Calabi complexes are locally quasi-isomorphic.

Boundary BFV theory and the Virasoro moment map

The variation of Chern–Simons theory on a manifold with boundary gives the boundary symplectic form

$$\Omega_{\partial} = \frac{k}{4\pi} \int_{\partial M} \langle \delta A \wedge \delta A \rangle.$$

A chiral polarization selects $A_{\bar{z}}$ as coordinate. The residual moment map is the flatness constraint.

Theorem 180.1 (Boundary current algebra). *Canonical quantization of the boundary symplectic form gives the affine current algebra at level k . Drinfeld–Sokolov reduction of the principal $\mathfrak{sl}_2$ constraint gives the Virasoro moment map*

$$T(z) = \frac{1}{2(k + h^{\vee})} : J^a J_a : (z) + \text{improvement},$$

with the classical Brown–Henneaux limit $c = 6k$.

Proof. The inverse of Ω_{∂} is the current Poisson bracket. Quantization gives the affine central extension. The BRST complex of the principal constraint eliminates the gauge directions and retains the improved quadratic current. Its classical central term is $6k$ in the gravitational normalization. $\square$

Matter stress tensor and the Einstein moment map

Let $\mathfrak{m}$ be the relative field algebra of a conformal pair. Projective covariance gives a Hamiltonian action of vector fields and therefore a moment map

$$\mu_{\text{matt}} : \mathfrak{m} \longrightarrow \Omega^{0,\bullet}(X, K_X^{\otimes 2})[-1].$$

Its classical component is the stress tensor.

Theorem 181.1 (Conservation from the master equation). *The coupled BV action*

$$S_{\text{tot}} = S_{\text{grav}} + S_{\text{matt}} + \langle h, \mu_{\text{matt}} \rangle$$

satisfies the classical master equation precisely when the matter moment map is equivariant. The component of that equation linear in the diffeomorphism ghost is

$$\nabla^\mu T_{\mu\nu} = 0.$$

Proof. The master bracket of the coupling with itself is the defect of equivariance. The bracket with the gravitational action is the infinitesimal diffeomorphism of the moment map. Their cancellation is the Hamiltonian equivariance identity. Pairing the identity with the diffeomorphism ghost and integrating by parts gives stress-tensor conservation. $\square$

One-loop gravity and analytic torsion

Expand Chern–Simons theory around an acyclic flat connection $\mathcal{A}$. Gauge fixing gives the Laplacians of the flat adjoint de Rham complex.

Theorem 182.1 (One-loop determinant). *The absolute value of the one-loop partition function is the square root of the Ray–Singer torsion of the flat adjoint bundle:*

$$|Z_{1\text{-loop}}(\mathcal{A})| = \tau_{\text{RS}}(M, \text{ad } \mathcal{A})^{1/2}.$$

The phase is the eta-invariant contribution fixed by the framing convention.

Proof. Bosonic and ghost Gaussian integrals give alternating determinants of the Laplacians in the flat de Rham complex. Their product is the analytic torsion. The odd signature operator supplies the determinant phase. This is the standard stationary-phase calculation for Chern–Simons theory [RS71, Wit88]. $\square$

The same determinant complex is obtained from the Einstein–Cartan, metric, and Calabi presentations through their chain equivalences. One-loop equality is therefore a consequence of quasi-isomorphism together with compatible determinant-line orientations.

BTZ, Cardy, and the area law

For a unitary modular boundary theory with left and right central charges c_L, c_R , the asymptotic entropy is

$$S_{\text{Cardy}} = 2\pi\sqrt{\frac{c_L}{6}\left(L_0 - \frac{c_L}{24}\right)} + 2\pi\sqrt{\frac{c_R}{6}\left(\bar{L}_0 - \frac{c_R}{24}\right)}.$$

For the BTZ black hole,

$$L_0 - \frac{c}{24} = \frac{(r_+ + r_-)^2}{16G\ell}, \quad \bar{L}_0 - \frac{c}{24} = \frac{(r_+ - r_-)^2}{16G\ell}, \quad c = \frac{3\ell}{2G}.$$

Theorem 183.1 (Cardy–Bekenstein equality). *Substitution gives*

$$S_{\text{Cardy}} = \frac{2\pi r_+}{4G}.$$

This is the Bekenstein–Hawking entropy of the BTZ horizon.

Proof. Each square root becomes $\pi(r_+ \pm r_-)/(4G)$. Their sum is $2\pi r_+/(4G)$. The modular asymptotic is Cardy’s theorem, and the geometric charges are the Chern–Simons holonomies of the BTZ solution [Car86, BnTZ92]. $\square$

Higher-spin holonomy and boundary charges

In $\mathfrak{sl}_N$ Chern–Simons gravity, a smooth Euclidean saddle is selected by the conjugacy class of the thermal holonomy. The invariant polynomials

$$\mathrm{Tr}(\mathrm{Hol}^s), \quad 2 \leq s \leq N,$$

fix the spin- s charges. Drinfeld–Sokolov reduction identifies the same charges with zero modes of the W_N currents.

Theorem 184.1 (Bulk–boundary charge equality). *For a principal $\mathfrak{sl}_N$ connection in Drinfeld–Sokolov gauge, the invariant polynomials of the spatial holonomy equal the classical W_N zero-mode Hamiltonians. Their Poisson brackets are the classical W_N brackets.*

Proof. Drinfeld–Sokolov gauge writes the connection as the principal raising operator plus lowest-weight components. The coefficients of the characteristic polynomial are the invariant polynomials and are polynomial functions of those components. Hamiltonian reduction gives the W_N Poisson structure on the same functions. $\square$

BOOK XXXVIII

**Anomalies, Hall sources, and
automorphic completion**

Four native anomaly classes

The construction carries four curvature classes.

1. A local BV anomaly lies in the degree-one local deformation complex.
2. A projective anomaly is the curvature of the determinant line over parameter space.
3. A framing anomaly is the transgression of a characteristic class to a three-manifold.
4. A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 185.1 (Curvature-killing extension). *Let κ be a closed central degree-two class in the deformation complex of a theory. Adjoining a degree-one generator u with $du = \kappa$ produces a central extension in which the lifted curvature vanishes. Tensoring with the inverse invertible field theory gives the geometric realization of the same cancellation.*

Proof. The extended differential squares to zero because $d\kappa = 0$ and κ is central. The new Maurer–Cartan equation contains the compensating field u . An invertible theory with curvature $-\kappa$ adds the opposite class under tensor product. $\square$

Hall and Calabi–Yau sources

Let $\mathcal{C}$ be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+ . A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Theorem 186.1 (Source-to-BV functor). *A morphism from the Hall boundary datum to a complete mixed boundary datum induces morphisms*

$$H^+ \longrightarrow A^{\text{ch}}, \quad Z_{\text{Hall}} \longrightarrow Z_{\text{mix}}(\mathbb{A}), \quad \mathbf{BVEnv}_{\text{Hall}} \longrightarrow \mathbf{BVEnv}_3^{\text{full}}(\mathbb{A}).$$

Compatibility with coproduct, pairing, and braiding propagates to the quantum line category. Compatibility with cyclic traces propagates to the quantum and modular amplitudes.

Proof. The Hall map preserves associative multiplication. Relative operadic induction extends it to the mixed colours and interface. Natural transformation of module actions gives the centre map. Functoriality of convolution deformation algebras and shifted cotangent gives the BV map. The remaining structures are natural under monoidal and cyclic functors. $\square$

For $X = K3 \times E$, Volume IV constructs the geometric BPS Borel $\mathfrak{b}_X$, the Igusa Borcherds Borel $\mathfrak{b}_\Delta$, and their completed Lie amalgam. Applying the enveloping functor gives the cocommutative Hopf algebra

$$H_{X \boxtimes \Delta} = \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{b}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q -deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

Theorem 186.2 (Amalgamated source envelope). *Relative mixed induction sends $H_{X\boxtimes\Delta}$ to a complete mixed boundary datum $\mathbb{A}_{X\boxtimes\Delta}$. Its universal BV envelope has two canonical retractions*

$$\mathbf{BVE}nv_3^{\text{full}}(\mathbb{A}_{X\boxtimes\Delta}) \longrightarrow \mathbf{BVE}nv_3^{\text{full}}(\mathbb{A}_X), \quad \mathbf{BVE}nv_3^{\text{full}}(\mathbb{A}_{X\boxtimes\Delta}) \longrightarrow \mathbf{BVE}nv_3^{\text{full}}(\mathbb{A}_\Delta).$$

A primitive comparison quotient induces the diagonal BV theory in which geometric BPS primitives and Igusa roots act as one field. The kernel of the quotient is the complete mixed interaction ideal.

Proof. Relative induction preserves coproducts and the common Cartan action, so the amalgamated Hopf algebra produces the amalgamated mixed datum. The Hopf retractions induce maps of mixed data. Functoriality of the centre, convolution deformation algebra, and shifted cotangent transgression gives the two BV retractions. A primitive comparison quotient is a quotient of mixed data, and functoriality carries its defining ideal to the corresponding ideal in the deformation and BV complexes. $\square$

The Igusa protected insertion

Let $\mathfrak{R}_\Delta$ be the Borchers–Ginzburg protected category of Volume III. Its ordinary Hall boundary is the quiver Hall category, its parity-refined line symmetry is the quantized Borchers superalgebra, its local–wrapped current object lies on the elliptic curve, and its first-order determinant is the oriented virtual-root Pfaffian.

Theorem 187.1 (Igusa insertion into the universal envelope). *The protected category $\mathfrak{R}_\Delta$ defines a complete mixed boundary datum on the elliptic curve. Put $\text{Den}_\Delta(Z) = D_5(2Z)$ and $J(Z) = Z/2$. Its universal BV envelope has primitive algebra $\mathfrak{g}_{\Delta_5}$, line symmetry $U_q(\mathfrak{g}_{\Delta_5})$, and Weyl-shifted one-loop determinant Den_Δ . The doubled half-coordinate trace is $-(J^* \text{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$ in the reduced enumerative normalization.*

Proof. The colored Ran current algebra supplies the chiral and transverse colours. The Hall category supplies ordered particle fusion and the interface to the three-Calabi–Yau completion. The quantum double supplies exchange. The root Dirac family supplies the determinant. The identities follow from the Borchers denominator and the primitive reduced $K3 \times E$ theorem. $\square$

The carrier lattice

The complete theory uses the carrier set

$$\mathfrak{C} = \{\text{diag}, \text{ch}, \perp, \text{int}, \text{ex}, \text{cyc}, \text{mod}, \text{BV}, \text{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\text{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S .

Theorem 188.1 (Carrier descent). *When S and T form a relation-excise pair, the square*

$$\begin{array}{ccc} \text{Str}_{S \cup T}(\mathbb{A}) & \longrightarrow & \text{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \text{Str}_T(\mathbb{A}) & \longrightarrow & \text{Str}_{S \cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. $\square$

Definition 188.2 (Relation-complete carrier cover). Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i \in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Čech homotopy colimit

$$\text{hocolim}_{\emptyset \neq J \subseteq I} \mathcal{I}_{\bigcap_{j \in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 188.3 (Power-set coherence). *The assignment*

$$S \longmapsto \mathrm{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i \in I}$ of U , the restriction map is an equivalence

$$\mathrm{Str}_U(\mathbb{A}) \xrightarrow{\sim} \mathrm{holim}_{\emptyset \neq J \subseteq I} \mathrm{Str}_{\cap_{j \in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. $\square$

Definition 188.4 (Exact-support deformation complex). For $S \subseteq \mathfrak{C}$, let $\mathrm{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S , and set

$$\mathrm{Def}_{=S} = \mathrm{cofib} \left(\mathrm{hocolim}_{T \subseteq S} \mathrm{Def}_{\leq T} \longrightarrow \mathrm{Def}_{\leq S} \right).$$

Thus $\mathrm{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S .

Theorem 188.5 (Carrier-support spectral sequence). *Let $\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}$ be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S|=p}} H^{p+q}(\mathrm{Def}_{=S}) \implies H^{p+q}(\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}).$$

A class on the S -summand is a deformation or curvature class whose first complete geometric carrier is S . The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its p th associated-graded quotient is the direct sum of the exact-support complexes $\mathrm{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. $\square$

Sixteen structural laws

The theory is governed by the following laws.

1. A geometric carrier determines the type of its operation.
2. The diagonal stores the complete principal-part filtration.
3. The interval stores ordered fusion.
4. The plane stores exchange.
5. The circle stores trace.
6. The stable curve stores modular sewing.
7. The transverse, associative-chiral, and chiral-Lie bars are distinct.
8. Chiral PBW compares the last two on enveloping objects.
9. The interface bar records the mixed distributive law.
10. The derived centre is the universal interior action.
11. The deformation algebra is the tangent algebra of the centre-extended datum.
12. Maurer–Cartan curvature is the field equation.
13. The L_∞ identity is the Bianchi identity.
14. Shifted cotangent is the universal classical BV envelope.
15. Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.
16. Einstein completion is projective matching followed by gravitational cotangent transgression.

Final synthesis

The construction proceeds by a single chain of functors:

$$\begin{aligned} \mathbb{A} &\xrightarrow{Z_{\text{mix}}} Z_{\text{mix}}(\mathbb{A}) \xrightarrow{\text{Def}} \mathfrak{l}_{\mathbb{A}}^{\text{full}} \xrightarrow{T^*[-1]} \mathbf{BVE}nv_3^{\text{full}}(\mathbb{A}), \\ \mathbf{BVE}nv_3^{\text{full}}(\mathbb{A}) &\xrightarrow{\text{projective matching}} \mathbf{EinBV}_3(A_L, A_R). \end{aligned}$$

Hall and Calabi–Yau sources enter on the left. Quantum groups enter through planar exchange and the line category. Automorphic products enter as characters of root-graded protected objects. Each arrow preserves the structure carried by its source and adds the structure specified by its geometric target.

BOOK **XXXIX**

**Technical appendices to the
preceding book**

The Chern–Simons normalization

With invariant pairing $\langle J_a, P_b \rangle = \eta_{ab}$, the Chern–Simons action is

$$S_{\text{CS}}[\mathcal{A}] = \frac{k}{4\pi} \int_M \langle \mathcal{A} \wedge d\mathcal{A} + \frac{2}{3} \mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A} \rangle.$$

For $A^\pm = \omega \pm \ell^{-1} e$ one finds

$$S_{\text{CS}}[A^+] - S_{\text{CS}}[A^-] = \frac{k}{2\pi\ell} \int_M \epsilon_{abc} e^a \wedge \left(R^{bc} + \frac{1}{3\ell^2} e^b \wedge e^c \right) + \int_{\partial M} (\cdots).$$

The equality with the Einstein–Hilbert normalization gives $k = \ell/(4G)$.

The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \dots, N - 1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m + 1) = (N - 1)(N + 1) = N^2 - 1.$$

Writing $s = m + 1$ gives one field at every spin $s = 2, \dots, N$.

Low-arity mixed signs

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \quad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\text{ch}} \oplus E_{\perp}) = \det(E_{\text{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

BOOK **XL**

The maximal synthesis

The theorem of the whole construction

Theorem 194.1 (Platonic Calabi–Yau quantum-group theorem). *For every integral Calabi–Yau quantum-group datum $\mathfrak{D}$, the following diagram is defined and commutes up to its canonical coherent homotopies:*

$$\begin{array}{ccccc}
 CC^*(C) & \xrightarrow{\text{FactEnv}_{\text{form}}} & \text{Obs}_{\mathcal{Q}} & \xrightarrow{\text{boundary}} & A_b \\
 \downarrow \text{moduli} & & \downarrow \text{OC} & & \downarrow Z_{\text{ord}} \\
 \mathcal{M}_C^{(2-d)\text{-symp}} & \xrightarrow{\text{Hall}} & H^+ & \xrightarrow{\text{paired double}} & D(H^+, H^0, H^-).
 \end{array}$$

A character functor sends the lower row to the corresponding Hall, PBW, Fock, determinant, or automorphic series. An open–closed complete boundary identifies the middle vertical map with an equivalence. A presentation-complete Hall comparison identifies the lower right quantum group with the quantum group of the geometric BPS sector.

Proof. The upper row is the formal-to-geometric realization and boundary restriction. The left vertical arrow is shifted-symplectic transgression followed by the oriented extension correspondence. The middle vertical arrow is the open–closed action. The lower row is Hall convolution followed by the paired Drinfeld double. Compatibility is the base-change and naturality content of the assembly theorem. The character statement follows from PBW and trace functoriality. The two equivalence clauses are the defining recognition criteria for open–closed completeness and presentation completeness. $\square$

The universal K3 quantum algebra

Let S be a K3 surface. The Nakajima Heisenberg correspondences, the Looijenga–Lunts–Verbitsky action, and the local ADE Yangian correspondences act on distinct geometric towers. Their common action algebra is defined as the complete algebra generated by their kernels inside the endomorphisms of the direct sum of these towers.

Theorem 195.1 (K3 action-envelope theorem). *The completed algebra generated by the Heisenberg, LLV, and local ADE kernels exists as a quotient of their completed amalgamated free product over the common Cartan and grading operators. It is initial among complete algebras receiving the three geometric actions with their actual mixed commutators. Its Tannaka coend reconstructs the corresponding coordinate bialgebra whenever the generated module category is rigid and carries a faithful continuous fibre functor.*

Proof. Complete filtered algebras admit degreewise amalgamated free products. Quotient by the kernel of the combined action map to the endomorphism algebra of the geometric tower. This quotient has the asserted universal property. Under rigidity and a faithful fibre functor, the coend construction gives the coordinate bialgebra and recovers the action category. $\square$

This object gives the phrase “K3 quantum algebra” a precise universal meaning. It preserves the three established geometric sources and stores their mixed commutators instead of identifying them by rank.

The central compact example

For $X = K3 \times E$, the universal Igusa object supplies the algebraic Hall–Borcherds target, its quantum group, its local–wrapped current object, and its automorphic line. A reduced oriented compact Hall source, when supplied, is compared with this target by the geometric recognition theorem. Thus the formal, algebraic, analytic, and automorphic layers are closed; the compact geometric content is concentrated in a single explicit realization datum.

Theorem 196.1 (Universal–geometric equivalence criterion). *Let $\mathbf{Geo}_{K3 \times E}$ be a reduced oriented $K3 \times E$ Hall datum with polarized Gram grading. A comparison*

$$\Phi_{\text{BPS}} : \text{Prim}(\mathbf{Geo}_{K3 \times E}) \longrightarrow \mathfrak{g}_{\Delta_5+}$$

extends to an equivalence of completed doubles, current factorization objects, centres, and determinant lines exactly when:

1. *it maps the geometric simple states to the real and imaginary simple generators;*
2. *it is surjective on primitives;*
3. *its parity-refined PBW character equals the Igusa root character;*
4. *it intertwines the coproduct, Hopf pairing, local–wrapped actions, and orientation line.*

Proof. The first condition gives a presentation morphism. The second and third give a primitive isomorphism by minimal-height PBW recognition. The fourth transports the additional structures. Completion is exact in every proper-height degree, hence the equivalence passes to the inverse limit. $\square$

Coda: geometry becomes algebra and preserves its origin

A Calabi–Yau category supplies duality. A BV theory supplies locality. A boundary supplies order. A second transverse direction supplies exchange. A moduli stack supplies states. Extension correspondences supply multiplication. A pairing supplies the double. A trace supplies a character. A Borchers lift supplies an automorphic denominator. The integrated datum preserves the source of every operation. This preservation is the simplicity of the subject: each geometric motion has one algebraic image, and the whole theory is their functorial composition.

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